

BEYOND STATISTIC PRESERVATION: CAUSAL EVALUATION OF COMPOSED FEDERATED-LEARNING DEFENSES

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ABSTRACT

A statistic can be preserved while attack suppression is not, and no test that admits a composition because one constituent already suppresses the attack can identify whether the downstream mechanism is still responsible for the composition’s suppression. The failure is structural, not statistical: such a gate conditions on a descendant of both the mechanism and the defense’s standalone strength, so the comparison identification needs is missing rather than noisy at any sample size. We therefore replace cross-defense comparison with within-defense intervention: intervene on the adversarial contribution a defense admits, do not infer it from a downstream statistic. Federated learning composes defenses never designed together, so upstream processing can destroy the *statistic* the downstream defense reads and requiring it to be preserved is the natural repair; the screen that follows is a consequence, not the contribution, settled algebraically and only for *positive per-client rescalings*. Across six cells the two designs disagree on four, and ignoring this costs not power but the sign: on the identical cell the outcome-gated ladder moves -0.272 (95% CI $[-0.422, -0.122]$) where the within-defense design moves $+0.098$ ($[+0.018, +0.178]$), intervals excluding zero and not overlapping. Preservation is a hierarchy, and we break its two lowest links by pinning every adversary’s coefficient so that the attack cannot simply be weakened instead. Losing decision invariance costs no suppression: disturbing Krum’s statistic flips its decision in up to 80% of rounds while the adversarial mass it admits never changes in any measured round, suppression staying inside our pre-registered ± 0.15 practical-equivalence margin. Holding decision and adversarial-contribution invariance does not deliver it: `cos_krum`’s selection is bit-identical at every rung while ASR falls 0.173. A test of this shape fails a second, independent way: suppression that is *emergent*, absent from both constituents and present in the composition, cannot satisfy an inherited-suppression gate, so recall is capped structurally, our own lowest-ASR composition being the witness. The negative replicates in a second invariance class, on a second dataset and architecture, and outside the theory’s rescaling family.

1 INTRODUCTION

A test that admits a composition because one constituent already suppresses the attack cannot identify whether the downstream defense is still responsible for suppressing it after composition, and the failure is structural rather than statistical: that gate removes the counterfactual identification requires. We therefore replace cross-defense comparison with within-defense intervention, identifying the effect of intervening on the upstream transform at fixed downstream defense and attack, not a causal model of federated learning. Two measured facts motivate the question: two defenses that each fail alone can jointly suppress an attack, FoolsGold $\rightarrow$ RFA reaching ASR 0.093 where FoolsGold alone reaches 0.732 and RFA alone 0.845; and conversely a composition can leave a downstream statistic bit-identical while the suppression that follows it moves.

Federated learning (McMahan et al., 2017) is vulnerable to data poisoning (Chen et al., 2017; Bagdasaryan et al., 2020), and an arms race has produced defenses (Fung et al., 2020; Pillutla et al.,

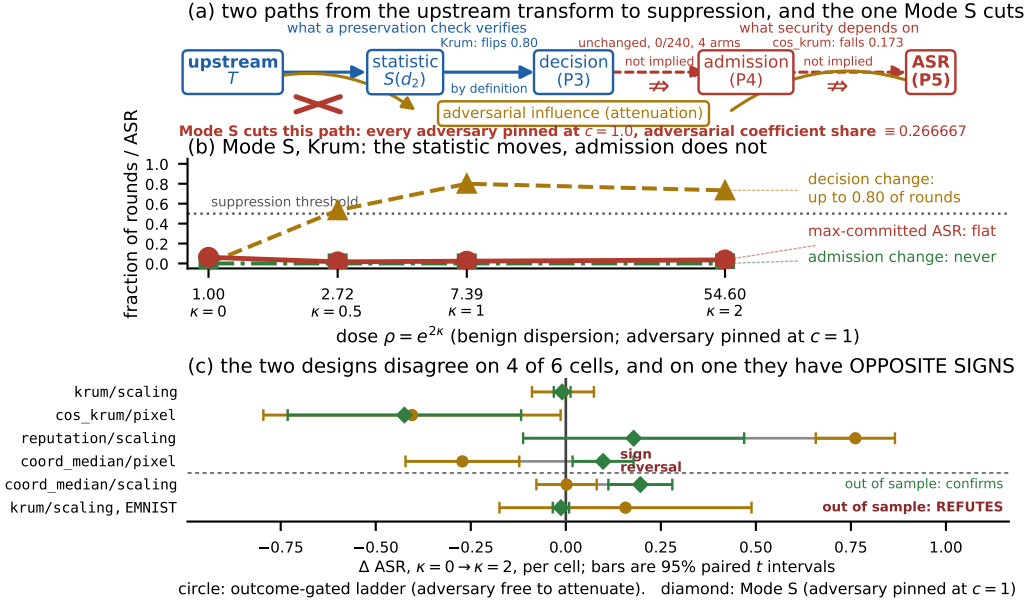


Figure 1: **The paper in one figure: the statistic moves, admission does not, suppression does not follow.** (a) A preservation criterion reasons along one chain, and its levels are operations rather than labels: (P1)/(P2) are $S(d_2)$ ’s value and the ordering it induces, (P3) what d_2 selects, (P4) whether adversarial input enters its aggregate, (P5) max-committed ASR; the first link holds *by definition*. The two lowest links are the ones we break, each broken arrow annotated with the arm that witnesses it. The lower arc is why they must be broken by intervening rather than comparing, and Mode S cuts it by construction; the three annotations are three scopes, not one experiment (App. J). (b) Mode S, Krum under model-scaling (§5): as ρ grows to 54.6 the *decision* flips in up to 80% of rounds while the adversarial mass it admits does not, and max-committed ASR stays inside the pre-registered ± 0.15 practical-equivalence margin ($|\Delta|=0.026$). Error bars here are Student- t 95% intervals on the levels ($n=5$). (c) Six cells, one contrast each, estimated two ways (App. E.3); each connector’s length is how much the answer moves when the attenuation channel is closed. The two designs disagree on 4 of 6 cells, and on `coord_median/pixel` they have opposite signs, -0.272 against $+0.098$. Above the dashed rule are the four cells the withdrawn mechanism was read off, so they are *training data, not evidence for it*; below are its two out-of-sample cells.

2022; Yin et al., 2018; Blanchard et al., 2017) with accuracy/robustness trade-offs and no dominant member (Appendix G). *Can we compose them for stronger robustness?* Composition does not even commute: FoolsGold $\rightarrow$ CoordMedian suppresses a pixel backdoor to ASR 0.076 while the reverse reaches 0.732, a rank aggregator emitting one vector rather than per-client updates, so it cannot be an *upstream* stage and the pair collapses to FoolsGold alone, bit-identically.

Why the question is sharper without a trusted coordinator. A contributor clips, a scheduler reweights by reputation, and whoever merges aggregates robustly: three independently developed mechanisms, nobody owning the stack, so the only check at each seam is that the next stage still reads what it expects. The identification problem this creates is defined independently of model scale, so we study it in a controlled federated setting and claim no transfer beyond it.

Contributions. We propose no new defense. **Preservation is a hierarchy, and statistic-level invariance is not a causal surrogate for security-level preservation.** The hierarchy is what a composition can preserve (Def. 1): the statistic’s value (P1), the ordering it induces (P2), the decision d_2 makes (P3), the adversarial contribution it admits (P4), suppression of the committed attack (P5). *Attack-suppression preservation under a fixed committed attack* is the object throughout, instantiated as ASR, never the statistic or the decision that follows. We claim nothing about confidentiality or availability. The question is scientific, not operational: we ask whether an evaluation can identify

that a mechanism caused an observed suppression, not whether a deployment can detect malicious clients; our instruments read adversary identity and are measurement devices, not defenses.

(1) Identification, and this paper’s primary contribution (§3). *Screen*, throughout, names a rule for choosing which compositions are worth running, never a predictor of security. A screen that keeps only pairs whose constituents already suppress the attack, then asks whether the composition preserved it, gates on a descendant of the quantity it is estimating, and **the consequence is non-identification, not bias**: one cell of the relevant 2×2 is empty for *every* defense menu, so no adjustment set, no reweighting and no sample size recovers it. A second limit follows on the same design and is proved at the same generality: emergent suppression, absent from both constituents, fails such a gate by definition, so **any evaluator using an inherited-suppression gate accepts a bounded blind spot for emergent synergy** (Prop. 2).

(2) Characterization: for which aggregator and transform pairs preservation is decidable before any run (§2). $(P1) \Rightarrow (P2)$ is definitional, $(P4) \not\Rightarrow (P3)$ and $(P4) \not\Rightarrow (P5)$ are not, and which FL statistics are invariant under positive per-client rescalings is decidable algebraically (Prop. 10, with margins on the weighting $\rightarrow$ aggregator class in Thm. 8). That characterization is where the technical work sits, and it is reusable without our screen: for a named aggregator under a positive rescaling, $(P1) - (P3)$ preservation is decidable a priori, with no run.

(3) Intervention: two dissociations, and a sign reversal (§5, §6). Two links in the hierarchy break, at fixed d_2 and attack: Krum’s decision flips in up to 80% of rounds with the adversarial mass it admits unchanged, and, conversely, a bit-identical decision and admitted mass coexist with a change in suppression. The demonstration is a sign inversion and two exact invariances, not attenuation and not small $n=5$ mean differences: on an identical cell the outcome-gated design estimates -0.272 where the intervention estimates $+0.098$. Collider bias is usually taught as attenuation, and here the confounded design does not lose power, it **reverses the direction** of the measured effect inside a live security-evaluation methodology. Reported as refutations: the cross-arm admission ordering we had predicted collapses, and our own lowest-ASR composition is a false negative of our own criterion.

(4) The screen, as a consequence and not a fourth contribution (§4, §6.3). Each clause of the evaluation protocol is forced by a term in the proposition, so the boundary arrives with the design that respects it rather than only a prohibition. **The criterion is an identification protocol, not a predictor**, and we price it as one: on the full menu we compare the three strategies an evaluator can actually run, no screening, statistic-preservation screening, and the mechanism criterion, in precision, recall, runs required, and whether the menu’s strongest composition survives. What we claim as new is the combination, not the causal observation alone: the FL preservation hierarchy, the algebraic invariance characterization of real aggregators, this impossibility result, the within-defense design it forces, and the measured dissociations. The paper is a methodological correction to how composed defenses are evaluated.

What this paper establishes, stated once. $(P1) - (P3)$ do not identify $(P5)$, $(P4)$ is causally closer to $(P5)$ than the statistic feeding it, and within-defense intervention separates a change in d_2 ’s decision from a change in the adversarial influence admitted. Both propositions contain no dataset, architecture or scale, so what carries beyond this paper is the design they license and *not* our screen. Thm. 8 and the invariance classes are proved for positive per-client rescalings into named rank and geometric aggregators, and bound no ASR. The experiments are four aggregators of three structural kinds under two committed attacks on CIFAR-10, one arm replicated on EMNIST-byclass ($n=3$ frozen, $n=5$ pooled), at $n=5$ seeds ($n=8$ for `cos.krum`).

What it does not establish. That $(P4)$ predicts security, or that the screen is a security predictor. Claims do not transfer upward: scope conditions in Table A2, evidence tiers in Table A1, limits in §H, (L1)–(L6). The criterion’s empirical validation is exploratory, and C0 was formulated *after* the first prospective suite and applied to it retrospectively (L4).

2 WHAT PRESERVATION COULD MEAN: A HIERARCHY OF FIVE LEVELS

What does it mean for a composition to preserve a defense? The word hides the ambiguity the rest of this paper turns on: at least three things can be preserved, what the downstream defense *reads* ($P1 - P2$), what it *decides* and what *survives* it ($P3 - P4$), and whether the attack is still *suppressed* ($P5$). They come apart, so we name the levels and ask of each whether it determines the one below.

Where that chain breaks is this paper’s subject: down to (P2) the implications are logical and need no experiment, below it they are causal and settled only by intervening. Fig. 1(a) and Fig. A6 (App. J) draw the chain with its two lower links broken.

To instantiate the hierarchy in FL we characterize the invariance classes, which is where the levels stop being a taxonomy and start being decidable: every per-client defense we consider acts as a positive scalar rescaling $T(u_i) = c_i u_i$, so the algebra fixes which level a given composition sits at (Prop. 10). Cosine similarity is exactly invariant, so (P2) holds for every $\cdot \rightarrow$ FoolsGold composition; coordinate ordering survives only under a bounded-ratio condition; consensus distance and residual magnitude are not invariant at all, so $\cdot \rightarrow$ Reputation compositions fail (P2); norm clipping does not blind FoolsGold, though it does blind Reputation.

Definition 1 (Preservation levels). *For an upstream transform T and a downstream defense d_2 reading statistic S :*

- (P1) **Value invariance:** $S(T(U)) = S(U)$.
- (P2) **Ordering invariance:** $S(T(U)) = \varphi(S(U))$ for some φ preserving the ordering S induces on clients.
- (P3) **Decision invariance:** the client or coordinate d_2 actually selects is unchanged.
- (P4) **Adversarial contribution (admission) invariance:** the adversarial input d_2 admits into its aggregate is unchanged. Admission is the pre-registered term throughout for how much adversarial input survives d_2 .
- (P5) **Attack-suppression preservation:** ASR stays below threshold, for the fixed committed attack the arm is run against.

We target (P4) $\nRightarrow$ (P3) in §5 and (P4) $\nRightarrow$ (P5) in §6.2. The conditions (C0)–(C3) the levels induce are in App. B and C.

Five words, fixed here and used in one sense throughout.

decision	which client or coordinate d_2 selects (P3)
admission	whether adversarial input enters d_2 ’s aggregate at all (P4)
influence	Λ_a , the weight the adversary’s direction carries in that aggregate
suppression	max-committed ASR, the security-level object (P5)
attenuation	the adversary’s own update shrunk upstream, reaching ASR without passing through S

Admission and influence are never interchangeable, and attenuation is the channel an intervention on T must close before either can be read.

(P4) is the first level at which adversarial influence enters the causal graph, which is why we intervene there rather than on the statistic. It is privileged for its semantics, not for sufficiency: what singles (P4) out is that its content is adversarial influence rather than d_2 ’s internal operation. It is instantiated per aggregator and is not one scalar comparable across defenses: Krum’s and `cos.krum`’s selected-client indicator, `coord.median`’s selected-coordinate share, reputation’s adversarial weight share. So our (P4) result is four within-family statements, not one cross-defense quantity, and we never difference or rank across them.

Quantitative margins for the weighting $\rightarrow$ aggregator case are a supporting result and stay in the appendix (Thm. 8, App. C.3); they certify mechanism preservation, they are safe at *both* extremes of the weight ratio so ρ alone mispredicts, and **heterogeneity closes the margin**, RFA’s collapsing from 3.24 to 1.01 at Dirichlet $\alpha=0.1$, where the bound goes vacuous (L5).

3 THE TESTABILITY BOUNDARY

Why can we not simply compare many defense pairs? The natural test keeps pairs whose constituents already suppress the attack and asks whether the composition preserves it. That is exactly where identification is lost, and it is this paper’s central claim. In words, before any notation: admit a pair only when one constituent already suppresses the attack, and the downstream defense is thereby guaranteed to suppress it alone, so no admitted pair exists in which that defense fails alone while the composition succeeds. **That missing comparison is the only one that would credit the suppression to the mechanism rather than to the downstream defense’s own strength.** *Whose protocol is this?* We do not claim the gate is common in published evaluations: it is the gate *our* own criterion applies, which is why the boundary costs us our own lowest-ASR composition (§6.3).

What exactly is the estimand? Fix a downstream defense d_2 and a committed attack a . The **treatment** T is the upstream per-client transformation applied before d_2 aggregates, with $T=\text{id}$ the control that is bit-identical to d_2 standalone. The **outcome** Y is $\text{ASR}(T, d_2, a)$. The **mechanism-preservation variable** $P \in \{0, 1\}$ records whether T preserves the statistic d_2 reads, at whichever level of Def. 1 is at issue; P is a deterministic function of T and d_2 's definition, not a measured covariate. The causal quantity is

$$\tau(T) = \mathbb{E}[Y \mid \text{do}(T), d_2, a] - \mathbb{E}[Y \mid \text{do}(T=\text{id}), d_2, a].$$

We do not estimate the causal effect of P itself. P is not manipulable apart from T , being a property of the pair, so “the effect of preservation” names no intervention. We estimate the effect of intervening on the upstream transformation while holding d_2 and a fixed, and what we report about P is whether $\tau(T) \neq 0$ occurs while P holds, and whether $\tau(T)=0$ occurs while P fails. Both occur (§5), which is what refutes P as a surrogate for Y . The design this rules out is the cross-defense contrast, which varies d_2 across a menu and so changes the conditioning event instead of intervening: that is the object of the proposition next.

Proposition 1 (Identification: constituent-effect contrasts cannot identify composition effects under inherited-effect gating). *Let a screen certify $(d_1 \rightarrow d_2)$ against attack a only if both (i) an **eligibility gate** $E(d_1, d_2, a) := \min_{d \in \{d_1, d_2\}} M(d, a) < \theta$ holds, where M and θ are the same outcome metric and threshold the screen predicts for the composition, and (ii) a **mechanism condition** $P(d_1, d_2)$ holds, determined by the two defenses' definitions alone. Suppose the screen is applied only where the downstream mechanism is load-bearing, $M(d_1, a) \geq \theta$. Consider estimating P 's contribution to $M(d_1 \rightarrow d_2, a)$ from contrasts across a defense menu $\mathcal{D}$. Then every certified pair satisfies $M(d_2, a) < \theta$, so of the four cells of $\{P, \neg P\} \times \{M(d_2, a) < \theta, M(d_2, a) \geq \theta\}$ the certified set occupies at most two, and **the cell needed to separate P from d_2 's standalone effectiveness is empty by the gate, for every $\mathcal{D}$. No contrast varying d_2 identifies P 's contribution. Identification requires holding d_2, a and therefore E fixed while intervening on the upstream transform: **the admissible designs are within-defense.*****

Nothing in this statement is about federated learning. M is any outcome metric, θ any threshold, P any condition on the two mechanisms' definitions, and $\mathcal{D}$ any menu; App. C.2 substitutes our protocol into each, term by term. The proof is short (App. C.1): the load-bearing condition excludes d_1 from attaining the minimum in (i), so certification forces $M(d_2, a) < \theta$ whatever $\mathcal{D}$ is. What takes work is deciding P (§2).

Is this not ordinary collider bias? Half of it is, and that half is textbook (Elwert & Winship, 2014; Pearl, 2009): we claim no credit for it. What the gate adds to that textbook structure, drawn beside ours in Fig. A2 (App. C.2), is that the criterion does not merely *risk* collider bias: it **removes the counterfactual support identification requires, at any sample size and any menu size**, a structural *positivity* violation rather than a confounding one, while reading as a reasonableness filter. The gate cannot be dropped either: without it the screen would certify pairs neither constituent suppresses, so wherever our conditions certify a pair, a low composition ASR is attributable to d_2 's standalone strength rather than to preservation, the reading we held ourselves (§6.1).

A screen of this shape fails a second, independent way, bounding what it can *find* rather than what it can be *validated* on.

Proposition 2 (Coverage: inherited-suppression screens cannot certify emergent synergy). *Call $(d_1 \rightarrow d_2)$'s suppression of a **emergent** if $M(d_1 \rightarrow d_2, a) < \theta$ while $\min_{d \in \{d_1, d_2\}} M(d, a) \geq \theta$. No emergent composition satisfies the eligibility gate of Prop. 1. Hence, writing $\mathcal{S}$ for the compositions with $M < \theta$ and $\mathcal{E} \subseteq \mathcal{S}$ for the emergent ones, any such screen's recall obeys $\text{recall} \leq 1 - |\mathcal{E}|/|\mathcal{S}|$, on every menu and every attack.*

Our own lowest-ASR composition is the witness: §1's FG→RFA, at max-committed ASR 0.093 ($n=30$, CI [0.062, 0.123]) against constituents at 0.732 and 0.845, is emergent and a false negative of our own criterion, and with $|\mathcal{E}|/|\mathcal{S}|=1/5$ the bound is 80%, above the 40% we report. **Any evaluator using an inherited-suppression gate must accept a structural blind spot for emergent synergy:** the bound is proved for every screen requiring a constituent to suppress the attack independently, and it splits a menu into *inheritance*, which we screen for, and *emergence*, a search problem we do not solve (App. D.3). A screen reaching $\mathcal{E}$ would have to read the composition's own suppression, the outcome it exists to avoid conditioning on: Prop. 1's boundary met from the other side. Our blind spot is one condition, C1, so the 40% is an *operating point*, not a ceiling (§6.3).

4 A CAUSAL EVALUATION PROTOCOL

If the comparison is inadmissible, what replaces it? Not a better criterion but a different design, and Prop. 1 determines it: do not select across defenses; hold the downstream defense and the attack fixed, intervene on the upstream transformation, and measure the causal channels separately. Each clause is forced by a term in the proposition rather than chosen for convenience.

- (1) **Hold d_2 and the attack fixed.** *Forced by* the gate $E(d_1, d_2, a)$: vary d_2 or a and the two sides of a contrast are conditioned on different events, so they are not comparable at any sample size.
- (2) **Intervene on the upstream transform.** *Forced by* the empty cell: variation in d_2 is exactly what the gate turns into missing support, so the intervention has to live on d_1 .
- (3) **Pin the adversarial coefficient channel.** *Forced by* what the proposition does *not* constrain: nothing prevents T from reaching the outcome without passing through d_2 's statistic at all, by attenuating whichever client carries the poison (Lem. 2).
- (4) **Measure the channels, not only the outcome.** *Forced by* what P asserts: a claim about a mechanism, so a contrast reporting only ASR cannot say which channel carried the effect, and every channel the intervention opens must be reported, including the accidental ones.

Operationally, before any of that. Name the downstream mechanism and its invariance class; state *which* level of Def. 1 is being claimed, since a claim that does not name its level is not checkable; check the upstream transform lies in the class the argument covers, because one that alters participation, carries cross-round state, or mixes clients is covered by no invariance argument we know of, ours included; and report the level at which the chain broke. Following the four clauses turned our own -0.272 into $+0.098$ on an identical cell (§5).

The protocol's boundary, measured rather than asserted. Every clause holds the attack fixed, so the protocol establishes **attack-suppression preservation under a fixed committed attack** and nothing wider, which is why we do not call the object "security". Adaptive security is a different estimand. One whitebox composition-aware *adaptive* stress test measures that boundary: solving the joint FoolsGold/RFA constraint knowing both stages raises $FG \rightarrow RFA$ to $ASR\ 0.167 \pm 0.069$, $3.7 \times$ its seed-matched 0.045 on the pixel arm it is built from. **Inherited robustness is not adaptive robustness**, and a protocol stated against an adaptive adversary is the sharpest thing this framework lacks (App. M).

5 STATISTIC PRESERVATION DOES NOT IDENTIFY ATTACK-SUPPRESSION PRESERVATION

Does disturbing what a defense reads actually cost suppression? The claim: it need not, and **the design that says otherwise gets the sign backwards**. We adjudicate the implication that losing (P3) means losing (P5) by oracle intervention on two arms in two invariance classes, not by a general law.

The instrument, as a counterfactual. The clean question asks what happens when the downstream statistic changes but the adversary cannot become weaker. **Mode S** enforces exactly that: it pins every adversary's coefficient at $c=1.0$ and spreads only the benign clients over $\rho = e^{2\kappa}$, holding the adversarial coefficient share constant at 0.266667 to within float32 read-back noise, closing Figure 1(a)'s attenuation path, so a rise in ASR here cannot be attenuation. §6's payload-share instrument is the other such device (Table A2, App. E.2).

Two readings of preservation, frozen against each other. *The statistic reading:* what matters is disturbance of d_2 's statistic, read off its *decisions*. *The admission reading:* what matters is the *adversarial input* d_2 admits. *Admission, influence and suppression* are used here in the one sense §2's five words fix (App. K). Mode S disturbs Krum's selection heavily while leaving the mass it admits untouched, so that arm adjudicates (Table A11): as ρ climbs to 54.6 the decision changes in up to 0.80 of rounds, the adversarial mass admitted in none, and ASR shows no evidence of the predicted increase, staying inside the pre-registered practical-equivalence margin ($p_{\uparrow}=0.70$, $|\Delta|=0.026 < 0.15$).

Table 1: **The four channels, all four aggregators, under Mode S**; endpoint change $\kappa=0 \rightarrow \kappa=2$. Δ agg. is the relative displacement of the aggregate the defense *emits*, $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$, so it is what enters training and *not* the value of a scoring statistic. Δ adm. is the *support* of the admitted adversarial mass and $\Delta\Lambda_a$ is the separate influence question, the weight the adversary’s direction carries; the two are never interchangeable. Admission never moves in any measured round of any arm. The table does *not* order the arms, $\Delta\Lambda_a$ and Δ ASR disagreeing in rank. Per-row provenance, the round counts, and the emitter are in App. E.2.

Aggregator	Kind	Δ agg.	Δ dec.	Δ adm.	$\Delta\Lambda_a$	Δ ASR
Krum	selector (pairwise distance)	0.892	0.733	0.000	0.000	-0.026
Reputation	weighted averager (cross-round state)	0.659	1.000	0.000	0.020	+0.178
Cosine-Krum	selector (cosine distance)	1.297	0.000	0.000	0.000	-0.425
Coord. median	coordinate-wise order statistic	0.670	0.482	0.000	0.033	+0.098
Krum (EMNIST-byclass)	selector (pairwise distance)	0.842	0.643	0.000	0.000	-0.017
Krum, score-only	selector, magnitude channel closed	0.986	0.733	0.000	0.000	-0.023

Here practical equivalence is established, not merely consistent: TOST against that frozen ± 0.15 margin, an evaluation-design tolerance and not a claim that 15 ASR points are universally negligible, gives $p=0.0014$ at $n=5$, and a pre-registered top-up to $n=20$ confirms it at $\Delta = -0.010$, $[-0.032, +0.012]$, half-width 14.7% of the margin (App. L, which scores every arm this way, including the one the frozen point rule passes and TOST does not). **The statistic reading is refuted:** statistic and decision disturbance need not change suppression once the adversarial coefficient share is held fixed. *The zero is a floor here*, Krum admitting no adversary at any rung, so unchanged admission is not by itself decisive, and we claim no per-channel decomposition of Δ ASR (§ E.2.1).

The table contains its own positive control, and it is what makes the negative readable, since “we moved something and nothing happened” cannot distinguish a real dissociation from an instrument too blunt to move. Both arms whose *influence* channel moves, reputation and coord.median, are arms whose suppression moves with it, and every arm whose *decision* moves at $\Delta\Lambda_a=0.000$ stays inside the practical-equivalence margin. The converse fails, and cos_krum is why: its ASR moves at $\Delta\Lambda_a=0.000$ (§6.2’s converse witness). So the instrument does register suppression changes when adversarial influence changes ($\Delta(P4) \neq 0 \Rightarrow \Delta\text{ASR} \neq 0$), and registers none from a decision change alone, *even at 0.733 of rounds*. Read across all four structural kinds rather than off Krum’s admission floor, that licenses one general statement: *large decision changes need not change adversarial influence or attack suppression* (§6.2; App. E.2).

What the instrument does not hold fixed, and the pre-registered control that closes it. Mode S does not identify the effect of statistic disturbance in isolation: it identifies the effect of the upstream intervention while holding the adversarial coefficient share fixed, which is what lets statistic and decision disturbance be *dissociated* from suppression rather than shown to be inert. Mode S pins the *adversarial* coefficients and nothing else: the dose also moves Krum’s pairwise distances, *which* benign client it selects, and the magnitude and geometry of the update actually aggregated, that third channel reading Δ agg. = 0.892 here.

Score-only Krum closes that third channel: Krum scores the transformed stack and so makes the same selection, but the *untransformed* selected update is aggregated, preserving the decision channel and removing the magnitude change. Its rules were frozen before any score-only ASR existed. We find $\Delta = -0.023$ ($0.062 \rightarrow 0.039$) against -0.026 uncontrolled on the identical cell, at the same Δ dec. of 0.733, so the central claim is instrumented against two independent confounds, attenuation and magnitude (App. E.2, § S9).

A pre-registered replication that refutes the positive half. coord.median under the pixel backdoor is *rank-discordant*: of the interpretable arms it has the smallest decision change (0.482) and the largest admission-level change ($\Delta\Lambda_a=0.033$), so the two readings predict opposite rank positions for its rise, and thresholds were frozen against the two published rises (confirm if $\Delta > +0.178$, refute if $\Delta < +0.150$). We find $\Delta = +0.098$, which **refutes the admission ordering**: the arm with the largest admission-level change rose *less* than one with a smaller change, and the decision reading does not predict it either ($R^2=0.239$ pooled). What replicates is the negative on

a coordinate-wise order statistic: a decision change of 0.482 costs no suppression by the frozen margin, at accuracy 0.698. Identity ASR here is 0.443, so equivalence is to the *identity rung*, and the rise is positive in 5/5 seeds (post-hoc): small and consistent, not zero.

The confounded design gets the causal sign wrong. On this identical cell (matched aggregate displacement, matched decision change, matched clean accuracy, Table A3), closing one channel reverses the sign of the measured effect, and Fig. 1(c) is that contrast: the outcome-gated ladder of §6.1, whose adversary is free to attenuate, gives $\Delta\text{ASR} = -0.272$ ($p_{\downarrow} = 0.0015$; paired 95% interval $[-0.422, -0.122]$), where the Mode S intervention of §5, which pins that adversary at $c=1$, gives $+0.098$ on the same cell ($[+0.018, +0.178]$). The confounded design does not lose power, it gets the sign wrong, and that is the one result to carry away: a criterion calibrated against that contrast would have been calibrated against the confound. **Downstream-statistic preservation is not sufficient to identify attack-suppression preservation** (Prop. 11): *within the rescaling class and defense family measured here*, no predicate of the downstream statistic alone determines the admitted adversarial mass, and none of that mass alone determines suppression: this is the claim behind the title’s first clause, established by counterexample (Table A2). A preserved statistic is not thereby uninformative; it is not *sufficient*, remaining evidence about what d_2 reads and none about what it suppresses.

Six cells, not one (App. E.3). One cell cannot separate a property of the evaluation design from one of this aggregator and attack, so we ran the contrast on six. The two designs disagree on four of them, across two attacks, two datasets and two architectures, and `coord_median` under *model scaling* is one ($+0.002$ against $+0.196$), so the reversal is not specific to the pixel backdoor. A mechanism we had frozen for *when* they disagree is refuted by one of the two out-of-sample cells and withdrawn, which is why §4 says to intervene rather than consult a rule.

6 WHAT THE BOUNDARY LICENSES, AND WHAT WE FOUND

Does the pattern hold beyond the adjudicating arm, and what does a screen then buy? We validate on CIFAR-10 with a 4-layer CNN, $N=10$ clients and $f=0.2$ adversarial under the two committed attacks of §1, at $n=5$ seeds per cell ($n=8$ for `cos_krum`, $n=3$ frozen and $n=5$ pooled on EMNIST-byclass); ASR means *max-committed ASR*, and the rest of the protocol is fixed in the supplement (§ S6). The core evidence is a seed-matched intervention on one cell, not a mean comparison across cells, and Table A1 bands it into three unequal strengths, with any cross-arm ordering and the exact (P4)→(P5) magnitudes in the weakest and nothing resting on them.

6.1 THE CONFOUNDED LADDER THAT MOTIVATED THE INSTRUMENT, AND THE SIGN IT GOT WRONG

Our earlier suites tried to validate ordering preservation by *choosing a different downstream defense*, the move Prop. 1 rules out. Varying only the *dose* at fixed d_2 and attack ($\kappa=0$ bit-identical to d_2 standalone) refutes that suite’s pooled prediction that disturbance predicts each arm’s rise: $\rho_s = +0.351$, one-sided $p=0.091$ against a frozen $p < 0.05$, and in three of four arms ASR *falls* as the dose rises (80 runs; Table A10). **The fault is ours and is now measured:** fixing $\text{mean}(c)=1$ leaves relative weights free, so a large spread mostly *dilutes* whichever client carries the poison: attenuation, not statistic disturbance (§ S7, § S10).

6.2 THE DISSOCIATION HOLDS ACROSS FOUR AGGREGATORS AND REPLICATES ON A SECOND DATASET

Applying the instrument to that ladder, most of the one confirmation was attenuation: with the adversary’s share pinned, reputation’s rise at $\rho=54.6$ shrinks from $+0.762$ to $+0.178$, and what remains tracks a small *admission* change, not the large decision change (App. E.2).

The Krum dissociation replicates on a second dataset and architecture: a *pre-registered* rerun on EMNIST-byclass/simple_cnn moves the decision on 0.643–0.714 of rounds with the adversary never admitted and suppression inside the frozen ± 0.15 margin, $\Delta = -0.017$ at the frozen $n=3$ and -0.013 at the pooled $n=5$ beside CIFAR-10’s -0.026 ; every reading survives any margin above 0.034, itself read off that $n=5$ interval (the supplement § S12).

The dissociation also replicates outside the rescaling family the theory covers: masking a random 0.8 fraction of each *benign* client’s coordinates at fixed L_2 norm leaves adversarial updates *exactly* untouched, so nothing we prove predicts it: the decision changes in 0.467 of rounds at admitted adversarial mass 0.000 and suppression does not follow, neither pre-registered refuting branch firing (App. E.2.1). One mask is not the class of all non-rescalings.

Two arms support the account rather than carry it. `cos_krum`’s selection is bit-identical at every rung yet ASR *falls* 0.173 at the first accuracy-gate-passing rung, from an identity 0.584 at which it never suppresses (one-sided $p=0.044$, $n=8$, 5/8 seeds), the converse witness; and pinning payload share at fixed dispersion separates the theorem’s two regimes, confirmed for `coord_median` and refuted for reputation (App. E.2, D, Table A13).

6.3 WHAT THE SCREEN CANNOT FIND, AND WHAT A CONTROLLED EVALUATION SHOULD RUN INSTEAD

What does the screen’s own failure show? Prop. 2’s prediction is paid out on the three strategies an evaluator can actually run, priced on our 42 ordered pairs at the same seeds and at the 10 runs per pair plus the 60 standalone runs C1 requires (App. D.4):

Strategy	selected	precision	recall	mean ASR	runs	reaches FG→RFA
no screen (all pairs)	42	12%	100%	0.743	420	yes
C2 alone (statistic preservation)	14	36%	100% [†]	0.588	140	yes
C1∧C2 (the criterion)	2	100%	40%	0.242	80	no

[†] development set; 11 of the 14 are in-sample calls.

The screen fails exactly where Prop. 2 says it must, and that is what the row is for. It buys 100% precision for a fifth of the runs and pays by discarding FG→RFA, the lowest-ASR composition in the menu at 0.045 on that common scoring (0.093 at $n=30$, §F). **Statistic-preservation screening, the rule this paper argues is not a causal surrogate, is the one that keeps it.** The C2 row is weaker than its label too, its in-sample calls (table note) resting on per-pair judgement rather than algebra and no low-ASR pair lying among the 3 that Prop. 10 settles (App. D.4, which also prices both against size-matched random draws). The recommendation splits by purpose: to find a menu’s strongest composition, do not screen on inherited suppression at all; to reach a defensible verdict on a small set cheaply, screen and report the recall you gave up. Of two prospective tests the first prices the menu and the second the criterion, the latter a census of the 3×3 region our invariance result settles exactly, committed at 96f8856, where all four registered cells came out as predicted; **that label does not survive more seeds**, the C1 input certifying the selected pair reading 0.300 over the frozen 3 seeds and 0.584 at $n=8$, where C1 fails and the criterion certifies nothing (§D.1, App. D.5).

7 CONCLUSION

The finding. Preserving a defense statistic is not sufficient to identify preserved attack suppression, and each direction has a measured witness: an intervention changes a defense’s statistic and decision substantially without changing the adversarial mass admitted or the suppression that follows, and conversely a bit-identical statistic and decision coexist with a change in suppression. The reason is structural: an outcome-gated cross-defense comparison conditions on the outcome it seeks to explain, which no additional data repairs (Prop. 1), and a screen of that shape cannot find emergent suppression either, our own lowest-ASR composition being the witness (Prop. 2).

What changes for an evaluator, without using our criterion at all: *do not* read a preserved statistic or an unchanged decision as evidence that suppression survived (Table 1); *do not* select the pairs you compare on the outcome you are about to attribute, the design that reversed our own sign; *do* hold d_2 and the attack fixed, intervene upstream, measure (P4) rather than the statistic feeding it, and report the level at which the chain broke. Reading (P4) requires adversary identity, so this identifies suppression preservation rather than screening for it (App. I). The open question: **can (P4) be estimated without knowing which clients are malicious?**

ETHICS STATEMENT

This paper studies poisoning attacks in order to evaluate defenses against them, so its subject matter is dual-use. Three things bound the risk. Every attack we run is already published (model-scaling (Bagdasaryan et al., 2020), pixel-pattern backdoors (Chen et al., 2017), DBA (Xie et al., 2020)) and we introduce no new attack or new attack capability. All experiments are simulations on public benchmark datasets (CIFAR-10, EMNIST-byclass, UCI Spambase) with no human subjects, no personal data and no deployed system involved. Our headline result is defensive and cautionary rather than offensive: it tells practitioners that a preserved downstream statistic is *not* evidence that a composed defense stack still suppresses an attack, which removes a false assurance rather than supplying an attacker with one. The instruments we build (Mode S, the payload-share instrument, score-only Krum) require knowing which clients are malicious and are therefore measurement devices, not deployable defenses; we state this at each use so that no reader mistakes them for a recommendation.

REPRODUCIBILITY STATEMENT

Every claim in the paper is tagged *general*, *formal* or *empirical* at the point it is made, and Table A2 states the scope conditions on each. For the formal claims, Propositions 1 and 2 are stated in full in the body and proved in the appendix, and Proposition 10 and Theorem 8 are stated and proved in the appendix and referenced from the body. For the empirical claims: each pre-registered arm has a decision rule git-committed before the corresponding runs existed, with the commit hash recorded in the text and a runner that refuses to start unless its own recorded commit matches; every table cell is emitted by a generator that reads only frozen artifacts rather than being transcribed; sample sizes, seed ranges and per-seed values are given for every arm, with per-seed detail in the appendix and the supplement; and the figures are recomputed per seed by the same scripts. Where a number was revised we report the revision and its cause rather than the corrected value alone, and where a derived condition turned out to be empty on our data we report it as empty rather than omitting it (§C.3). **No code or data package accompanies this submission**, and throughout this paper “the supplement” means the companion PDF only. What will be released on publication is specific rather than a general promise: every experiment runner and measurement script, the `results/` JSON artifacts every reported number is read from, the simulator itself, the generators that emit the figures, and the 16 pre-registration documents fixing the decision rules for every pre-registered arm. Everything this paper’s tables and figures are computed from runs from the repository root on Python 3 with `numpy`, `scipy`, `matplotlib`, `torch` and `torchvision`. The analysis entry points behind them are **idempotent**: they read the frozen artifacts and rewrite them byte-identically, so every number we report is a deterministic function of a frozen artifact rather than of a re-run, and no table cell can drift between builds. What the paper itself provides is the pre-registration commit hashes, the frozen decision rules they fix, and per-seed values for every arm, which is what is needed to check the reported numbers against a reimplementation.

AI USE STATEMENT

In this work, we used generative AI tools as a writing and analysis assistant: for prose editing and revision of the manuscript, for $\LaTeX$ restructuring, and for writing verification and analysis scripts that check the manuscript against the frozen result artifacts. We did not use generative AI tools to generate research ideas, to formulate the hypotheses, to design the experiments or pre-registrations, or to produce any experimental result, and no claim, number, figure or table in this paper was authored by an AI tool. We have reviewed all AI-assisted work: AI-assisted prose was checked against the underlying artifacts sentence by sentence, and AI-written analysis scripts were run against frozen inputs whose outputs are reproducible independently of the tool. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

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A THE THREE TIERS OF EVIDENCE

Three tiers of evidence, of very unequal weight, never pooled. Every empirical claim about the preservation hierarchy or the criterion is tagged with one of these three words at the point it is made, so a reader can look up what licenses it instead of reconstructing it from hedges.

B PRELIMINARIES AND THE PRESERVATION LEVELS

Federated learning. N clients each hold a local dataset $\mathcal{D}_i \sim P_i$; the server learns $w^* = \arg \min_w \sum_i \frac{|\mathcal{D}_i|}{|\mathcal{D}|} F_i(w)$. In FedAvg (McMahan et al., 2017), each round the server broadcasts w^t , selected clients run local SGD, and the server aggregates $w^{t+1} = w^t + \frac{1}{|S_t|} \sum_{i \in S_t} \Delta_i^t$.

Poisoning and robust aggregation. An adversary controlling a fraction f of clients can flip labels (Chen et al., 2017), inject backdoor triggers (Bagdasaryan et al., 2020; Xie et al., 2020), or scale malicious updates to dominate aggregation (Bagdasaryan et al., 2020). The server may replace FedAvg with a Byzantine-robust rule: Krum (Blanchard et al., 2017), Trimmed Mean, Coordinate-wise Median, Norm Clipping (Sun et al., 2019), or RFA (geometric median) (Pillutla et al., 2022), each imposing an accuracy/robustness trade-off.

Game theory. A normal-form game is $G = (\mathcal{N}, \{\mathcal{S}_i\}, \{u_i\})$; an equilibrium is *pure* when every player’s strategy puts all its mass on one action and *mixed* otherwise. **No claim of this paper is game-theoretic:** this is needed only to read the residual scale check of Appendix N.1, and the type-space and Bayes-Nash machinery an earlier version used is not needed here.

C THE SCREEN, THE IDENTIFICATION BOUNDARY, AND THE PROOFS

The paper’s central practical question is *which* defense pairs compose well under deterministic joint application. Specifically, the criterion screens for *mechanism-preserving composition*: when one constituent defense is already individually effective (C1), can the other defense be added without breaking that suppression? This is the operationally relevant question (the criterion is not a general characterization of all composable pairs, but a sufficient condition for *mechanism preservation* among pairs where at least one component has known effectiveness), never a sufficient condition for low ASR. We answer it first, with a checkable criterion, and treat

Table A1: **The three tiers of evidence, of very unequal weight and never pooled**, and the *orthogonal* strength axis. Every empirical claim in the paper carries one of the three tier words at the point it is made; the tiers are never combined into a single verdict. The lower panel is a second, independent axis: what a claim’s *design* licenses (upper panel) and how strongly the *statistics* support it are different questions, and a Causal-tier claim can still sit in the weakest band.

tier	design	what it licenses
Causal	within-defense intervention at fixed d_2 , attack and C1 input: Mode S, Mode A (§E.2)	statements about the <i>mechanism</i> , scoped to these cells
Replication	EMNIST-byclass, second dataset <i>and</i> architecture ($\Delta=-0.017$), and <code>coord.median</code> , a second invariance class (+0.098)	“replicated on a second dataset and architecture”, never “generalizes”
Screening	the outcome-gated 42-pair suite	evaluation prioritization only ; by Proposition 6 it cannot identify the mechanism’s contribution at any sample size
Statistical strength , an axis orthogonal to the tiers above. Read this before quoting any number.		
Strong	the invariance results (Prop. 10, Thm. 8) and the identification boundary (Prop. 6), which are <i>proofs</i> and carry no n ; <code>cos_krum</code> ’s exact score-only identity, which is bit-identical rather than estimated; the pre-registered Mode S intervention, frozen before any of its ASR existed; and the <i>direction</i> of the negative, which replicates on a second dataset and architecture and outside the rescaling family	
Moderate	the ASR practical-equivalence results at $n=5$, one of which was topped up to $n=20$ under a pre-registered clause and held ($\Delta=-0.010$); at $n=5$ these are TOST verdicts against a frozen margin, not tight estimates	
Weak, exploratory	any <i>cross-arm quantitative ordering</i> , which the admission reading predicted and which Table 1 refutes; and the exact (P4)→(P5) <i>magnitudes</i> , for which we claim the implication and not the size. Nothing in the paper’s argument rests on this band	

the alternative, temporal mixing, as outside the scope of this paper. **But the practical question is not what the paper establishes:** what it establishes is that preservation of the statistic the screen checks is not causally tied to the security the screen predicts, so the criterion below should be read as the application §D.3 scopes it to (an ordering over an evaluation budget) and not as a deployment recommendation.

Observe that each robust aggregation defense d operates through an *admission signal* $\sigma(u)$ applied to client update u : it accepts, clips, or down-weights u based on whether $\sigma(u)$ exceeds a threshold. Crucially, the admission signal varies by defense family:

- **Weighting defenses** (reputation, FoolsGold): $\sigma(u)$ = distance from consensus (ℓ_2) or pairwise similarity.
- **Clipping defenses** (NormClip): $\sigma(u) = \|u\|_2$.
- **Rank-based defenses** (TrimmedMean, CoordMedian): $\sigma_k(u)$ = coordinate-wise rank of u_k .
- **Geometric defenses** (RFA): $\sigma(u)$ = distance from geometric median.

When two defenses are applied sequentially (d_1 transforms the updates, d_2 aggregates the result), d_1 ’s transformation may *corrupt* d_2 ’s signal or vice versa.

Criterion 3 (Signal-Preservation Composability). *Let d_1 and d_2 be defense mechanisms. The following sufficient conditions guarantee **preservation of d_2 ’s suppression mechanism under composition, conditional on the standalone suppression C1 asserts**. They are **not** an independent guarantee of low composition ASR: Theorem 8, which proves them for one class, bounds no ASR. Stated once so the scope cannot be misread:*

$$C1 \wedge C2 \wedge C3 \Rightarrow d_2\text{'s suppression mechanism survives composition, levels (P1)–(P3) of Def. 2.}$$

$$C1 \wedge C2 \wedge C3 \not\Rightarrow \text{low composition ASR.}$$

The conditions are of two different kinds, and we group them accordingly, because conflating an empirical screen with a mechanistic property is the main source of confusion about what the criterion claims.

Stage 0: applicability. *Before either kind of condition is informative, the question the criterion asks has to be live.*

(C0) The downstream mechanism is load-bearing. $\text{ASR}(d_1, a) \geq 0.5$ for at least one committed attack a . *C2 and C3 concern preservation of d_2 ’s admission signal, so they say nothing when d_1 alone*

already suppresses everything: the composition inherits that suppression, destroying d_2 's signal costs nothing, and a correct prediction would be correct for a reason the criterion did not supply. The asymmetry is deliberate: a strong d_2 is the intended case, since asking whether upstream processing breaks an effective downstream defense is the framework's whole question, whereas a strong d_1 makes the question moot. C0 was **formulated post-hoc** and its provenance is in Remark 5.

The empirical screening condition. This one is measured rather than read off definitions, and it is the gate the identification boundary is about.

(C1) Standalone suppression. Some constituent suppresses the committed attack on its own: $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a) < 0.5$, quantified per attack. With C0 this forces the suppressing constituent to be d_2 (Lemma 1), which is exactly what makes the gate a descendant of the quantity the screen is trying to estimate (Proposition 6).

Mechanism-preservation conditions. These are what Theorem 8 proves for the weighting $\rightarrow$ aggregator class, and they are determined by d_1 's and d_2 's definitions alone, no experiment required. **What we recommend is an evaluation-prioritization criterion, $C1 \wedge C2$, because that is the whole of what our evidence supports.** We do call it a screen throughout, in the ordinary sense of a cheap test run before an expensive one. **We disclaim predictive validity explicitly, here at first use:** the criterion orders candidate compositions for evaluation, and it does not predict whether a composition is secure. The pooled two-channel model over 26 cells is unconfirmed ($R^2=0.239$), and our own pre-registered replication **refuted** the cross-arm admission ordering the criterion would need in order to predict (§E.2); recall is 40% (Proposition 7), so the negative direction carries no information either. **It is never a security screen, and a PASS is an evaluation ordering rather than a deployment decision** (Table A2). C3 is retained as a **margin diagnostic** rather than a gate, because on our data it never binds on a pair $C1 \wedge C2$ already admits and shows no incremental predictive value (§D.3). The pre-registered labels of §D were frozen with C3 in the gate and are not relabelled; the demotion is a recommendation going forward, supported by a post-hoc factorial ablation, and it changes no reported number: $C1 \wedge C2$ selects exactly the same two in-distribution pairs.

(C2) Signal preservation. d_1 's per-client transformation T_1 does not destroy d_2 's admission signal: $\sigma_2(T_1(u_a)) \approx \sigma_2(u_a)$ for malicious updates u_a .

(C3) Boundary preservation. d_1 's transformation does not compress the adversarial–benign separation below d_2 's discrimination threshold. Formally: if d_2 suppresses a by detecting $\tau_{d_2}(u_a) > \theta$ (triggering trimming/rejection), then $\tau_{d_2}(T_1(u_a)) > \theta$ must hold, i.e., d_1 must not map adversarial updates from above to below d_2 's rejection boundary.

When C1 holds but C2 fails (signal destruction), composition is no better than d_2 alone. When C1 and C2 hold but C3 fails (boundary compression), d_2 retains its admission signal direction but its threshold no longer fires: the attack passes because d_1 's rescaling places adversarial magnitudes within d_2 's acceptance region.

Writing C2 as preservation of d_2 's admission signal elides a distinction our own experiments later force, so we name the levels before relying on the condition. They are the paper's main conceptual instrument, and every claim below is stated at one of them.

Definition 2 (Preservation levels). For an upstream transform T_1 and a downstream defense d_2 reading admission signal σ_2 :

- (P1) **Value invariance:** $\sigma_2(T_1(U)) = \sigma_2(U)$.
- (P2) **Ordering invariance:** $\sigma_2(T_1(U)) = \varphi(\sigma_2(U))$ for some φ preserving the ordering σ_2 induces on clients. This is the level C2 is stated at (made explicit in (2)); the distinction from (P1) matters, because exact values are sufficient but never necessary.
- (P3) **Decision invariance:** the client or coordinate d_2 actually selects is unchanged.
- (P4) **Adversarial contribution (admission) invariance:** the adversarial input d_2 admits into its aggregate is unchanged. Admission is the term we use throughout for how much adversarial input survives d_2 , and it is a **level, not one measurable quantity:** for a selector it is the indicator that an adversarial update is selected at all; for a weighted averager, the adversarial share of the weight mass; for a coordinate-wise order statistic, the adversarial share of selected coordinates, which is why Table A12's influence column is aggregator-specific by necessity. The committed pre-registrations say "admission", so that word is retained in every frozen rule name and no scored quantity is relabelled.
- (P5) **Attack-suppression preservation:** ASR stays below threshold.

(P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) hold, and hold by definition: equal values induce the same ordering, and a rule reading only that ordering cannot decide differently. **These three are logical implications under the definitions above, not empirical claims about how defenses behave,** and nothing is asserted by proving them: their content is that they stop at (P3). **Below (P3) the chain breaks, and it breaks in both directions.** Both breaks are measured. Holding decision and admission invariance does not deliver suppression: *cos.krum*'s selection is bit-identical at every dose and its admitted adversarial mass is 0.000 throughout, yet its ASR falls

0.173 by the first rung alone through the magnitude channel, at $p=2.72$, where clean accuracy moves only $0.599 \rightarrow 0.567$ and our accuracy gate passes (paired $t=-1.976$, one-sided $p=0.044$, falling in 5/8 seeds). Across the whole ladder it falls 0.425, but that endpoint sits below our own accuracy floor, so **the gate-passing rung is the witness we rest on** (§E.2). Losing decision invariance does not cost suppression: Krum’s selection flips in up to 80% of rounds under an intervention that pins the adversarial mass, and its ASR stays inside the pre-registered practical-equivalence margin ($|\Delta|=0.026 < 0.15$) (§E.2). A criterion stated at (P1)–(P2) therefore constrains security in neither direction. **(P4) is where the security content sits, but we do not claim it is sufficient for (P5), and it is a level rather than one universal scalar security quantity.** It is instantiated per aggregator by the measure that aggregator’s own rule exposes, Krum’s and `cos_krum`’s selected-client indicator, `coord_median`’s share of selected coordinates, reputation’s adversarial weight share (`experiments/build_channel_table.py`), so **(P4) names what to measure, not a single number comparable across defenses.** It is the first level that refers to adversarial influence on the aggregate rather than to the downstream defense’s internal decision, and `cos_krum` is our own witness that (P4) $\nRightarrow$ (P5): admission identically 0.000 at every rung while suppression moves. **So the hierarchy answers “which level does a security claim have to be stated at?” and not “which level predicts ASR?”:** no level in it is offered as a predictor.

Remark 4 (Which level C2 asserts, and which one is load-bearing). At $\rho = 54.6$ the dose ladder of §E changes Krum’s decision in 86.7% of rounds while changing its admission of adversarial input in 13.3%, and Krum’s suppression does not deteriorate at all. A selection that flips from one benign client to another disturbs the statistic maximally and the mechanism not at all. **Terminology, once and for all: “mechanism” always means the downstream defense’s discriminative operation (what separates adversarial from benign updates) and never its security behaviour, since that implication is precisely what this paper refutes.** We reserve “mechanism preservation” for the formal property Theorem 8 establishes, saying statistic or decision preservation where only the weaker property holds. C2 above is the (P2) condition (which is what makes it checkable from definitions alone) but the load-bearing reading is (P4). **This is the paper’s central tension: our theorem operates at (P1)–(P3), while the experiments indicate security behaviour is decided at (P4);** §E.2 puts the two against each other in an experiment where they predict opposite outcomes, and Figure A6 is that argument in one picture.

Remark 5 (C0’s provenance: an applicability condition formulated post-hoc). C0 sits in Criterion 3 as a stage-0 applicability condition because Lemma 1 is stated over $C0 \wedge C1$, but **it does not have the same evidential status as C1–C3**, and we record that here rather than let its promotion obscure it. It is **vacuous on the 7-defense menu of §D**, where the strongest single defense reaches only 0.519, and we formulated it only after the prospective suite of §D.1 introduced a defense for which it binds. It is a lesson from prospective evaluation, applied there retrospectively; it is not part of the criterion validated on the 42 pairs, and none of the accuracy figures in §D depend on it.

A scope condition on the criterion itself, not a limitation of our runs. C1 and every prediction built on it are defined against a *committed* adversary, one whose attack is fixed before the composition is chosen. An adversary who observes the composition can re-target whichever channel the screen certifies, so **what the screen bounds is inherited robustness under a fixed attack, never robustness against an adaptive one.** Our single whitebox composition-aware stress test is consistent with that reading ($0.045 \rightarrow 0.167$ seed-matched on the pixel arm the attack is built from, $3.7\times$; §F).

C.1 THE TESTABILITY BOUNDARY

C0 and C1 together constrain what any validation of this criterion can look like. We did not anticipate this; it turns out to explain the central weakness of our own prospective suite (§D.1, L4), and (once stated in general form) to *derive* the class of within-defense designs in which C2’s contribution is identifiable (§E). Its content is narrower than “a general identification boundary” and worth stating precisely: **under inherited-effect gating**, cross-defense contrasts cannot identify the mechanism’s contribution, because the gate conditions on a descendant of both the mechanism and the outcome. What we offer as the transferable contribution is not that negative but **the admissible design it derives**, set out below and run in §E.2.

Lemma 1 (Testability boundary). *Let $(d_1 \rightarrow d_2)$ satisfy C0 and C1, and let a^* be a committed attack witnessing C0, so $\text{ASR}(d_1, a^*) \geq 0.5$. Then $\text{ASR}(d_2, a^*) < 0.5$: the downstream defense suppresses a^* on its own.*

Proof. C1 at a^* gives $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a^*) < 0.5$. By C0 the minimum is not attained at d_1 , hence it is attained at d_2 . $\square$

Nothing in the argument is specific to federated learning, to backdoors, or to our defense menu. Lemma 1 is **the enabling observation for the FL instance**; the substantive claim is the general one, and we state it as such.

Table A2: **Scope of claims: three kinds, and the one place each scope condition is stated. Claims do not transfer upward:** the formal guarantee does not certify the general framing, and the empirical figures certify neither level above them. This table is the conservation record for every caveat in the paper; §H restates the same conditions individually as (L1)–(L6).

kind	content	status, and what it does <i>not</i> license
General	An evaluation gated on the outcome it predicts cannot identify the mechanism’s contribution, and admissible designs are within-defense; an inherited-suppression gate cannot certify emergent synergy (Prop. 6, Prop. 7).	Proved, and not specific to FL, to backdoors, or to our conditions. Licenses no claim that any composition is secure, and none that mechanism preservation implies suppression. Recall is bounded, not achieved: the bound is 80% from $ \mathcal{E} / \mathcal{S} =1/5$, and 40% is what we measure (L2).
Formal	Mechanism preservation as <i>invariance</i> under positive per-client rescalings $T(u_i) = c_i u_i$, $c_i > 0$, into specific rank and geometric aggregators, with explicit margins (Thm. 8, Lem. 2, Cor. 2, Prop. 10, Prop. 9).	Proved on that class only: a <i>laboratory</i> for composition, not a model of FL defenses in general. A d_1 that alters participation, carries cross-round state, mixes clients, or aggregates first admits no such T and is covered by nothing here . Bounds no ASR. C0 was formulated post-hoc and applied retrospectively to arms frozen before it (Rem. 5, L4). Regime A’s condition is not operationally checkable, S not being a stable per-attack constant (L5). Cor. 2’s bound goes vacuous as $m \rightarrow 1$, which heterogeneity realizes.
Empirical	§E.2’s interventions and §D–§D.1’s 42-pair menu: CIFAR-10, CifarCNN/ResNet18, $N \in \{10, 100\}$, $K \in \{5, 10\}$, $f \in [0.05, 0.6]$, Dirichlet $\alpha \in \{0.1, 0.5\}$, star topology , three attack families; one arm on EMNIST-byclass/simple.cnn.	Measured at $n=5$ seeds per cell ($n=8$ for <code>cos.krum</code> , $n=3$ frozen and $n=5$ pooled post-hoc on EMNIST-byclass, $n=30$ for the FG→RFA characterization), all fixed before the runs, so intervals are wide and no null is read as flatness without a pre-registered practical-equivalence margin. ASR is thresholded at 0.5, the level at which the trigger fires on the majority of triggered inputs; a cell below the 0.35 clean-accuracy floor is <i>uninterpretable</i> , not suppressed. Dataset transfer is tested on the flagship arm alone: “replicated on a second dataset and architecture”, never “generalizes” (L1). That one arm is a <i>channel</i> replication and carries no screen: no single defense measured on EMNIST-byclass suppresses the committed pixel backdoor alone (minimum 0.609; the menu measured there excludes <code>foolsgold</code> , <code>reputation</code> and the cosine mirrors), so C1’s per-attack precondition fails on that arm and the strategy comparison is out of scope over the measured menu rather than merely untested (L1). “Refuted” always means a pre-registered <i>directional</i> hypothesis was rejected against a rule frozen for it, never that a mathematical claim was disproved. The $\gamma \approx 0.66$ boundary is architecture-conditional (L1); the strongest development-set figures are <i>fit</i> , not prediction (L2).
Both instruments are oracles	Mode S and Mode A build c_i from adversary identity, as does Δ_c (§E.2, supplement §S10).	They identify a causal channel and are not screens: no deployed defense knows which clients are adversarial, and estimating (P4) without that knowledge is the open problem of §I. Admission is a <i>direction to measure</i> , not a validated predictor: the pooled two-channel model is unconfirmed over 26 cells ($R^2=0.239$) and its cross-arm ordering was refuted by our own replication (L6). C1 and everything built on it assume a committed adversary, so what is bounded is <i>inherited</i> robustness under a fixed attack, never adaptive robustness .

Table A3: **The confounded ladder against the Mode-S instrument, on the one arm that has both.** Same downstream defense, same attack, same seeds, same nominal dose, and the two designs share their $\kappa=0$ cell *numerically* (mean and worst per-seed gap 0.00e+00, asserted at 10^{-12}). At the top rung they are matched on aggregate displacement, decision change and mean clean accuracy, so **the ladder’s fall is not an accuracy collapse**; they differ chiefly in the **adversarial coefficient channel**, and the ASR effect **changes sign**. **The one non-match, disclosed:** aggregate adversarial influence Λ_a moves in *both* designs, 0.081 against 0.033 ($2.4\times$), so neither column is a clean statistic-only intervention, which is why the contrast is “differ chiefly in the coefficient channel”, never “only in it”. *Vocabulary:* this row is the $\Delta\Lambda_a$ column of the channel table (the adversarial share of selected coordinates, the same field in the same artifact) and **not** its Δ adm. column, which counts rounds where the *support* of the adversarial mass changes and reads 0.000 on every row: what moves here is how much weight the adversary’s direction carries, not whether it enters at all. This is a two-column contrast on one arm: it locates where the two designs differ, and attaches no p -value to the difference between the columns. Every cell recomputed from per-seed and per-round rows by `build_comparability_table.py`; nothing here is retyped.

	Confounded ladder dose_kappa	Instrument (Mode S) doseS_kappa
downstream d_2	coord_median	coord_median
attack	committed pixel	committed pixel
seeds	42–46	42–46
nominal dose	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6
benign coefficients	mean 1 over <i>all</i> clients	mean 1 over <i>benign</i> clients
adversarial coefficients	drawn from the same spread	pinned $c = 1.0$ exactly
adv. coefficient share	0.267 $\rightarrow$ 0.187	0.266667, constant
Δ_c	+0.000 $\rightarrow$ -0.336	0 by construction (read back 9e-08)
aggregate displacement	0.000 $\rightarrow$ 0.722	0.000 $\rightarrow$ 0.670
decision change	0.000 $\rightarrow$ 0.487	0.000 $\rightarrow$ 0.482
adversarial influence $\Delta\Lambda_a$	0.000 $\rightarrow$ 0.081	0.000 $\rightarrow$ 0.033
mean clean accuracy	0.767 $\rightarrow$ 0.690	0.767 $\rightarrow$ 0.698
ASR	0.443 $\rightarrow$ 0.170	0.443 $\rightarrow$ 0.541
ASR effect	-0.272 ($p_{\downarrow} = 0.0015$)	+0.098 ($p_{\uparrow} = 0.069$)

Proposition 6 (Why constituent-effect contrasts cannot identify composition effects under inherited-effect gating; Proposition 1 of the body, restated with proof). *Let a screen certify $(d_1 \rightarrow d_2)$ against attack a only if both (i) an **eligibility gate** $E(d_1, d_2, a) := \min_{d \in \{d_1, d_2\}} M(d, a) < \theta$ holds, where M and θ are the same outcome metric and threshold the screen predicts for the composition, and (ii) a **mechanism condition** $P(d_1, d_2)$ holds, determined by the two defenses’ definitions alone. Suppose the screen is applied only where the downstream mechanism is load-bearing, $M(d_1, a) \geq \theta$. Consider estimating P ’s contribution to $M(d_1 \rightarrow d_2, a)$ from contrasts across a defense menu $\mathcal{D}$. Then every certified pair satisfies $M(d_2, a) < \theta$, so of the four cells of $\{P, \neg P\} \times \{M(d_2, a) < \theta, M(d_2, a) \geq \theta\}$ the certified set occupies at most two, and **the cell needed to separate P from d_2 ’s standalone effectiveness is empty by the gate, for every $\mathcal{D}$. No contrast varying d_2 identifies P ’s contribution. Identification requires holding d_2, a and therefore E fixed while intervening on the upstream transform: **the admissible designs are within-defense.*****

Proof. The load-bearing condition excludes d_1 from attaining the minimum in (i), so certification forces $M(d_2, a) < \theta$; both cells with $M(d_2, a) \geq \theta$ are therefore unpopulated whatever $\mathcal{D}$ is, and P is never observed against a downstream defense that fails on its own. Any estimator of P ’s contribution from such contrasts is unidentified. $\square$

A second boundary: what such a screen cannot find. The gate that limits what a screen can be *validated* on also limits what it can *cover*, which converts the most obvious objection to our criterion, that its recall is poor, into a scope result.

Proposition 7 (Inherited-suppression screens cannot certify emergent synergy). *Call $(d_1 \rightarrow d_2)$ ’s suppression of a **emergent** if $M(d_1 \rightarrow d_2, a) < \theta$ while $\min_{d \in \{d_1, d_2\}} M(d, a) \geq \theta$. No emergent composition satisfies the eligibility gate of Proposition 6. Hence, writing $\mathcal{S}$ for the compositions with $M < \theta$ and $\mathcal{E} \subseteq \mathcal{S}$ for the emergent ones, any such screen’s recall obeys $\text{recall} \leq 1 - |\mathcal{E}|/|\mathcal{S}|$ on every menu and every attack.*

Proof. Emergence is the negation of the gate, so each member of $\mathcal{E}$ is rejected while belonging to $\mathcal{S}$: a false negative. The bound is the resulting recall. $\square$

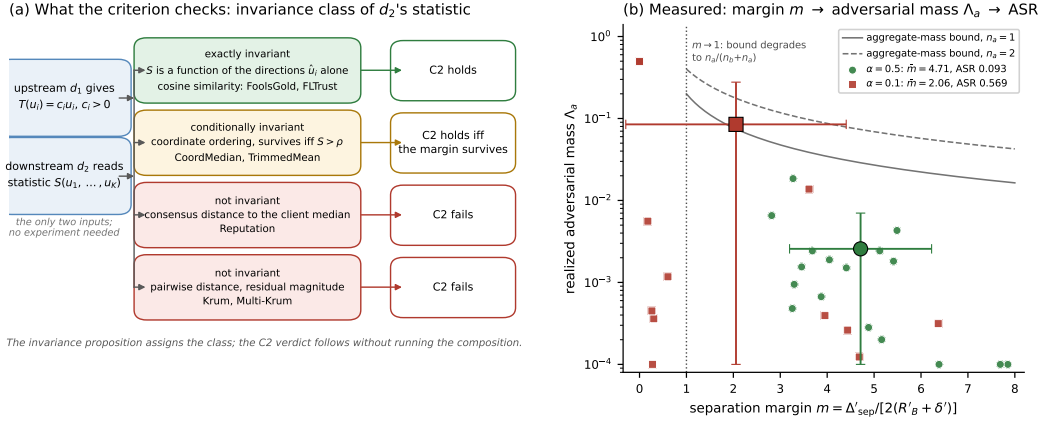


Figure A1: **The criterion is a mechanism test, and the mechanism is measurable.** (a) What C2 checks: the upstream transform T and the statistic S that d_2 reads determine an *invariance class* (Proposition 10), and the class determines the verdict: no composition needs to be run. This is why norm clipping does not blind FoolsGold but does blind Reputation. (b) The measured chain for FG→RFA, per round, on the transformed points RFA actually receives. Each marker is one round with $w_a > 0$ (rounds where FoolsGold zeroes every adversary are Lemma 2 and carry no payload); large markers are mean $\pm$ s.d. Corollary 2’s bound holds in every round, and the $\alpha=0.1$ collapse is visible as a collapse in the theorem’s own quantity: m falls from 4.71 ± 1.51 ($n=17$ rounds) to 2.06 ± 2.35 ($n=12$), with several rounds at $m < 1$ where the bound degrades to $n_a / (n_b + n_a)$, and realized Λ_a rises from 0.003 to 0.085 (max 0.50) as composition ASR goes $0.093 \rightarrow 0.569$. Values below 10^{-4} are drawn at 10^{-4} .

Mechanism–effectiveness confounding: Proposition 6 in our instance. The boundary has an uncomfortable consequence for validation, and it is worth spelling out concretely. In *every* in-scope pair the criterion certifies, d_2 already suppresses the attack that d_1 fails. A certified composition’s low ASR can therefore always be attributed to d_2 ’s standalone strength (C1) rather than to d_1 having preserved d_2 ’s mechanism (C2, C3): the two are confounded **by the conditions themselves**, not by an unlucky choice of defense menu. We name this **mechanism–effectiveness confounding** and offer it as a methodological observation about validating composition criteria of this shape, applicable beyond our own experiments.

In particular it rules out the experiment we previously named as decisive: a d_2 whose admission signal is invariant under positive rescaling but which is *weak in isolation* cannot satisfy C1 in scope, so no such certified pair exists to test. The achievable substitute is **per-attack complementarity**: a d_2 that suppresses the attack d_1 fails while remaining weak on a *different* committed attack, so its max-committed ASR stays high and it cannot suppress the composition single-handedly. Reputation is such a defense (0.017 on model-scaling, 0.842 on the pixel backdoor), and §D.2 builds the test on that observation.

Argument sketch. (C1 sufficient) If at least one of d_1, d_2 individually suppresses a , composition inherits that suppression provided C2∧C3 preserve the signal chain. C1 is sufficient, not necessary: two individually-weak defenses may suppress synergistically through complementary mechanisms (e.g., partial downweighting + geometric rejection), but predicting such synergy requires the quantitative margins below. The criterion is therefore *conservative* in the direction this sketch argues for: it may reject working compositions. **It is not a guarantee in the other direction, and nothing here establishes one.** That no clean C1∧C2∧C3 PASS is a false positive is a *measured* property of 42 pairs (§D.3), not a consequence of the argument above; the criterion has no established predictive validity ((L2), §H), and against an adversary that solves both constituents’ constraints at once a certified pair’s ASR rises $3.7\times$ (§F).

(C2 role) Suppose d_2 individually suppresses a via signal $\sigma_2(u_a) > \theta$. If d_1 ’s transformation T_1 destroys this signal ($\sigma_2(T_1(u_a)) \not\approx \sigma_2(u_a)$), d_2 cannot detect the attack. Example: NormClip as d_1 equalizes all norms to $\leq \tau$, so reputation’s distance signal $\sigma_2(u) = \|u - \hat{\mu}\|$ collapses; adversarial and benign updates become equidistant from consensus. This is measured, not hypothetical: NormClip→reputation leaves model-scaling ASR at 0.807 ± 0.048 and the pixel backdoor at 0.842 ± 0.047 over 5 seeds, against reputation alone at 0.017 (per-seed values in §M.1). Quantitatively: C2 holds when relative signal distortion $\Delta_{\text{signal}} := |\sigma_2(T_1(u_a)) - \sigma_2(u_a)| / \sigma_2(u_a) < \delta_2$ for a defense-specific tolerance δ_2 .

(C3 role) Even when d_1 preserves the *direction* of d_2 ’s signal (C2 holds), it may compress magnitudes below d_2 ’s threshold. Example: FoolsGold as d_1 downweights adversarial clients by $\alpha \ll 1$. TrimmedMean’s coordinate-wise rank signal still distinguishes adversarial coordinates (C2 holds), but after rescaling by α ,

per-coordinate values $\alpha \cdot u_{a,k}$ fall within the trimming range: the coordinate-wise extreme that would trigger trimming is no longer extreme. Quantitatively: C3 holds when the margin ratio $\Delta_{\text{margin}} := (\tau_{d_2}(T_1(u_a)) - \theta) / (\tau_{d_2}(u_a) - \theta) > 0$, i.e., adversarial scores remain above d_2 's rejection boundary after d_1 's transform.

(Sufficiency) If C1 holds, C2 holds ($\Delta_{\text{signal}} < \delta_2$), and C3 holds ($\Delta_{\text{margin}} > 0$), then d_2 detects and suppresses u_a even after d_1 's transformation. For the weighting→aggregator class, Theorem 8 instantiates these margins explicitly: $\rho < r^*$ for the rank aggregators, and the transformed-point separation condition $\Delta'_{\text{sep}} > 2(R'_B + \delta')$ for the geometric one, which Corollary 1 restates in a form that can be checked without running the composition. $\square$

C.2 FROM THE FEDERATED PROTOCOL TO THE ABSTRACT STATEMENT, TERM BY TERM

Proposition 6 is stated over an abstract screen precisely so that nothing in it rests on federated learning, and Lemma 1 is its instance under our criterion. That correspondence is what carries the result into this paper and out of it again, so we display it rather than assert it. Figure A2 is the same object drawn as a graph; Table A4 is the same object written as a substitution.

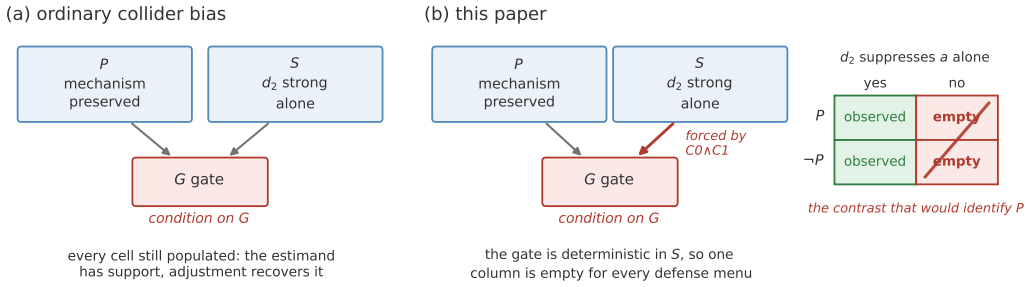


Figure A2: **Why the gate is not ordinary collider bias: a structural *positivity* violation induced by inherited-effect screening, which is why the remedy is a design and not an estimator.** This is the figure for §3's argument. (a) Conditioning on a descendant G of both the mechanism P and the defense's standalone strength S makes them co-vary: every cell stays populated, so an adjustment recovers the estimand in principle. (b) Here $C0 \wedge C1$ make the gate *deterministic* in S (Lemma 1), so the column the identifying contrast needs is empty for every defense menu, and no adjustment set, no reweighting and no sample size restores it (Proposition 1).

Table A4: **The abstract statement and our criterion are the same object under one substitution.** Left: the symbols of Proposition 6, which mentions no dataset, architecture, attack family or defense. Middle: what each becomes under Criterion 3. Nothing in the middle column is new here; the table is a reading aid for a correspondence the two statements already have.

Proposition 6	Our federated instance	Stated at
outcome metric $M(d, a)$	ASR(d, a), the committed attack's success rate under defense d	Criterion 3
threshold θ	0.5, the level at which the trigger fires on the majority of triggered inputs	Table A2
load-bearing condition $M(d_1, a) \geq \theta$	(C0), $\text{ASR}(d_1, a) \geq 0.5$ for some committed a	Rem. 5
eligibility gate E : $\min_d M(d, a) < \theta$	(C1), $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a) < 0.5$, quantified per attack	Criterion 3
mechanism condition $P(d_1, d_2)$	(C2) $\wedge$ (C3): $\sigma_2(T_1(u_a)) \approx \sigma_2(u_a)$, and the margin survives T_1	Prop. 10, Thm. 8
menu $\mathcal{D}$	the pairs $(d_1 \rightarrow d_2)$ a suite draws from a defense library	§D
certification $M(d_1 \rightarrow d_2, a) < \theta$	the criterion's PASS label on a pair	§D.1
the empty cell of the 2×2	$\neg P$ against a d_2 that fails alone: no admitted pair has one	Lem. 1
holding d_2, a, E fixed	Mode S and Mode A: same d_2 , attack and seeds, intervene on T	§E.2

Two features of the substitution carry the whole argument, and both are legible in the table rather than argued for. **First, E and the quantity the screen predicts are the same measurement.** C1 is an ASR comparison at 0.5 and the PASS label is an ASR prediction at 0.5, so the gate is not an unlucky covariate that a larger menu

would decorrelate: it is a function of the outcome. **Second, P is decided without running anything.** C2 and C3 are read off d_1 's and d_2 's definitions, which is what Prop. 10 makes algebraic, so no version of the design recovers identification by measuring P more carefully. The two together are what force the within-defense design of §E.2: with d_2 , a and the seeds fixed, E is fixed too, and the intervention on T moves P alone.

What the mapping does not license. It shows our criterion is *an* instance, not that every composition screen is one. A screen whose gate reads something other than the outcome it predicts, a per-client norm statistic say, has a different E and falls outside Proposition 6 entirely; the proposition is a statement about outcome-gated screens, ours included. Nor does the mapping transfer any *empirical* claim: the middle column names measured quantities, and their scope is Table A2's, not the proposition's.

C.3 A RESTRICTED THEOREM: $C2 \wedge C3$ FOR WEIGHTING $\rightarrow$ RANK AND WEIGHTING $\rightarrow$ GEOMETRIC COMPOSITIONS

What follows is a boundary theorem, and deliberately not a security theorem. It bounds what an *invariance* argument about composition can establish: preservation of the downstream defense's discriminative mechanism, levels (P1)–(P3) of Definition 2, and it says nothing whatever about suppression, which lives at (P4)–(P5). We state that in the same breath as the theorem because the whole point of the paper is the gap between the two: the theorem is the best case for statistic-level reasoning, proved with explicit margins, and §E.2 then shows that even exact invariance neither implies nor preserves suppression. Read it as a *formal* claim in the sense of Table A2: proved, on one class, bounding no ASR.

The criterion above is an operational heuristic. For the class of compositions that matter most in practice, a per-client *weighting* defense d_1 (reputation, FoolsGold) followed by a *rank-based* or *geometric* aggregator d_2 (TrimmedMean, CoordMedian, RFA), which contains the flagship FoolsGold $\rightarrow$ RFA, C2 and C3 can be *proven* rather than asserted, with an explicit tolerance on how aggressively d_1 may reweight. The key structural fact is that d_2 's admission score is scale-equivariant, so it commutes with d_1 's reweighting up to a bounded ratio; the composition fails only when d_1 compresses that ratio past a margin set by d_2 's separation.

Theorem 8 (Bounded reweighting preserves the downstream discriminative mechanism). *Consider a round with n_b benign and n_a adversarial updates under honest majority $n_b > n_a$. Let d_1 be a per-client reweighting $T_1(u_i) = w_i u_i$ with $w_i \geq 0$, and define the ratio over strictly positive weights, $\rho := \max_i w_i / \min_{i: w_i > 0} w_i$. Suppose d_2 suppresses attack a through a client admission score $\sigma_2(\cdot; U)$ that is scale-equivariant, $\sigma_2(c u_i; c U) = c^k \sigma_2(u_i; U)$ for all $c > 0$ and some degree $k \geq 0$. Assume $w_a > 0$ for adversarial clients; the $w_a = 0$ case is Lemma 2. The theorem establishes preservation of d_2 's suppression mechanism; it does not bound ASR. Then:*

1. **(Rank aggregators; $C2 \wedge C3$ hold up to a margin.)** If $d_2 \in \{\text{TrimmedMean}, \text{CoordMedian}\}$, so σ_2 is a coordinate-wise rank, then d_2 's per-coordinate suppression of a is preserved when two conditions hold. (i) **Count majority:** $n_a < K/2$ for CoordMedian, and $n_a \leq \lfloor \beta K \rfloor$ for TrimmedMean with β -trimming. (ii) **Signed ordering:** on each suppression-relevant coordinate k , the adversarial transformed value lies strictly beyond every benign one, $w_a u_{a,k} > \max_b w_b u_{b,k}$ (upper tail) or $w_a u_{a,k} < \min_b w_b u_{b,k}$ (lower tail). Sufficient magnitude condition for (ii): if $\text{sign}(u_{a,k})$ is common to all adversaries and $|u_{a,k}| > \rho \cdot \max_b |u_{b,k}|$, then (ii) holds; writing $r^* := \min_k \min_b |u_{a,k}| / |u_{b,k}|$ for the coordinate-wise separation margin, this is $\rho < r^*$. Magnitude alone is not sufficient without the sign assumption: an adversarial coordinate can be large in absolute value yet sit between the benign values. At $\rho = 1$ ranks are preserved exactly, recovering Proposition 9.
2. **(Geometric median / RFA; $C2 \wedge C3$ hold up to a margin.)** The composition hands RFA the transformed points $u'_i := w_i u_i$ and RFA runs its standard unweighted geometric median on them; it is not a weighted geometric median, and w_i enters only by moving the points. All quantities must therefore be defined on the transformed set. Let μ'_B be the geometric median of the benign u'_b alone, μ' that of all of U' , $R'_B := \max_b \|u'_b - \mu'_B\|$ the transformed benign radius, and $\delta' := \|\mu' - \mu'_B\|$ the displacement caused by including the adversarial points. Suppression is preserved when

$$\Delta'_{\text{sep}} := \min_{a,b} \|u'_a - u'_b\| > 2(R'_B + \delta'),$$

under which every adversarial residual $\|u'_a - \mu'\|$ strictly exceeds every benign residual $\|u'_b - \mu'\|$, so RFA's inverse-residual weight on a stays below the benign floor.

3. **(Norm-equalizing d_1 ; $C2$ fails: the degenerate case.)** If instead d_1 maps every client norm to a common value (NormClip with τ below all client norms), then any σ_2 keyed on update norm or consensus distance becomes constant across clients, so σ_2 no longer separates a and $C2$ fails. This is not a bounded reweighting: the induced per-coordinate weight ratio is unbounded, so no ρ -margin applies.

Parts (1) and (2) are sufficient conditions for d_2 to preserve the downstream suppression ordering, not to exclude adversarial contributions outright. Neither is a sufficient condition for low ASR, and part (2) orders residuals per client only; it constrains the aggregate adversarial coefficient solely through Corollary 2.

Proof. (1) TrimmedMean/CoordMedian act on coordinate k through the rank of $w_i u_{i,k}$ among $\{w_j u_{j,k}\}_j$; a is suppressed on k because its transformed coordinate is extreme (trimmed, or not selected by the median). Given (ii), at most n_a of the extreme order statistics are adversarial, so with (i) the selected value is benign. For the magnitude condition, $w_a > 0$ gives $\rho \geq \max_b w_b/w_a$, so $|u_{a,k}| > \rho \max_b |u_{b,k}|$ implies $|w_a u_{a,k}| > w_a \rho \max_b |u_{b,k}| \geq (\max_b w_b) \max_b |u_{b,k}| \geq \max_b |w_b u_{b,k}|$; with a sign common to all adversaries this places $w_a u_{a,k}$ strictly beyond every benign value on that side. Scale-equivariance ($k=0$ for pure rank) makes the uniform case $\rho=1$ exact. (2) All quantities are on the transformed points $u'_i = w_i u_i$, which is what RFA receives. Every benign residual obeys $\|u'_b - \mu'\| \leq R'_B + \delta'$ by the triangle inequality. For adversarial u'_a , $\|u'_a - \mu'_B\| \geq \Delta'_{\text{sep}} - R'_B$, hence $\|u'_a - \mu'\| \geq \Delta'_{\text{sep}} - R'_B - \delta'$. Demanding that this exceed the benign bound gives $\Delta'_{\text{sep}} - R'_B - \delta' > R'_B + \delta'$, i.e. the stated condition. No weighted-majority assumption is needed, and none would be appropriate: RFA weights every input point equally and the w_i act only by relocating the points. Our implementation runs Weiszfeld for 50 iterations at tolerance 10^{-6} , so the argument transfers to it. (3) Clipping to a common τ sends $\|u_i\| \mapsto \tau$ for all i , so $\sigma_2(u_i) = \|u_i\|$ (NormClip-as- d_2) or the consensus distance $\|u_i - \bar{\mu}\|$ collapses to a client-independent constant; separation vanishes, which is precisely C2-failure. $\square$

The condition as stated cannot be checked before running the composition, and that is fixable. Part (2) is stated on the transformed points, and $\delta' = \|\mu' - \mu'_B\|$ is by definition the displacement that including the adversarial points causes: evaluating it requires executing the very composition the criterion is meant to screen. In a screening criterion that is a real defect. It is also removable, because the displacement is controlled by the benign spread and the client counts alone.

Corollary 1 (Checkable separation margin). *In the setting of Theorem 8(2), write $S'_B := \sum_b \|u'_b - \mu'_B\|$ and let $\bar{R}'_B := S'_B/n_b$ be the mean transformed benign residual. Under honest majority $n_b > n_a$,*

$$\delta' \leq \frac{2S'_B}{n_b - n_a} = C_A \bar{R}'_B, \quad C_A := \frac{2n_b}{n_b - n_a},$$

and therefore the separation condition of Theorem 8(2) holds whenever

$$\Delta'_{\text{sep}} > 2R'_B + 2C_A \bar{R}'_B. \quad (1)$$

No quantity in (1) is computed from the composed aggregate.

Proof. μ' minimizes $F(x) := \sum_i \|x - u'_i\|$, so $F(\mu') \leq F(\mu'_B)$. Write $t := \delta'$. By the triangle inequality each benign term obeys $\|\mu' - u'_b\| \geq t - \|u'_b - \mu'_B\|$ and each adversarial term obeys $\|\mu' - u'_a\| \geq \|\mu'_B - u'_a\| - t$, so $F(\mu') \geq n_b t - S'_B + \sum_a \|\mu'_B - u'_a\| - n_a t$. Since $F(\mu'_B) = S'_B + \sum_a \|\mu'_B - u'_a\|$, the inequality $F(\mu') \leq F(\mu'_B)$ gives $(n_b - n_a)t \leq 2S'_B$, which is the displacement bound. Substituting it into $\Delta'_{\text{sep}} > 2(R'_B + \delta')$ yields (1). $\square$

Honest majority is exactly what the geometric case needs, and it enters on the displacement, not on the weight ratio. C_A diverges as $n_a \rightarrow n_b$, so Corollary 1 is the precise point at which $n_b > n_a$ becomes indispensable, while the theorem itself needs no majority assumption on the *weights*. An earlier version of this paper summarized part (2) as a weight-ratio bound $\rho < n_b/n_a$; that summary was wrong, and the correction is not cosmetic, because the majority condition belongs on δ' instead. **Measured at Dirichlet $\alpha=0.5$:** the displacement bound holds in all 17 rounds with $w_a > 0$, and is conservative there by a factor of at least 10.5 (mean 215), because the realized displacement is set by a force balance among the residuals rather than by the worst case a bound must cover. Condition (1) is satisfied in 16 of those 17 rounds: 15/15 at $n_a=1$, where $C_A = 2.67$ and the requirement at the measured mean $\bar{R}'_B/R'_B = 0.429$ is $\Delta'_{\text{sep}} > 4.29 R'_B$ against a measured $\Delta'_{\text{sep}}/R'_B$ never below 5.88; and 1/2 at $n_a=2$, where $C_A = 6.00$ tightens the requirement past what one of the two rounds supplies. **The heterogeneity limit recorded for Corollary 2 applies here too:** at $\alpha=0.1$ condition (1) holds in only 5 of 12 positive-weight rounds. The checkable form certifies less exactly where the condition it replaces also stops binding, and it does not rescue the guarantee there.

A further reduction to ρ alone is vacuous on our runs, and we report it as vacuous. The natural next ask is the analogue of part (1)'s $\rho < r^*$: a threshold on ρ computable from the *pre-composition* round geometry, so that no pilot is needed at all. Such a threshold exists formally. Bounding $\Delta'_{\text{sep}} \geq w_{\min}[\Delta_{\text{sep}} - (\rho - 1)N_A]$ and $R'_B \leq 2w_{\min}[\rho R_B + (\rho - 1)\|\mu_B\|]$, where Δ_{sep} , R_B and μ_B are the untransformed analogues and $N_A := \max_a \|u_a\|$, makes $w_{\min}$ cancel and turns (1) into a linear inequality in ρ with a critical value ρ^* depending only on $(\Delta_{\text{sep}}, R_B, \|\mu_B\|, N_A, n_a, n_b)$. **It is empty on our data.** It requires $\Delta_{\text{sep}} > 4(1+C_A)R_B$, and the measured $\Delta_{\text{sep}}/[4(1+C_A)R_B]$ clears 1 in exactly 1 of the 17 positive-weight rounds. Its maximum is 1.015 at $\alpha=0.5$, attained in that $n_a=1$ round where $C_A = 2.67$ makes the requirement loosest, with mean

0.754 over those 15 rounds and a maximum of only 0.389 over the two $n_a=2$ rounds; at $\alpha=0.1$ the maximum is 0.853. Consequently $\rho^* \leq 1$ in 16 of 17 rounds at $\alpha=0.5$ and in all 12 at $\alpha=0.1$, against a measured ρ of at least 1.034 and 1.015 in the two settings respectively; in the one round where ρ^* does exceed 1 it reaches only 1.006, against a measured ρ of 1.196 in that round. The reduced condition is therefore met in *no* round, which is a stronger negative than a merely conservative threshold. The looseness is in the composition of the two bounds rather than in either one separately, and closing it would need a curvature assumption on the benign configuration that we are not prepared to assume. **We therefore claim no pilot-free screen for the geometric class**, and the calibration protocol below prices it accordingly.

What part (2) does not assume. Recorded explicitly, because the step invites readings it does not support: there is no honest-majority assumption on the *weights*, no requirement that adversarial updates lie outside the benign convex hull, and the count majority $n_b > n_a$ is used in part (2) only through Corollary 1, nowhere in the theorem’s own proof.

Theorem 8(2) orders residuals *per client*: each adversarial point receives less inverse-residual weight than each benign point. That alone does not bound the adversary’s *total* share of the aggregate: several adversaries could each be down-weighted yet collectively hold substantial mass. The separation margin controls this, but only quantitatively:

Corollary 2 (Bounded aggregate adversarial influence). *Let $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ be the realized separation margin, so Theorem 8(2) reads $m > 1$. Every benign residual is at most $D_b := R'_B + \delta'$ and every adversarial residual at least $D_a := \Delta'_{\text{sep}} - R'_B - \delta' = (2m - 1)D_b$. Bounding $\sum_j 1/d_j$ below by the benign terms alone and each adversarial term above by $1/D_a$,*

$$\Lambda_a = \sum_a \lambda_a \leq \frac{n_a/D_a}{n_b/D_b + n_a/D_a} = \frac{n_a}{n_b(2m - 1) + n_a},$$

so μ' is a convex combination whose adversarial share is at most Λ_a : writing $\mu' = (1 - \Lambda_a)\bar{u}_B + \Lambda_a\bar{u}_A$ for its benign and adversarial parts, the displacement from the benign-only combination $\bar{u}_B$ is at most $\Lambda_a\|\bar{u}_A - \bar{u}_B\|$.

The bound is only as good as the margin, and **the separation condition alone is not enough**: as $m \rightarrow 1$ it degrades to $n_a/(n_b + n_a)$, permitting the adversary 40% of the mass at $n_a=2$, $n_b=3$. A margin comfortably above 1 does the work. We measure $m \geq 2.82$ in every round with $w_a > 0$ (mean 4.71 over those rounds), capping the adversarial share at 5.1% for $n_a=1$ and 12.6% for $n_a=2$. The realized figure is far tighter than the bound: measuring Λ_a directly, it never exceeds 1.8% in any round, and the bound is satisfied in all 17 measured rounds with $w_a > 0$. With Lemma 2 this completes the picture, and the mechanism survives for *opposite* reasons: at $w_a=0$ the adversaries hold most of the Weiszfeld mass ($\Lambda_a = 0.71$ measured) but carry no payload; at $w_a > 0$ they carry payload but hold at most 12.6%.

This is the one place in the paper where the theory predicts a number and the measurement lands inside it, so we state the pair plainly: **bound 12.6%, realized $\leq 1.8\%$** , in every measured round with $w_a > 0$. **And it is also the one place where heterogeneity makes the guarantee measurably empty.** The bound is a function of the realized margin m alone, and at Dirichlet $\alpha=0.1$ RFA’s **all-round mean margin collapses from $m=3.24$ to $m=1.01$** , exactly the limit at which Corollary 2 degrades to $n_a/(n_b+n_a)$ and certifies nothing, with two of three certified compositions crossing the ASR threshold there (§D.6(E)). So the theorem’s explanatory value is genuine and measured in *both* directions: it predicts a number that holds where heterogeneity is moderate, and its own quantity records where it stops holding. **That is not a robustness check; it is one of the paper’s three central empirical findings** (§E.2, §I).

Λ_a is level (P4) in measurable form, but (P4) is a level, not a single metric. Before the instantiation, one clarification the rest of the paper depends on: **(P4) is a conceptual level of the hierarchy and is instantiated defense-specifically**, because different aggregator families have different notions of what it means for adversarial input to enter the aggregate at all. A *selector* admits or does not, so its (P4) is an indicator in $\{0, 1\}$; a *weighted averager* has an adversarial coefficient share in $[0, 1]$; a *coordinate-wise order statistic* has a fraction of selected coordinates that are adversarial. There is no single scalar all three compute, and this is why the influence column of Table A12 is **necessarily aggregator-specific** rather than a common measurement across rows: a fact about the level, not a shortcoming of the table. With that said: Definition 2’s levels are not equally meaningful. (P1)–(P3) are properties of a statistic and of a decision rule reading it; **(P4) is the first level whose content is the adversary’s influence on the aggregate**, and for a geometric aggregator Λ_a is exactly that level made measurable: Corollary 2 bounds it, Lemma 2 drives it to a benign-only convex combination, and §E.2’s instruments intervene on it directly. So the negative results do not leave the theory empty-handed: **the level the evidence points to is one the theory can bound, just not from a (d_1, d_2) definition pair alone**, which is what a screen would need. That gap is the open problem we state in §I. **Three quantities are distinguished throughout and never used interchangeably**: *admitted adversarial mass*, whether an adversarial update enters d_2 ’s aggregate at all, binary for a selector and a coordinate fraction for an order statistic; the *adversarial coefficient share*, the fraction of the upstream coefficient vector’s total mass held by adversaries, which is what the instruments of §E.2 pin; and *aggregate adversarial influence* Λ_a , the weight the adversary’s

direction actually carries in the aggregate. Unqualified, “admission” means the first. (Symbols are collected in Appendix K.)

Lemma 2 (Zero-weight annihilation). *Zero-weight clients are not removed from the composition: d_2 receives them as zero vectors and still sees all K points. Our implementation runs the ε -smoothed Weiszfeld iteration ($\varepsilon = 10^{-8}$), whose fixed point is $\mu' = \sum_i \lambda_i u'_i$ with $\lambda_i \propto 1/\max(\|u'_i - \mu'\|, \varepsilon)$: a convex combination for every configuration, including the singular case $\mu' \in U'$ that a zero-weight client at the origin can create, so no non-degeneracy assumption is required. If $w_a = 0$ for every adversarial client, then $u'_a = 0$ and, writing $\Lambda_a := \sum_a \lambda_a$, the adversarial terms vanish identically: $\mu' = \sum_{b \text{ benign}} \lambda_b u'_b = (1 - \Lambda_a) \bar{u}$ with $\bar{u}$ a convex combination of benign transformed updates. The aggregate is a contraction by $1 - \Lambda_a \in (0, 1)$ of a benign-only weighted average: the attack direction never enters μ' . The origin points’ only effect is that contraction: a reduction in effective step size, i.e. a utility cost, not a security failure. The same holds for CoordMedian, where a zeroed coordinate carries no payload but shrinks the aggregate when selected.*

Why the positive-weight restriction matters. Defining ρ over all weights would make it infinite in every round: FoolsGold’s normalization subtracts the minimum weight, so 2 of 5 clients receive exactly zero (measured, $N=10, K=5$). A bound of the form $\rho < 4$ would then never be satisfiable, and Theorem 8(2) would be vacuous on the very composition it is stated over. Restricting ρ to strictly positive weights and handling $w_a=0$ separately via Lemma 2 is what makes the analysis apply to the algorithm we actually run.

Operational calibration protocol. To evaluate the criterion for $(d_1 \rightarrow d_2)$: **C1** is available from standard single-defense baselines (zero cost). **C2/C3 for the theorem-backed class** (FG or reputation as d_1 ; RFA, CM, TM as d_2): for the rank aggregators, Theorem 8(1)’s bound $\rho < r^*$ needs only ρ and the coordinate-wise separation, both read off the C1 baseline runs, so this case really is free. **For the geometric case an earlier version of this paper claimed the same, and that was false:** Corollary 1 removes the one quantity that would have required running the composition, but the benign spread that replaces it is measured on the transformed points, so RFA carries the same short-pilot cost as the other families and we no longer exempt it. **C2/C3 for other families:** one short pilot (5–10 rounds, $\sim 10\%$ of full cost) measuring $\sigma_2(T_1(u_a))$ vs. $\sigma_2(u_a)$ suffices. The full 50-round experiment runs only for criterion-PASS candidates. $\delta_2 = 0.5$ separates all PASS from C2-FAIL cases on the development set; insensitive to choice: any $\delta_2 \in [0.3, 0.7]$ yields identical classifications on all 42 pairs.

Proposition 9 (Rank-invariance of coordinate-wise aggregators). *Let d_2 be a coordinate-wise rank aggregator (TrimmedMean or CoordMedian) with admission statistic $\sigma_k(u) = \text{rank}_k(u)$ among the received updates $\{u_1, \dots, u_m\}$. Let $T_1(u_i) = \alpha \cdot u_i$ be a uniform positive rescaling ($\alpha > 0$, same for all clients). Then:*

1. $\sigma_k(T_1(u_i)) = \sigma_k(u_i)$ for all i, k , i.e., C2 and C3 hold trivially.
2. Under heterogeneous rescaling $T_1(u_i) = \alpha_i \cdot u_i$ with $\alpha_i \neq \alpha_j$, rank preservation fails when rescaling reorders coordinates across clients: $\exists k$ s.t. $\text{rank}_k(\alpha_i u_{i,k}) \neq \text{rank}_k(u_{i,k})$.

Proof. (1) Rank is invariant to positive scalar multiplication: $u_{i,k} > u_{j,k} \iff \alpha u_{i,k} > \alpha u_{j,k}$ for $\alpha > 0$. (2) Counterexample: let $u_{1,k} = 3, u_{2,k} = 2$, with $\alpha_1 = 0.1, \alpha_2 = 1$. Then $\alpha_1 u_{1,k} = 0.3 < \alpha_2 u_{2,k} = 2$, reversing the rank. $\square$

Corollary 3. *CoordMedian (pure rank selection) is robust to C3 boundary compression under any uniform rescaling but vulnerable under heterogeneous client-specific rescaling (e.g., FoolsGold or reputation weights). This predicts that compositions $X \rightarrow \text{CoordMedian}$ will show intermediate ASR when d_1 applies heterogeneous weights: consistent with observed $\text{NC} \rightarrow \text{cm}$ (0.433) and $\text{rfa} \rightarrow \text{cm}$ (0.485), both near the 0.5 threshold.*

Proposition 9 concerns rank under uniform rescaling. The sharper question is which admission signals survive heterogeneous per-client rescaling, since that is what every per-client defense in our menu applies: NormClip ($c_i = \min(1, \tau/\|u_i\|)$), reputation ($c_i \propto e^{-d_i/s}$) and RFA-reweighting ($c_i \propto 1/\max(\|u_i - \mu'\|, \varepsilon)$) are all of the form $T(u_i) = c_i u_i$ with $c_i > 0$ strictly, each positive by construction. FoolsGold is of the same form with $c_i \geq 0$ but attains $c_i = 0$ in every round (its shift-and-clamp gives the least-trusted client weight exactly zero, 2 of 5 clients here), so it falls outside the strict hypothesis below and is governed by Lemma 2 instead: the same distinction we draw for FLTrust’s ReLU zeros in §D.1. The strictly-positive case has a sharp answer, and it partitions the menu:

Proposition 10 (Invariance classes under positive per-client rescaling). *Let $T(u_i) = c_i u_i$ with $c_i > 0$, strictly: the form taken by NormClip ($c_i = \min(1, \tau/\|u_i\|)$), Reputation ($c_i \propto e^{-d_i/s}$) and RFA-reweighting ($c_i \propto 1/\max(\|u_i - \mu'\|, \varepsilon)$), each positive by construction. FoolsGold is deliberately not in that list: its shift-and-clamp gives the least-trusted client weight exactly 0 in every round (2 of 5 clients here), so a zero coefficient is not a positive rescaling and FoolsGold-as- d_1 falls outside the hypothesis, governed by Lemma 2 rather than by (a)–(c). Then:*

- (a) **Cosine similarity is exactly invariant:** $\cos(c_i u_i, c_j u_j) = \cos(u_i, u_j)$. FoolsGold’s weight vector is a function of the pairwise-cosine matrix alone, so $w(T(U)) = w(U)$ identically, and **C2 holds for every $\cdot \rightarrow \text{FoolsGold}$ composition.**

- (b) **Coordinate ordering is not invariant in general**; it survives under the bounded-ratio condition of Theorem 8(1).
 (c) **Consensus distance and residual magnitude are not invariant**. Heterogeneous c_i can invert their ordering, so $\cdot \rightarrow \text{reputation}$ compositions fail C2, and $\cdot \rightarrow \text{RFA}$ needs the separation condition of Theorem 8(2).

Proof. (a) $c_i u_i / \|c_i u_i\| = u_i / \|u_i\|$ for $c_i > 0$, so every pairwise cosine, each client’s normalized maximum similarity, and hence every FoolsGold weight is unchanged. (c) Counterexample for consensus distance: with $u_1=(1, 0)$, $u_2=(0, 1)$, $u_3=(3, 3)$ the coordinate-wise median is $(1, 1)$ and u_3 is farthest; under $c = (1, 1, 0.1)$ the median becomes $(0.3, 0.3)$ and u_3 becomes the *closest* point, reversing the ordering reputation depends on. $\square$

This corrects a natural but wrong intuition. Norm clipping does *not* blind FoolsGold: it cannot, by (a), though it does blind reputation, whose signal is consensus distance rather than similarity. We verified (a) in the shipped implementation: FoolsGold’s weight vector changes by at most 3×10^{-8} (float32 rounding) under NormClip, across 1000 synthetic trials and live FL rounds in which 1–2 of 5 clients were genuinely clipped (raw norms up to 24.5 against $\tau=5$). Note this also means Theorem 8(3)’s norm-equalizing degenerate case requires τ below *all* client norms; at our $\tau=5$ only 1–2 of 5 clients clip per round, so norms are not equalized and that case does not describe our runs.

C.4 THE FRAMEWORK IN GENERAL FORM

Positive rescaling is the instantiation that matters for current defenses, but the structure is more general, and stating it separates what is essential to the framework from what is an artifact of today’s defense menu. Write d_2 ’s admission mechanism as a statistic $S(U)$ of the round’s updates together with a decision rule that consumes only the *ordering* S induces over clients, which are extreme, which are far from consensus, which are down-weighted. Let T be d_1 ’s per-client transform. Then $(d_1 \rightarrow d_2)$ is **mechanism-preserving** exactly when

$$S(T(U)) = \varphi(S(U)) \quad \text{for some } \varphi \text{ preserving the ordering } d_2 \text{ acts on,} \quad (2)$$

which is C2; C3 additionally asks that φ not compress the quantitative gap below d_2 ’s operating threshold. Proposition 10 is the case $T(u_i) = c_i u_i$: for $S = \text{cosine similarity}$, $\varphi = \text{id}$, so (2) holds unconditionally; for $S = \text{coordinate value}$, φ is a per-client positive scaling, order-preserving only under the bounded-ratio condition $\rho < \tau^*$ of Theorem 8(1); for $S = \text{consensus distance or residual magnitude}$, no φ exists, because the reference point (median, geometric median) moves with T , so $S(T(U))$ is not a function of $S(U)$ at all. Three consequences worth stating. First, this is why the conditions are checkable without running an experiment: one reads S off d_2 ’s definition and T off d_1 ’s, and the question becomes algebraic. Second, it identifies what a genuinely new instantiation would require: an S and a non-rescaling T , e.g. a d_1 that projects updates onto a subspace or applies a coordinate-wise mask, for which (2) would need to be re-derived rather than inherited. Third, it marks the framework’s real boundary: a d_1 whose transform is not per-client (one that mixes updates across clients, or carries cross-round state) admits no T of this form, and nothing here applies to it.

D WHAT THE SCREEN BUYS, AND WHAT IT DOES NOT

Criterion 3 predicts for each two-defense pair whether composition achieves low max-committed ASR. All experiments in this section and §F use one deployment regime throughout (CIFAR-10, $N=10$ clients, $K=5$ per round, adversarial fraction $f=0.2$, 50 rounds, 4-layer CNN), with committed attacks `model_scaling` and `backdoor_pixel`. **Convention:** unless stated otherwise, *ASR* means *max-committed ASR* (the maximum over the two committed attacks of the per-attack mean across seeds) thresholded at 0.5: the level at which the trigger fires on the majority of triggered inputs, so that “suppresses” means the backdoor is not the model’s dominant response; adaptive-attack, order-sensitivity and per-seed figures are labelled explicitly where they differ. A second gate is needed because a destroyed model has a low ASR for the wrong reason: every cell in §E–§E.2 must also hold mean clean accuracy ≥ 0.35 (our **accuracy floor**), below which we call the cell *uninterpretable* rather than suppressed. **Sample sizes are small and were fixed before the runs:** $n=5$ seeds per cell in the interventional suites ($n=8$ for `cos_krum`, $n=3$ frozen for the second-dataset replication, which also reads at a pooled, post-hoc $n=5$, §E.2), $n=30$ for the FoolsGold $\rightarrow$ RFA characterization, so intervals are wide, and no null result is read as flatness unless it clears a pre-registered practical-equivalence margin. **“Refuted” means one thing throughout:** a pre-registered *directional* hypothesis was rejected against the rule frozen for it, never that a mathematical claim was disproved. Adversaries are the fixed client identities $\{0, \dots, fN - 1\}$ and each round samples K participants uniformly without replacement from all N , with no stratification, so the per-round adversary count is *hypergeometric* rather than fixed: for $N=10$, $f=0.2$, $K=5$, $P(n_a=0) = P(n_a=2) = 56/252 = 0.222$ and $P(n_a=1) = 140/252 = 0.556$ (measured over 100 rounds: $0.19/0.54/0.27$). Two consequences the data bear out: CoordMedian’s count-majority condition $n_a < K/2 = 2.5$ holds for *every* attainable $n_a \in \{0, 1, 2\}$, i.e. in 100% of rounds, while TrimmedMean’s

budget $\lfloor \beta K \rfloor = \lfloor 0.2 \cdot 5 \rfloor = 1$ is exceeded whenever $n_a=2$: 22.2% of rounds, ≈ 11 of 50. That is a second, count-based reason FoolsGold $\rightarrow$ TrimmedMean fails (0.508) while FoolsGold $\rightarrow$ CoordMedian succeeds (0.107) under an identical upstream transform; we retain the pre-registered C3-FAIL label, the budget violation being an additional mechanism rather than a relabelling. We validate in two waves. The **development set** (18 targeted pairs, Table A5) spans all criterion prediction categories and was used to refine the criterion (adding C3). The **held-out set** (24 pairs, Table A6) was *pre-registered*: all 24 predictions were committed to version control (commit 7e4d9b1) *before* any held-out experiment was run, eliminating post-hoc classification concerns. The criterion predicts LOW (<0.5) or HIGH (≥ 0.5) max-committed ASR. The verdict categories are:

Table A5: Composability criterion validation (max across committed_scaling and committed_pixel; $\pm$ is per-seed std where reported). Threshold: LOW < 0.5 , HIGH ≥ 0.5 . **Seed counts differ by row and are marked**: unmarked rows are $n=3$, * is $n=5$, $\dagger$ is $n=30$.

Pair	Failure mode	Predicted	Max ASR	Correct?
<i>PASS: C1$\wedge$C2$\wedge$C3 all hold (predicted LOW)</i>				
foolsgold $\rightarrow$ rfa †	—	LOW	0.093 ± 0.082	✓
foolsgold $\rightarrow$ coord_median*	—	LOW	0.107 ± 0.034	✓
reputation $\rightarrow$ coord_median*	—	LOW	0.423 ± 0.155	✓
<i>C1-FAIL: neither defense individually suppresses all attacks (predicted HIGH)</i>				
reputation $\rightarrow$ trimmed_mean †	C1 (pixel)	HIGH	0.679	✓
reputation $\rightarrow$ rfa	C1 (pixel)	HIGH	0.825	✓
reputation $\rightarrow$ foolsgold	C1 (pixel)	HIGH	0.807	✓
reputation $\rightarrow$ norm_clip	C1 (pixel)	HIGH	0.813	✓
norm_clip $\rightarrow$ foolsgold ‡	C1	HIGH	0.887	✓
<i>C2-FAIL: d_1 destroys d_2's signal direction (predicted HIGH)</i>				
norm_clip $\rightarrow$ reputation	C2	HIGH	0.819	✓
rfa $\rightarrow$ reputation	C2	HIGH	0.882	✓
<i>C3-FAIL: d_1 compresses boundary below d_2's threshold (predicted HIGH)</i>				
norm_clip $\rightarrow$ trimmed_mean	C3	HIGH	0.780	✓
norm_clip $\rightarrow$ rfa	C3	HIGH	0.853	✓
rfa $\rightarrow$ trimmed_mean	C3	HIGH	0.678	✓
rfa $\rightarrow$ coord_median	C3	HIGH	0.485	×
foolsgold $\rightarrow$ trimmed_mean	C3	HIGH	0.508	✓
norm_clip $\rightarrow$ coord_median	C3	HIGH	0.433	×
<i>DEGEN: d_1 has no per-client transform (predicted HIGH)</i>				
fedavg $\rightarrow$ norm_clip*	DEGEN	HIGH	0.935	✓
fedavg $\rightarrow$ trimmed_mean*	DEGEN	HIGH	0.846	✓
Overall: 16/18 = 89%. HIGH direction: 13/15 correct (87%). Under the tightened C1 the certified PASS set is 2/2 correct; fg $\rightarrow$ rfa is reclassified as a synergy false negative.				
‡ Reclassified from C2-FAIL: by Proposition 10(a) clipping preserves FoolsGold's similarities exactly, so C2 holds and the pair fails C1. Prediction and outcome unchanged.				

PASS precision: 100%, all compositions the criterion predicts to work *do* work. Under the tightened per-attack C1 (§C) the certified in-distribution set is 2 pairs, fg $\rightarrow$ cm (0.107) and rep $\rightarrow$ cm (0.423), both correct; fg $\rightarrow$ rfa (0.093) is no longer certified and is reported below as a false negative of our own criterion. This is the practitioner-relevant result: the criterion's positive predictions are reliable, at the cost of 40% recall. Overall development-set accuracy: **16/18 = 89%**. Two misses, both in the HIGH direction:

One certified pair is not resolved below threshold. Rep $\rightarrow$ CM's interval $[0.231, 0.616]$ crosses 0.5, and one of its five seeds reaches 0.665. Its LOW label therefore rests on a mean whose uncertainty admits the HIGH side, and we report it as correct on the mean while flagging it as the one certified pair whose margin is not statistically established. The certified in-distribution set is 2/2 correct either way, but only fg $\rightarrow$ cm is certified *and* resolved below threshold at the seed budget we can afford. Clean accuracy on the three cells is 0.50, 0.68, 0.76, all well clear of the collapse regime.

Two boundary conditions, stated alongside the primary result. Both are developed in full later, but neither should be met late. (i) **Extreme heterogeneity breaks two of the three PASS pairs.** At $\alpha=0.1$, fg $\rightarrow$ rfa reaches 0.569 and rep $\rightarrow$ cm 0.548; only fg $\rightarrow$ cm survives, at 0.433. What makes this more than a negative datapoint is that the failure appears in the theorem's own quantity: RFA's separation margin $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$, measured on the transformed points, collapses from an all-round mean of 3.24 to

1.01 (exactly the $m \rightarrow 1$ boundary at which Corollary 2’s bound degrades to 40% adversarial mass), so a practitioner running the 5–10 round pilot would measure $m \approx 1$ and decline to deploy. We emphasise that this is a post-hoc measurement of a known failure, not a pre-registered prediction of it; full sweep and caveats (bimodality at $n=3$, accuracy collapse to 0.20–0.48) in §D.6(E). (ii) **Mechanism preservation is not adaptive-attack robustness.** A criterion-aware adversary solving FoolsGold’s and RFA’s constraints jointly (decorrelated coordinate blocks to evade the similarity test, ϵ -ball projection to stay inside the geometric median’s radius) raises $\text{fg} \rightarrow \text{rfa}$ from 0.045 (seed-matched, and on the pixel arm the attack is built from) to 0.167 within our evaluated composition-aware attack family (§F). Theorem 8 bounds what the preserved mechanism admits, not what an adversary reshaping its update to satisfy both constraints at once can do, and the $3.7\times$ increase is the size of that gap on our flagship pair.

Held-out validation (pre-registered): 22/24 = 92% (*Screening*: an evaluation ordering, and by Proposition 6 not an identification of the mechanism’s contribution). Table A6 reports the 24 pre-registered pairs. Combined across both waves: **38/42 = 90.5%**. **Read against its base rate that figure is nearly uninformative, which is why it appears here, where it is computed, and in no summarizing position.** Only 5 of the 42 pairs are low-ASR (the 12% base rate of §D.3), so the constant predictor that calls every pair HIGH already scores $37/42 = 88.1\%$; 90.5% is one pair better than that, and the tally is carried by correct negatives. The informative quantities are precision 2/2 and recall 40% (§D.3). The two held-out misses are both LOW-predicted pairs that were flagged as *speculative* in the pre-registration (neither is a clean $C1 \wedge C2 \wedge C3$ PASS):

Table A6: Held-out pre-registered validation (24 pairs, $n=5$ seeds; max across committed_scaling and committed_pixel). All predictions committed to version control (commit 7e4d9b1) before experiments. Threshold: LOW < 0.5, HIGH $\geq$ 0.5. Only the two LOW predictions are shown individually; the 22 HIGH predictions (all correct) are summarized.

Pair	Category	Predicted	Max ASR	Correct?
<i>Speculative LOW predictions (not clean $C1 \wedge C2 \wedge C3$ PASS)</i>				
fedavg $\rightarrow$ coord_median	DEGEN	LOW	0.519	×
foolsgold $\rightarrow$ norm_clip	C1 (uncertain)	LOW	0.927	×
<i>HIGH predictions (22 pairs: 6 CM-as-d_1, 6 TM-as-d_1, 4 fedavg-as-d_1, 3 weighting-d_1, 3 RFA-as-d_1; range 0.70–0.94)</i>				22/22 ✓
Held-out: 22/24 = 92%. Combined with development set: 38/42 = 90.5%.				
Constant-HIGH baseline on the same 42 pairs: 37/42 = 88.1%.				

Rep $\rightarrow$ tm: composition dominates despite C1-FAIL. Reputation $\rightarrow$ TrimmedMean is correctly classified as C1-FAIL (pixel ASR 0.679 > 0.5), yet it is the paper’s main composition result. The resolution: the criterion predicts *max-committed* ASR, while the deployment-relevant metric is *oracle* ASR (0.316). The oracle cannot exploit the pixel gap because scaling (which reputation *does* suppress) dominates its utility. The criterion thus correctly identifies that rep $\rightarrow$ tm does not suppress every committed attack, while the game-theoretic analysis shows it is strategically dominant.

Tightening C1 costs us our own flagship, and reveals a synergy case. Under the per-attack quantifier in C1, **fg $\rightarrow$ rfa no longer certifies**: neither FoolsGold (0.732) nor RFA (0.845) suppresses the pixel backdoor individually, and FoolsGold’s 0.200 against model-scaling is hollow because it comes with 10% clean accuracy. The composition nonetheless reaches 0.093. This is genuine *synergy* (suppression neither constituent possesses) and Proposition 7 proves that *no* inherited-suppression screen can certify it, so this is a scope boundary rather than a tuning failure. We therefore report our lowest-ASR composition as a false negative of our own criterion. The certified in-distribution set is fg $\rightarrow$ cm (0.107) and rep $\rightarrow$ cm (0.423), both correct: precision 2/2, recall 2/5 = 40%.

D.1 A PROSPECTIVE TEST ON A NEVER-RUN DEFENSE MENU

Result: 13/20, and the shortfall is in our experimental design. At the same 0.5 threshold: TP = 8, FP = 0, TN = 5, FN = 7 (Table A7). **The suite carried the sanity gate our out-of-sample battery lacked**, pre-registered with the labels: a predicted-FAIL pair counts only if the attack clears ASR 0.5 against FedAvg, which both committed attacks do (0.667 scaling, 0.768 pixel), so a pair cannot be scored correct because the attack failed on its own. Every predicted-LOW pair came out LOW, 8/8, and that figure is far weaker than it looks, because **FLTrust suppresses both committed attacks on its own** (max-committed ASR 0.048 at 0.575 clean accuracy), the first defense in our menu that does. All 13 pairs containing FLTrust land in [0.041, 0.078], within 0.030 of FLTrust alone, in either slot. The two slots fail for different reasons, and the distinction matters:

The two slots fail differently, and the difference is the finding. (i) *FLTrust as d_1* (7 pairs, 2/7). d_1 alone already suppresses, so the composition inherits that suppression whatever d_2 does; all five predicted-HIGH misses have real C2/C3 failures, but destroying d_2 ’s mechanism is costless when d_2 was never load-bearing.

This is C0, invisible on the original 42 pairs because no defense there suppresses alone (the minimum single-defense max-committed ASR is CoordMedian’s 0.519, and the lowest ASR among all 39 non-PASS pairs is 0.433), so we count these 7 pairs out of scope *retrospectively*: a weaker claim than a pre-registered exclusion. (ii) *FLTrust as d_2 (6 pairs, 6/6), but the prediction is nearly an identity*. FLTrust gates on the scale-invariant $\cos(u_i, g_{\text{sv}})$ and then renormalizes every update to $\|g_{\text{sv}}\|$, discarding magnitude entirely, so a positive per-client rescaling is *exactly* cancelled: four of the six pairs are bit-identical to FLTrust alone or differ only within seed noise. The single genuine deviation is $\text{FG} \rightarrow \text{FLTrust}$ (0.064), the one d_1 producing exact *zeros* rather than positive rescalings and hence a Lemma 2 interaction. This arm is a consistency check between theory and implementation, **not** a falsifiable test: every d_1 in our menu is a positive rescaling and Proposition 10(a) permits no other outcome. (iii) *Neither defense is FLTrust (7 pairs, 5/7)*: the only arm that could have surprised us. Five predicted-HIGH pairs are correct (0.870–0.943), though only 2 of the arm’s 7 pairs compose two real mechanisms once the pass-through and MultiKrum degeneracies are accounted for (below); both misses are diagnosable and neither is a mechanism error: $\text{fg} \rightarrow \text{krum}$ measures 0.078 at 9.9% clean accuracy (the collapsed-model artifact, not suppression), and $\text{krum} \rightarrow \text{cm}$ measures 0.450 against a 0.5 threshold as a degenerate pair reducing to CoordMedian alone, whose own baseline is 0.519. The arm contains *zero* predicted-LOW pairs, so it bears on specificity only. Per-pair numbers for all three arms are in the supplement (§ S1).

What the suite does and does not establish. Counting only in-scope pairs gives 11/13; the arm that carried real risk of falsification gives 5/7 with both misses explained. But it is not the prospective positive the framework needs, because **it contains none**: by (i) and (ii) the 8 predicted-LOW pairs are 6 near-identities plus 2 out-of-scope inheritances.

We withdraw the three matched contrasts we previously drew here, which we had read, d_1 fixed while d_2 moved from FLTrust to a Euclidean rule, as a 19–20× ASR gap turning on exactly the property Proposition 10 identifies. Two ran against MultiKrum, which at $K=5$ selects *all five* participants and is bit-identical to FedAvg (1.9×10^{-6} per aggregation; a genuine selector again at $K=10$), so they carried no downstream defense at all. Substituting Krum does not repair them, and that is the finding: all three Krum arms take their max-committed ASR from the pixel backdoor, which Krum *fails to suppress standalone* (0.583), so the gap measures C1 failure and not disturbed invariance, and $\text{nc} \rightarrow \text{krum}$ is bit-identical to Krum alone on all three seeds because under that attack no client exceeds $\tau=5$ and d_1 is the exact identity. Counting the pass-through d_1 rows too, 8 of the 20 rows reduce to a single defense rather than the 5 labelled DEGEN at freeze, and arm (iii) composes two real mechanisms in 2 pairs, not 7 (App. S2). The frozen labels and the 13/20 score stand; what we retract is our reading of them. The confound is accordingly worse than we reported: in this menu the invariant-signal d_2 is also the only d_2 that suppresses alone, so C1 and C2 co-vary. Lemma 1 explains why no other choice of defense would have helped: in scope, $\text{C0} \wedge \text{C1}$ force d_2 to suppress a^* alone, so the confounding is structural. Breaking it requires holding the suppression principle and standalone effectiveness fixed while varying only the invariance class, which is the design of §D.2; that section’s Δ analysis is the only contrast in either paper holding C1 exactly fixed, and it is where our C2 evidence now rests.

D.2 BREAKING THE CONFOUND: A MATCHED METRIC-SWAP ABLATION

Lemma 1 says mechanism–effectiveness confounding cannot be escaped by picking a different defense off a menu: in scope, $\text{C0} \wedge \text{C1}$ force the invariant-signal d_2 to be one that suppresses a^* alone. So we break the confound by construction instead, holding the suppression *principle* fixed and varying only the invariance class of the statistic d_2 reads.

The design’s premise was refuted by its own baselines. Matching failed in both arms, for two independent reasons, and because the C1 inputs are measured before the freeze we learned this before the suite ran: the only reason it can be reported as a finding rather than a post-hoc excuse. `cos_reputation` does not merely weaken relative to its Euclidean original, it collapses, from Reputation’s 0.017 on model-scaling to **0.982**. `cos_krum` reaches 0.078 on model-scaling, which passes C1’s numeric threshold, but at **0.146** clean accuracy, the hollow-suppression artifact already documented for FoolsGold (0.200 at 0.10) and $\text{fg} \rightarrow \text{krum}$ (0.078 at 0.099), so C1 fails under the accuracy gate we apply everywhere else. The mechanism is visible in the baselines rather than asserted: undefended FedAvg under model-scaling is itself at 0.100 accuracy, so Krum’s 0.577 is *restoring* utility the attack destroyed and `cos_krum`’s 0.146 fails to. **Magnitude information is what lets this defense family handle a magnitude attack**: a direction-only mirror of a Euclidean scorer cannot see a scaling attack at all, so removing the magnitude dependence removes the suppression with it: exactly the coupling the pre-registered contingency anticipated, stated there as the possibility that “invariance and standalone effectiveness may be inherently coupled” for this family. On the magnitude-neutral pixel backdoor the mirror does survive (0.300 at 0.598), but there its control Krum fails C1 (0.583), so those arms are unmatched in the opposite direction and the pixel cells cannot carry the test either.

The consequence is that **the confound re-materialized inside an experiment built specifically to break it**, which we report as the round’s methodological finding rather than as a failed run. Lemma 1 establishes that mechanism–effectiveness confounding is structural in scope; this is what trying to escape it looks like when the mechanism being varied is the mechanism the defense relies on. Two responses were available and we

Table A7: Prospective suite: 20 pairs, labels frozen at commit 262cf35 before any run. Max-committed ASR over 2 attacks $\times$ 3 seeds; threshold 0.5. Categories are verbatim from the pre-registration, including its own uncertainty hedges.

Pair	Frozen category	Pred.	Max ASR	Acc	Hit
<i>In scope (C0 holds): d_2 load-bearing</i>					
Rep $\rightarrow$ FLTrust	certified	LOW	0.046	0.577	✓
NC $\rightarrow$ FLTrust	certified	LOW	0.046	0.576	✓
RFA $\rightarrow$ FLTrust	certified	LOW	0.047	0.574	✓
Krum $\rightarrow$ FLTrust	DEGEN	LOW	0.048	0.574	✓
MKrum $\rightarrow$ FLTrust	DEGEN	LOW	0.048	0.574	✓
FG $\rightarrow$ FLTrust	certified	LOW	0.064	0.547	✓
FG $\rightarrow$ Krum	C1-fail	HIGH	0.078	0.099	×
Krum $\rightarrow$ CM	DEGEN	HIGH	0.450	0.770	×
Krum $\rightarrow$ RFA	DEGEN	HIGH	0.870	0.780	✓
MKrum $\rightarrow$ RFA	DEGEN	HIGH	0.870	0.780	✓
RFA $\rightarrow$ MKrum [†]	C1-fail	HIGH	0.875	0.782	✓
Rep $\rightarrow$ Krum	C1-fail	HIGH	0.880	0.530	✓
NC $\rightarrow$ MKrum [†]	C1-fail	HIGH	0.943	0.761	✓
<i>Out of scope (C0 violated): FLTrust as d_1 suppresses alone at 0.048</i>					
FLTrust $\rightarrow$ CM	certified-uncertain	LOW	0.041	0.505	✓
FLTrust $\rightarrow$ MKrum [†]	C2-fail	HIGH	0.045	0.576	×
FLTrust $\rightarrow$ TM	C3-fail-uncertain	HIGH	0.046	0.529	×
FLTrust $\rightarrow$ NC	C2-fail	HIGH	0.047	0.576	×
FLTrust $\rightarrow$ RFA	certified	LOW	0.048	0.512	✓
FLTrust $\rightarrow$ Rep	C2-fail	HIGH	0.048	0.518	×
FLTrust $\rightarrow$ Krum	C2-fail	HIGH	0.078	0.361	×

Full suite 13/20 (TP 8, FP 0, TN 5, FN 7); in scope 11/13; non-FLTrust arm 5/7.

[†]MultiKrum at $K=5$ keeps all five participants, so it is bit-identical to FedAvg and these rows carry no downstream defense; MKrum $\rightarrow$ FLTrust and MKrum $\rightarrow$ RFA repeat their Krum rows exactly. 8 of 20 rows reduce to one defense; frozen labels unchanged (App. S2).

took neither. We did not switch to a pixel-only protocol, which would have made `cos.krum` pass C1 and manufactured two predicted-LOW cells: choosing the attack after seeing the C1 inputs is attack-shopping regardless of the resulting labels. We did not adjust the accuracy gate to admit 0.146, which would have produced a predicted-LOW pair by weakening the very criterion that catches hollow suppression elsewhere. The pre-registration’s second non-negotiable (“no switching of committed attack, threshold, seed count, or accuracy gate”) forecloses both, and either would have bought a positive test at the cost of the property that makes the test worth running.

Outcome: 8/8 correct, and zero positive tests. Because C1 binds for all eight pairs (on model-scaling through the two baseline failures above, on the pixel backdoor by design) every frozen label was HIGH, so the suite bears on specificity only, as a consequence of inputs measured before the freeze rather than a choice made afterwards. We ran all 8 pairs $\times$ 2 attacks $\times$ 3 seeds anyway, because a suite whose labels are all one class is still falsifiable in the direction that matters: a single LOW cell would have been a false positive against a criterion claiming 100% precision. All eight came out HIGH (8/8 correct, TP = 0, FP = 0, TN = 8, FN = 0, max-committed ASR spanning 0.508–0.935) with labels frozen in `pre_registration_metric_swap.md` at commit 356897b, timestamped before `results/metric_swap/` existed and reproduced verbatim in the runner’s PAIRS list so scoring cannot drift. We state the limitation in the same breath as the result: the suite contains **no falsifiable prospective positive test**, and 8/8 is not validation of the framework’s predictive claim: eight correct negatives on pairs our own conditions could only have called HIGH is weak evidence, and it supports a claim about specificity, not prediction.

One prediction in the suite was genuinely two-sided, and it is the round’s only prospective positive result. `cos.krum`’s C1 verdict on model-scaling turns entirely on whether the accuracy gate is applied (0.078 passes the numeric threshold; 0.146 accuracy fails the gate), so rather than choose a reading afterwards we froze *both*, with the resolution rule fixed in advance: accuracy-gated $\Rightarrow$ HIGH, numeric-only $\Rightarrow$ LOW, and a LOW outcome at accuracy ≥ 0.35 would have recorded our two primary labels as misses. Both pairs came out **HIGH** (NormClip $\rightarrow$ `cos.krum` 0.508, RFA $\rightarrow$ `cos.krum` 0.566), refuting the secondary reading and vindicating the accuracy gate against a pre-registered alternative specification of itself on data neither of us had seen. The weaker half is flagged in the supplement (§ S3): the NormClip arm sits barely above threshold and it is the RFA arm that carries the conclusion.

A post-hoc contrast that is not confounded. The cross-pair scoring above is confounded in exactly the way Lemma 1 predicts, but the frozen suite happens to contain a comparison that is not, and it is the strongest C2 evidence in either paper. Compare a composition against *the same* d_2 's own standalone ASR on the same attack, under two different upstream transforms. C1 is then identical by construction (same downstream defense, same attack, same measured standalone effectiveness on both sides), so the residual $\Delta = \text{ASR}(d_1 \rightarrow d_2, a) - \text{ASR}(d_2, a)$ cannot be attributed to d_2 being a stronger or weaker defense. What does vary is how much the upstream transform actually disturbs d_2 's statistic, and that quantity was measured on the mechanism itself (§D.2, `results/cos_invariance_check.json`) rather than inferred from any ASR. The frozen label scoring is fixed at $n=3$ and must not move, so we topped the six powered cells up to $n=5$ in a separate directory, with the tested direction fixed in the top-up script before the new seeds ran (`results/metric_swap_topup/`). For Krum under model-scaling (standalone 0.061 at 0.577 accuracy): NormClip changes Krum's selection in 0/9 rounds and the composition reaches 0.062 ($\Delta = +0.001$, per-seed 0.120/0.018/0.047/0.058/0.068); RFA changes it in 6/9 rounds and the composition reaches 0.779 ($\Delta = +0.718$, per-seed 0.974/0.116/0.982/0.904/0.916): a gap of +0.716 between two compositions sharing a downstream defense, an attack, and a baseline, and at $n=5$ it is statistically resolved (Welch $t = 4.28$, two-sided $p = 0.012$; paired $t = 4.60$, $p = 0.010$; Mann-Whitney one-sided $p = 0.008$), which it was not at $n=3$.

The pattern holds across the suite once cells without power are excluded: Δ is observable only where standalone d_2 genuinely suppresses, since a baseline already above 0.5 shows $\Delta \approx 0$ by ceiling effect whatever the transform did. Restricting to standalone $\text{ASR} < 0.5$ at accuracy ≥ 0.35 leaves six cells, and at $n=5$ measured disturbance still splits them perfectly: the three undisturbed cells (0/9) give $\Delta \in [+0.001, +0.190]$, the three disturbed ones ($> 0/9$) give $\Delta \in [+0.718, +0.878]$: a separation of +0.527 with no overlap. Two of the three undisturbed cells are `cos_krum` on the pixel backdoor, where the mirror suppresses standalone (0.300 at 0.598 accuracy) and carries that suppression through under *both* upstreams. Here the two extra seeds *weaken* the confirming half rather than strengthen it, and we report it as measured: Δ moves from +0.000 at $n=3$ to +0.190 under NormClip and from -0.027 to +0.147 under RFA, leaving both compositions still below threshold (0.490 and 0.447 at 0.61 accuracy) but no longer intact. Exact invariance of the *selection* therefore does not by itself carry suppression through unchanged: the magnitude channel of Lemma 2 is a second route by which Δ can grow, and the ladder's arm 3 (§E) is built to measure exactly that prospectively rather than describe it post-hoc. The ten excluded cells are listed rather than dropped silently in the supplement (§ S3).

The suite contains a second such contrast, on Reputation under model-scaling (standalone 0.017 at 0.777 accuracy), and the two are informative in opposite ways. NormClip disturbs Reputation's weight ordering in 3/9 rounds and the composition reaches 0.806; RFA disturbs it in 9/9 and the composition reaches 0.895. The gap is only +0.089 (both arms are far above threshold, because Reputation's ordering is disturbed under *both* upstreams) but within-arm variance is small, all five seeds move in the predicted direction (per-seed deltas +0.149/ +0.089/ +0.030/ +0.061/ +0.115), and it is resolved at both $n=3$ and $n=5$ (Welch $t = 3.06$, two-sided $p = 0.018$; paired $t = 4.31$, $p = 0.013$). So the two contrasts trade off exactly as the disturbance measurements imply: Krum has the large effect because one of its upstreams leaves it untouched, and Reputation was resolvable even at $n=3$ because neither upstream does.

This is post-hoc, and we do not present it otherwise. It was not pre-registered, it re-uses a suite frozen for a different purpose, the two extra seeds per cell were added *after* seeing the $n=3$ result, and the six-cell split is a description of that suite rather than a test on new data. The top-up cut in both directions, which is why we report all of it: it resolved the Krum gap (Welch $p = 0.16$ at $n=3$, where per-seed paired deltas +0.854/ +0.097/ +0.935 left one seed not separating, to $p = 0.012$ at $n=5$) and at the same time degraded the invariant control's Δ from ≈ 0 to +0.19. What the analysis does have, which neither the FLTrust contrasts of §D.1 nor the cross-pair scoring has, is C1 held exactly fixed and a disturbance measurement taken on the statistic rather than read back from the outcome. We offer it as the cleanest available piece of C2 evidence and as the shape a properly prospective version of this test would take: fix a downstream defense that suppresses the attack, then vary the upstream along measured disturbance rather than along defense identity. Table A8 sets out, claim by claim, what this design can and cannot conclude.

D.3 AN APPLICATION: SCREENING AS EVALUATION PRIORITIZATION

What follows is a consequence of the identification boundary, not a claim the paper argues for. Proposition 6 says a criterion gated on the outcome it predicts cannot be validated by cross-defense contrast; what such a criterion can still do is *order an evaluation budget*, and that is the only use we claim for it. Proposition 7 bounds what it can find. Read the numbers below as a measurement of a prioritization heuristic's cost, not as validation of the framework.

Does every condition earn its place? C3 does not: reported as a finding. Scoring all 42 pairs on C1, C2 and C3 *independently* rather than recording only the first failure: against a 12% base rate of low-ASR pairs, C1 alone selects 4 at 50% precision, C2 alone 14 at 36%, and $C1 \wedge C2$ selects 2 at 100%. **The surviving C2 failures are exactly the non-invariant signal class:** after the Proposition 10 reclassification every one of

Table A8: **What this design can and cannot conclude.** The middle column is about *identifiability*, not statistical power: a “no” marked *structural* cannot be repaired by more seeds, defenses or attacks (Proposition 6). Every entry is scoped to the measured setting of §D–§E.2.

question	identifiable?	answer, and its evidence
Does disturbing d_2 ’s statistic change its <i>decision</i> ?	measured	Yes: Krum’s selection flips in up to 0.80 of rounds
Does it change the <i>admitted adversarial mass</i> ?	by construction	No: unchanged in every one of the 60 measured rounds
Does it change <i>suppression</i> ?	instrumented	No: $ \Delta =0.026$, inside the frozen margin (Prop. 11(a))
Does <i>adversary-correlated</i> reweighting change suppression?	instrumented	Yes, non-monotonically: single-peaked (Mode A)
Does admitted mass <i>order</i> outcomes across defenses?	pre-registered	No, refuted: +0.098 against a frozen +0.150
Is (P4) <i>sufficient</i> for (P5)?	control arm	No: <code>cos.krum</code> admits 0.000 while ASR falls
Can (P4) be screened <i>without</i> adversary identity?	no, open	Both instruments are oracles: the stated open problem
Does C1 identify C2’s contribution?	no, structural	Empty cell by the gate, for every menu (Prop. 6)
Does any of this certify <i>deployed</i> security?	no, scope	Committed attack only; one adaptive test breaks it

them is a $\cdot \rightarrow \text{reputation}$ pair ($\text{nc} \rightarrow \text{rep}$ 0.819, $\text{rfa} \rightarrow \text{rep}$ 0.882, $\text{fg} \rightarrow \text{rep}$ 0.798), consensus distance being the signal a positive rescaling does not preserve, while all three non-degenerate $\cdot \rightarrow \text{FoolsGold}$ pairs are C1 failures instead. **C3 has no demonstrated incremental predictive value in this study:** alone it selects 35 pairs at 9% precision, below the base rate, and it never binds on a pair $\text{C1} \wedge \text{C2}$ already admits. So **we demote C3 from a gate to a margin diagnostic and recommend $\text{C1} \wedge \text{C2}$ as the criterion**, retaining C3 on theoretical grounds only: Theorem 8 shows margin compression is a real failure mode, and the $\alpha=0.1$ regime below is precisely a margin collapse. **The demotion changes no number**, $\text{C1} \wedge \text{C2}$ selects the same two pairs at the same precision (recomputed by `analyze_screening_cost.py`), which is why we report it as a finding about the criterion rather than a revision of results. *The pre-registered PASS/FAIL labels and the 16/18 and 22/24 tallies were frozen with C3 in the gate and are not relabelled*; this ablation is post-hoc and labelled as such, and the recommendation is for use going forward.

If the goal is to find emergent compositions, drop C1: a recommendation we price but do not validate. Proposition 7 says an inherited-suppression gate cannot certify a composition whose suppression neither constituent possesses, so an evaluator whose object is discovery rather than prioritization should not run C1 at all, and this is a consequence of the proposition rather than a finding of ours. Two designs follow, and §D.4 prices both on our own menu. **Sample the ungated menu** when the budget allows: 42 pairs at 12% precision reaches every low-ASR pair by construction, and it is the only design with no structural blind spot. **Otherwise gate on the mechanism condition alone**, C2 without C1: it admits 14 of the 42 at 36% precision, a third of the ungated cost, and it reaches FoolsGold $\rightarrow$ RFA where every C1-bearing variant misses it. **The caveat is ours and it is not small:** C2-alone’s recall on this menu is 100%, but 11 of those 14 admissions rest on a per-pair call made in sample rather than on algebra, and all 5 low-ASR pairs are among the 11 (§D.4), so that recall is a development-set number and we offer the design as a recommendation rather than as a validated procedure. Finding emergence efficiently is a search problem, and nothing here solves it: what the propositions establish is which designs cannot solve it.

D.4 THE SCREENING FRONTIER: PRECISION, RECALL AND EVALUATION COST

40% recall is an operating point, not a verdict on the conditions. Scoring C1, C2 and C3 independently over the 42 pairs traces a frontier, and the criterion the paper reports is its high-precision end. Emitted by `experiments/build_screening_frontier.py`, which joins the two existing generators and introduces no number of its own.

Neither screen is noise, and neither is a predictor. Both C2 alone and $\text{C1} \wedge \text{C2}$ beat a size-matched random draw from the same 42 pairs ($P \leq 0.005$ at 10^5 resamples against the menu’s 12% base rate, seeded `rng(42)`; emitted by `experiments/blind_selection_analysis.py`). Beating a random draw is the whole of what that establishes: it separates the screens from chance and not from each other, and it says nothing about either one’s ability to predict suppression.

Variant	selected	precision	recall	runs	reaches FG→RFA
no screen (all pairs)	42	12%	100%	420	✓
C3 alone	35	9%	60%	350	✓
C2 alone	14	36%	100%	140	✓
C2 algebraic only [†]	3	0%	0%	30	—
C2 $\wedge$ C3	8	38%	60%	80	✓
C1 alone [†]	4	50%	40%	40	—
C1 $\wedge$ C3 [†]	4	50%	40%	40	—
C1 $\wedge$ C2 [†]	2	100%	40%	20	—
C1 $\wedge$ C2 $\wedge$ C3 [†]	2	100%	40%	20	—

[†] additionally needs the 60 standalone runs before the condition can be evaluated at all.

[‡] the same screen restricted to the pairs whose C2 verdict Prop. 10 settles algebraically.

The blind spot is attributable to exactly one condition, and the theory named it before the ablation was run. Every C1-bearing variant misses FoolsGold→RFA and every C1-free variant reaches it: its row is C1 *false*, C2 *true*, C3 *true*, max-committed ASR 0.045. C1 is the inherited-suppression gate, and Prop. 7 proves no gate of that form can certify an emergent composition, so the miss is a *predicted* property of one condition rather than an unexplained failure, and it is tradeable: C2 alone recovers every low-ASR pair in the menu at 36% precision. **We report the high-precision end because false positives deploy an insecure stack**, not because the conditions cannot be operated at higher recall.

The C2 column is not all one kind of evidence, and the high-recall claim depends on the weaker kind. We describe C2 as read off the two defenses’ definitions, and for two of its four branches that is literally what happens: a d_2 reading a cosine is exactly invariant under any positive per-client rescaling (Prop. 10(a)) and a d_2 reading a consensus distance, a residual magnitude or a norm is not (Prop. 10(c)). The remaining branch is *conditional*: `coord.median` and `trimmed.mean` preserve a coordinate ordering only under a separation condition, and Thm. 8(2) gives `rfa` an analogous one, so `analyze_condition_ablation.py` marks all three None and `c2.holds` defers to a *per-pair call*, the stored category. Nothing in the C2 column distinguishes a verdict reached by algebra from one reached by that call, so we census it (`c2.provenance`, same generator): of the 14 pairs C2 admits, 3 **rest on algebra** and 11 **on the per-pair call**, the 11 being every `coord.median`, `trimmed.mean` and `rfa` row. **And all 5 low-ASR pairs are in the 11**: 0 of 5 are algebraically settled. Hence the [‡] row above. **Worse for us, all 11 are development-set pairs**, so those calls were made *in sample*, while the criterion was being written; they are not pre-registered, and the generator derives the split from which results artifact holds each pair rather than taking our word for it. **Two consequences we accept rather than repair.** First, C2 alone as we score it *is not a pure statistic-preservation screen* for those three d_2 classes, so the body’s contrast between it and C1 $\wedge$ C2 is a contrast between two screens that both borrow from the same in-sample call, not between algebra and measurement. Second, *within this menu* the region where our own invariance result decides the question contains no low-ASR pair at all, so the algebraically settled sub-region cannot discriminate any strategy from any other *here*, and the 100% recall of C2 alone is carried entirely by in-sample judgement. **The second consequence is the one we then removed**, by enlarging the algebraic region until it closed rather than by relabelling this one: the 3 pairs of the [‡] row are one column of a 3×3 census, and completing it prospectively puts a low-ASR pair inside the algebraic region and separates the two strategies more sharply there than anywhere in this table (§D.5). The *first* consequence stands and needs an invariance theorem for the conditional classes, which we do not have; what neither needs is a relabelling, and no number in this appendix or the body changes. This is (L5) in concrete form: the conditions are not always readable off a (d_1, d_2) pair.

What the screen saves, in runs rather than in pairs. `experiments/analyze_screening_cost.py`: unscreened, the 42 pairs under 2 committed attacks at 5 seeds are 420 runs; the reported criterion evaluates 16 cells, 80 runs, of which 60 are the standalone measurements C1 needs. That is 340 runs avoided, 81% **of runs**. We give the run-level figure rather than the pair-level 95.2%, which overstates the saving by ignoring C1’s own standalone cost. **The estimate is in runs, not hours**: no per-run timing telemetry exists in any results file, so any wall-clock figure would be a reconstruction.

D.5 A CLOSED CENSUS OF THE ALGEBRAICALLY SETTLED REGION, PRE-REGISTERED

The [‡] row above is one column of a 3×3 box, and the two thirds of that box which had never been run are what this wave measured. The region where mechanism preservation is settled on *both* sides is small and enumerable: d_1 emitting strictly positive per-client coefficients $T(u_i) = c_i u_i$, so Prop. 10(a) applies, against a d_2 reading an exactly scale-invariant statistic. Over this project’s defenses and the metric-swap mirrors that region is $\{\text{norm.clip}, \text{rfa}, \text{reputation}\} \times \{\text{foolsgold}, \text{cos.krum}, \text{cos.reputation}\}$, exactly 9 ordered pairs, of which 7 were already measured at the identical protocol and 2 were unspent. Enumeration is what turned a planned 20-pair sample into a census: 118 **ordered pairs of this project are already spent**, and once the admissibility constraints (a non-degenerate d_1 , and C0)

are imposed only 6 candidates remain anywhere. A census needs no sampling argument, and an all-negative census is as informative as a positive one, which is why this design was worth 24 runs and 4.87 h. Pre-registered at 96f8856 with per-pair C0–C3 verdicts, a frozen scoring rule and a two-sided resolution table (experiments/pre-registration-wave3-emergent.md); scored by experiments/analyze_wave3-emergent.py, which refuses to run against an uncommitted prereg.

$d_1 \downarrow$ $d_2 \rightarrow$	foolsgold	cos.krum	cos.reputation
norm_clip	0.887 HIGH	0.508 HIGH [†]	0.935 HIGH [†]
rfa	0.914 HIGH	0.566 HIGH	0.861 HIGH
reputation	0.807 HIGH	0.263 LOW*	0.787 HIGH*

Max-committed mean ASR over seeds 42, 43, 44; LOW iff < 0.5 .

* the two unspent cells, labels frozen at 96f8856 before the output directory existed.

[†] norm_clip's clip is inactive on the pixel arm (max update norm $3.043 < \tau=5$), so those pixel cells are bit-identical to d_2 alone; both verdicts are carried by the scaling arm, where it is not.

foolsgold is excluded from the d_1 column by measurement, and that measurement changed the design before any prediction was frozen. It was going to supply two more C1-fail $\wedge$ C2-true pairs. But foolsgold normalizes with `weights - weights.min()`, forcing at least one coefficient to *exactly* zero every round, which puts it outside Prop. 10(a)'s strict $c_i > 0$ hypothesis and under Lemma 2 instead. Measured through the shipped transform over 3 seeds $\times$ 3 rounds: 2/5 exact zeros every round, and both downstream cosine statistics moved in 9 of 9 rounds. Synthetically at 1000 trials the effect is entirely due to the zero: strictly positive coefficients change cos.krum's selection 0/1000 times, one exact zero changes it 471/1000. reputation by contrast emits 0/5 zeros every round at a positive-part ratio up to 18839.8, and under that same extreme rescaling in the same rounds the two cosine statistics moved 0/9 while non-invariant krum's selection moved 8/9 and reputation's own ordering 9/9: **a contrast between statistics on identical inputs, not between conditions.** So the two foolsgold $\rightarrow$ cos.* pairs were entered as *measured-C2-fail controls* rather than as members of the census. **Had C2 been assumed from the definitions instead of measured, this wave would have reported two mislabelled pairs** (experiments/verify_wave3-invariance.py, results/wave3-invariance-check.json).

#	pair	cell	pred.	measured
1	reputation $\rightarrow$ cos.krum	certified (C1 $\wedge$ C2)	LOW	0.263 LOW ✓
2	reputation $\rightarrow$ cos.reputation	C1-fail $\wedge$ C2-true	HIGH	0.787 HIGH ✓
3	foolsgold $\rightarrow$ cos.krum	measured-C2-fail control	HIGH	HIGH [§] (raw 0.086) ✓
4	foolsgold $\rightarrow$ cos.reputation	measured-C2-fail control	HIGH	0.824 HIGH ✓

[§] scored HIGH *only* by the frozen DEGENERATE-ACC rule: mean clean accuracy $0.255 < 0.35$ on the arm supplying the max, so its raw 0.086 is a collapsed model and not a suppressed one.

4/4 as registered, including the pair we flagged uncertain, and the near-identity check came back clean on every arm (0 of 3 seeds identical to d_2 alone on all 8 arms; max per-seed differences 0.073 to 1.000), so these are compositions and not d_2 in disguise. Pair 1 was the live risk and the prereg said so: its LOW rests on two *different* attacks being suppressed by two *different* constituents, reputation holding model-scaling (0.017 at accuracy 0.777) and cos.krum the pixel backdoor (0.300 at 0.598), and on the scaling arm the certification and the two nearest measured analogues (norm_clip $\rightarrow$ cos.krum 0.508, rfa $\rightarrow$ cos.krum 0.566) pointed *opposite* ways. It came out LOW on both arms (0.064 scaling, 0.263 pixel). **So the one low-ASR pair in the closed census is a pair neither constituent suppresses alone**, and it is the pair C1 $\wedge$ C2 selects.

Both strategies priced over the same closed census, which costs no further runs because C1 is defined on standalone numbers that already exist.

strategy over the 9-cell census	selected	precision	recall	runs
A: statistic preservation (C2 alone)	9	11%	100%	90
B: the criterion (C1 $\wedge$ C2)	1	100%	100%	10

In this region C2 alone is not a screen at all: it admits everything, because everything here satisfies it by construction. **That inverts the body's trade**, where B bought precision by giving up recall; here, at the frozen seed count, B dominates A on precision, recall and cost at once, and disclosure (v) below is the qualification that phrase needs. It is also the answer to the objection of §D.4 that the algebraically settled sub-region cannot discriminate strategies: it could not, at 3 pairs; at 9 it discriminates them completely.

Five disclosures, none of which the numbers above carry on their face. (i) B's 100% is a gated reading. C1 requires suppression at mean clean accuracy ≥ 0.35 , the gate applied throughout §D.2, and it flips exactly

two census cells: `cos_krum` alone reaches ASR 0.078 on model-scaling but at accuracy 0.146, barely above CIFAR-10’s 0.10 chance, so it is a destroyed model rather than a defended one. Read at the bare 0.5 threshold, C1 additionally admits `norm_clip` $\rightarrow$ `cos_krum` and `rfa` $\rightarrow$ `cos_krum`, both HIGH, and B selects 3 for 33% precision. The gate is the paper’s pre-existing convention rather than a knob turned for this table, and `analyze_wave3_emergent.py` emits both readings. (ii) *The one positive is $n=3$ and wide:* pair 1’s pixel arm is 0.000/0.107/0.681 per seed, so one of three seeds sits above the threshold and the LOW verdict is the frozen mean rule applied to a spread that a fourth seed could move. We do not add seeds after seeing this. (iii) *The controls are weaker evidence than intended.* Every `foolsgold-as-d1` scaling arm sits at chance accuracy (0.100), and pair 3 is HIGH only through the accuracy gate, its raw ASR being 0.086, and its scaling arm tripped the frozen “ASR identically 0.0 across all 3 seeds” flag, which on a model at 0.100 accuracy means the attack had nothing left to succeed against; so pairs 3 and 4 establish that the C2 probe is not inert, which is their stated role, but they are not independent confirmations of a HIGH prediction. (iv) *7 of the 9 cells are retrospective.* Only the two `reputation` cells were blind; the other seven were read from frozen artifacts, with each transcription in the prereg checked against its source by the analyzer. So the census’s *closure* is prospective at the margin, not throughout, and pair 2’s scaling arm tripped the second of the wave’s two frozen sanity flags (clean accuracy 0.790 against the FedAvg reference 0.667). Both flags are reported, neither adjusted. (v) *The C1 input that certifies pair 1 is the lowest-seed reading of a high-variance cell.* C1 holds for pair 1 on the pixel arm because `cos_krum` alone measures 0.300 there, and that is the mean over the three frozen seeds 42–44 (0.003/0.269/0.630). The identical cell, same protocol and identity upstream, is measured twice more in this paper at larger seed counts, and the cell’s mean climbs monotonically with the count: 0.490 at $n=5$ (Table A10, arm 3’s $\kappa=0$ rung, which adds 0.864/0.686) and 0.584 at $n=8$ (Table A11, which adds 0.578/0.845/0.795), the last being the reading §E.2 states as `cos_krum` never suppressing. Six of the eight seeds are above 0.5 and the frozen three are the two lowest plus the fourth-lowest. So B’s 100% precision and 100% recall here are contingent on the frozen seed count as well as on the accuracy gate of (i), and we state the contingency at the resolution the numbers support rather than rounding it against ourselves: 0.490 is still *below* the threshold, by 0.010, so C1 survives at $n=5$; at $n=8$ C1 fails for pair 1, B certifies nothing in this census, and its one low-ASR pair becomes another false negative of the criterion rather than its one hit. We report this instead of rescoring, because the pre-registration fixes the seed set at three and forbids adding seeds afterwards (`pre_registration_wave3_emergent.md`, §5). What the disclosure costs is the claim that B *dominates* A on this region, which holds at the frozen seeds and is not established beyond them.

D.6 OUT-OF-SAMPLE PASS VALIDATION

The development and held-out validation above tests predictions from the *same* attack families (model_scaling, backdoor-pixel) and scale ($N=10$, `cifar_cnn`) used to design the criterion. To confirm that PASS predictions generalize we pre-registered three out-of-sample tests (commit `ca627e5`, before any of them ran); all held, the accounting is in the summary below, and per-cell values for every test are in the supplement (§S4). We state plainly what this is and is not: it is evidence that the conditions keep identifying compositions which survive a change of attack family, scale and architecture; it is *not* an out-of-sample classification result, because every prediction in it is positive and the matching FAIL controls proved uninformative. Negative-direction evidence rests entirely on the in-distribution tiers.

(E) Heterogeneity: where the criterion breaks. Earlier heterogeneity sweeps in this work covered *single defenses* only; no composition had been run at any $\alpha \neq 0.5$. We ran all three PASS pairs at $\alpha \in \{0.1, 1.0, 10.0\}$ (3 seeds, both committed attacks, 50 rounds). Reporting max-committed ASR / clean accuracy:

Composition	$\alpha=0.1$	$\alpha=0.5$	$\alpha=1.0$	$\alpha=10.0$
fg $\rightarrow$ rfa	0.569 / 0.20	0.093 / 0.50	0.051 / 0.57	0.063 / 0.62
fg $\rightarrow$ cm	0.433 / 0.40	0.107 / 0.68	0.133 / 0.74	0.129 / 0.78
rep $\rightarrow$ cm	0.548 / 0.48	0.423 / 0.76	0.377 / 0.79	0.326 / 0.81

The criterion transfers cleanly across $\alpha \in \{0.5, 1.0, 10.0\}$, including the near-IID regime where FoolsGold might plausibly founder on legitimately similar benign updates; it does not, and $\alpha=10$ is indistinguishable from $\alpha=0.5$. **Under extreme heterogeneity ($\alpha=0.1$), two of the three PASS compositions cross the threshold** (fg $\rightarrow$ rfa 0.569, rep $\rightarrow$ cm 0.548); only fg $\rightarrow$ cm survives, at 0.433. Two caveats cut in opposite directions: fg $\rightarrow$ rfa at $\alpha=0.1$ is strongly bimodal across seeds (0.065–0.923), so at $n=3$ the direction is clear but the magnitude is not; and clean accuracy collapses to 0.20–0.48, so this regime stresses utility as much as security. Crucially Theorem 8 supplies a **diagnostic quantity that retrospectively explains this boundary** rather than predicting it: re-measuring on the transformed points at $\alpha=0.1$, RFA’s separation margin $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ collapses from an all-round mean of 3.24 (met in 68% of rounds) to 1.01 (met in 20%). Over the payload-carrying rounds alone (the ones Figure A1 plots) the same collapse reads 4.71 ± 1.51 ($n=17$) to 2.06 ± 2.35 ($n=12$), with the per-round minimum falling $2.82 \rightarrow 0.00$; the all-round means are lower because they average in the Lemma 2 rounds, where FoolsGold zeroes every adversary and m is recorded as 0. Either way

the collapse is measured, and the all-round figure is the one that reaches the vacuous limit. This sits exactly at the $m \rightarrow 1$ boundary where Corollary 2’s bound degrades to 40% adversarial mass: extreme heterogeneity inflates the benign radius R'_B until the benign cluster is no longer separated from the adversarial points, which is precisely the precondition the theorem requires. A practitioner running the 5–10 round pilot would measure $m \approx 1$ and decline to deploy, so the criterion is mechanistically diagnostic at this boundary, though we emphasise this is a post-hoc measurement of a known failure, not a pre-registered prediction of it.

The one axis not validated with composition experiments is *dataset* (CIFAR-100, with 100-class output): the conditions are dataset-agnostic in mechanism, but the downstream aggregator’s decision boundary may shift with the number of classes (L1).

E THE INTERVENTION EVIDENCE: THE CONFOUNDED LADDER, AND THE INSTRUMENT IT MOTIVATED

The controlled upstream transform. We introduce a synthetic d_1 whose only parameter is the *dispersion* of a positive per-client rescaling. For the K participants of a round, at position j in a (seed, round)-keyed pseudorandom permutation,

$$c_j \propto \exp\left(\kappa \cdot \left(\frac{2j}{K-1} - 1\right)\right), \quad \text{normalized so } \frac{1}{K} \sum_j c_j = 1. \quad (3)$$

Table A9: The four arms of the dose–response ladder, and the frozen predictions. The statistic each d_2 reads is, in order, summed pairwise Euclidean distance, distance to the client median, pairwise cosine, and coordinate ordering. *Standalone* is d_2 ’s own ASR (clean accuracy) on that attack, the C1 input, held fixed within every arm. *Measured disturbance* is the fraction of live rounds on which the rung changes that arm’s decision (for `coord_median`, the mean fraction of *coordinates* whose median-attaining client changes) measured before the freeze on the same raw updates, with no ASR computed (`results/dose_disturbance.json`). Shapes and decision rules are frozen at commit e711a95.

	arm (d_2 / attack)	class (Prop. 10)	standalone ASR (acc)	measured disturbance				predicted shape in κ
				$\kappa=0$	0.5	1.0	2.0	
1	krum / scaling	(c) not inv.	0.061 (0.577)	0.00	0.60	0.73	0.87	monotone rise
2	reputation / scaling	(c) not inv.	0.017 (0.777)	0.00	0.87	0.87	0.93	monotone rise
3	cos.krum / pixel	(a) exact	0.300 (0.598)	0.00	0.00	0.00	0.00	FLAT (control)
4	coord_median / pixel	(b) cond. inv.	0.443 (0.767)	0.00	0.25	0.38	0.49	rise, shallower

The abscissa is measured, not nominal. What matters to C2 is not ρ but what the rescaling does to the statistic d_2 actually reads, so before freezing any prediction we pushed live FL updates through the shipped transform and recorded each arm’s decision at every κ on the *same* raw updates, computing no ASR. That establishes three things before any outcome exists (Table A9): *there is a dose*: disturbance is exactly 0 at the identity rung and rises monotonically with κ in every non-invariant arm; *the class-(a) invariance claim holds in the shipped code*: `cos.krum`’s selection is unchanged in 0 of 120 cells at ρ up to 54.6; and *class (b) is graded rather than binary*, at roughly half the class-(c) disturbance (0.487 vs. 0.867/0.933 at $\kappa=2.0$), which is what arm 4’s prediction turns on.

Frozen predictions. The *primary* test is pooled: across all 16 (arm, κ) cells, the rise in ASR over that arm’s own $\kappa=0$ rung is predicted to be positively rank-correlated with the measured disturbance rate (Spearman, one-sided, $\alpha = 0.05$). Its predictor is a *measured property of the statistic*, not the identity of the defense and not its standalone effectiveness, both held fixed within every arm. The *secondary* predictions are the per-arm shapes of Table A9, tested by a one-sided Jonckheere–Terpstra trend test across the four ordered rungs at $n=5$, with arm 3’s “flat” scored as an *equivalence* claim at a 0.15 margin fixed and justified in advance (under a quarter of the +0.629 within-defense gap for Krum (the $n=3$ value on record at the freeze; the later top-up puts it at +0.716, so the margin is if anything conservative) and $0.300 + 0.15 = 0.450 < 0.5$, so “within the margin” and “still suppressing” coincide for this arm) and arm 4’s class-(b) signature being the *shallower slope*.

One prediction we deliberately do not make. Theorem 8(1)’s ordering-survival condition $\rho < S$ would give arm 4 a crossing rung fixed by a measured S , but **no frozen prediction here uses S** : we measured it and it is **not a stable per-attack constant**, ranging 0.00–12.59 on model-scaling (median 0.57) and 0.61–1.63 on the pixel backdoor (median 0.92) across the 12 rounds that sampled an adversary, so the $S = 10$ used illustratively in §C.3 is a stipulation rather than a measurement. This is a limitation of our own theorem that the dose-response measurement surfaced, reported as one (L5) rather than quietly left unchecked.

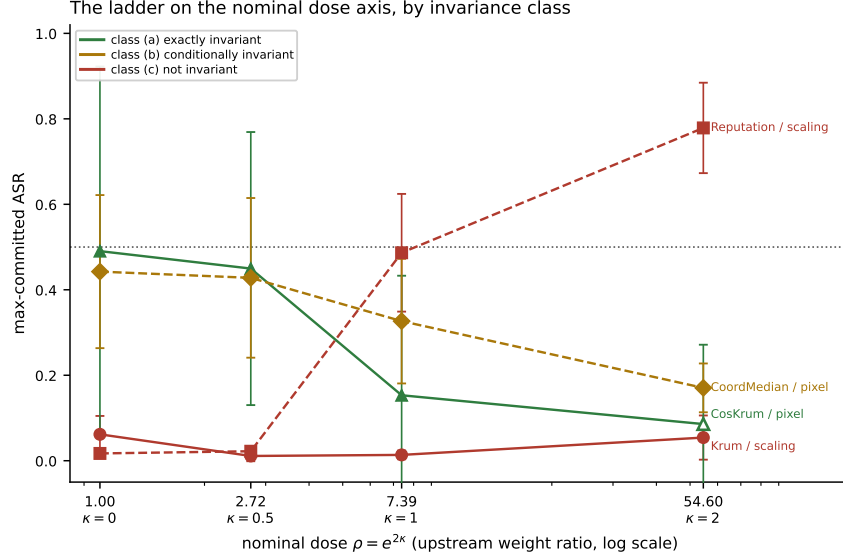


Figure A3: **The ladder and its frozen predictions.** The 16 cells on the nominal dose axis $\rho = e^{2\kappa}$ ($n=5$ per point, Student-t 95% intervals), coloured by Proposition 10 class. Of the two class-(c) arms whose statistics the framework’s fragility claim is stated over, only reputation rises; ASR *falls* with the dose in three of the four arms, against the predicted rises. Hollow markers fail the 0.35 clean-accuracy gate and are uninterpretable. Drawn by `paper/figures/dose_response_figure.py`, which recomputes every value from `results/` and imports the arm construction from the scoring script so the two cannot diverge.

E.1 OUTCOME OF THE CONFOUNDED DESIGN: ONE ARM CONFIRMED, TWO REFUTED, ONE INDETERMINATE, AND THE POOLED TEST FAILS

All 80 runs completed (16.5 h wall clock). Table A10 is the whole ladder; Figure A3 plots it on the nominal dose axis against the frozen shapes and Figure A4 on the independently measured disturbance axis with the real-transform anchors, scored by `analyze_dose_response.py` against the rules frozen at e711a95, every number recomputed per seed from `results/dose_response/`. The **primary** pooled test does not confirm: across the 16 cells, the rise in ASR over each arm’s own $\kappa=0$ rung correlates with the measured disturbance at Spearman $\rho_s = +0.351$, one-sided $p = 0.091$, so the frozen criterion ($\rho_s > 0$ at $p < 0.05$) is not met and **we report the primary prediction as refuted**. Of the four frozen shapes one is confirmed, two are refuted, and one is indeterminate; **in three of the four arms ASR falls as the dose rises**. This is the round’s headline result and it is a negative one. It is also the outcome the pre-registration anticipated and named in advance: “pooled correlation absent while individual arms move: the mechanism is defense-specific and measured disturbance is not a transferable predictor.”

Table A10: The dose–response ladder as measured, $n=5$ per cell, per-attack mean ASR with mean clean accuracy in parentheses. *Predicted* is the shape frozen at commit e711a95 before any run; $p_{\uparrow}$ is the one-sided Jonckheere–Terpstra probability for an *increasing* trend across the four ordered rungs, which is the only direction any frozen rule tests. Verdicts apply the frozen criteria unchanged (Table A9 and the supplement § S6); an arm satisfying neither its confirmation nor its refutation criterion is reported as *indeterminate* rather than scored in our favour. [†] below the 0.35 clean-accuracy gate: a collapsed model, not suppression. Student-t 95% intervals and per-seed values are in the supplement (§ S8).

arm (d_2 / attack)	predicted shape	mean ASR (clean accuracy)				JT verdict $p_{\uparrow}$
		$\kappa=0$	0.5	1.0	2.0	
1 krum / scaling	monotone rise	0.062 (0.57)	0.011 (0.65)	0.014 (0.68)	0.054 (0.62)	0.63 refuted
2 reputation / scaling	monotone rise	0.017 (0.78)	0.022 (0.78)	0.487 (0.77)	0.779 (0.70)	1.7×10^{-6} confirmed
3 cos_krum / pixel	FLAT (control)	0.490 (0.61)	0.450 (0.57)	0.153 (0.47)	0.086 (0.22) [†]	0.999 indeterminate
4 coord_median / pixel	rise, shallower	0.443 (0.77)	0.428 (0.76)	0.327 (0.75)	0.170 (0.69)	0.998 refuted

Arm 2 confirms: the ladder yields one prospective confirmation, while the pooled prediction is rejected. Reputation under model-scaling rises monotonically from 0.017 at the identity rung to 0.487 at $\rho = 7.39$ and 0.779 at $\rho = 54.6$ (Jonckheere–Terpstra $z = +4.64$, one-sided $p = 1.7 \times 10^{-6}$; permutation $p < 5 \times 10^{-5}$),

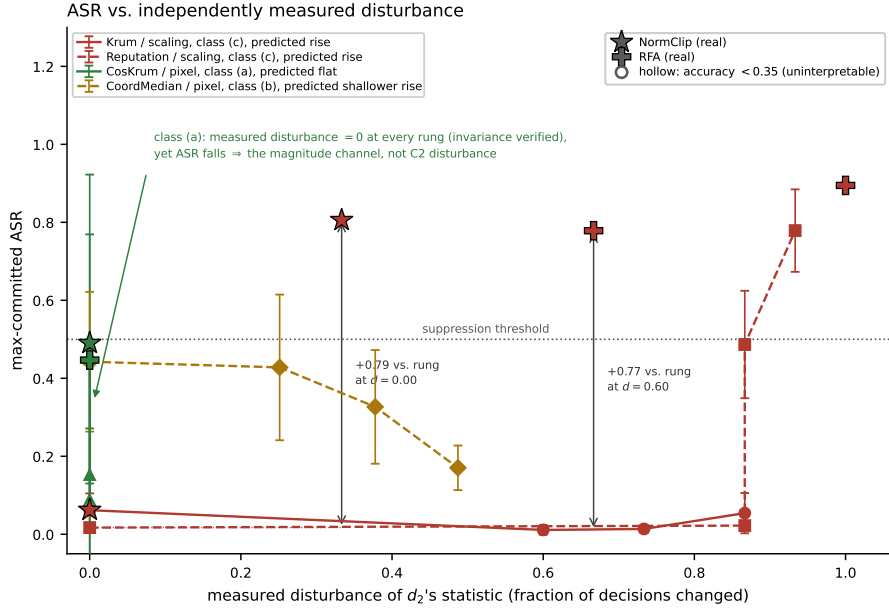


Figure A4: **Why a transform’s flip rate is not a sufficient statistic for what it does.** ASR against the *independently measured* disturbance of each arm’s own statistic ($n=5$, Student-t 95% intervals), the abscissa fixed before the freeze. Stars and crosses are the two *real* upstream transforms measured on these same (d_2 , attack) cells (NormClip, RFA), placed at their own measured disturbance rate: at matched disturbance they sit +0.77 and +0.79 ASR *above* the synthetic curve, while their realized weight ratios (mean 3.61 and 11.93, maximum 4.90 and 17.38) fall *inside* the ladder’s span. The class-(a) arm’s four rungs coincide at disturbance exactly 0 (0/120 selections changed, maximum relative score drift 1.1×10^{-4}) and its ASR falls anyway: Lemma 2’s magnitude channel, not C2 disturbance.

while clean accuracy moves only $0.777 \rightarrow 0.701$, so what is lost is suppression, not the model. Same d_2 , same attack, same standalone baseline at every rung: nothing about effectiveness varies, and $55\times$ dispersion in a *positive per-client rescaling* (a transform the criterion treats as benign for invariant statistics) carries a defense from 0.017 across the 0.5 threshold. Two qualifications keep this in proportion. It validates the *destruction* half of C2’s mechanism (a non-invariant statistic, actually disturbed, loses suppression); the *preservation* half (invariance $\Rightarrow$ suppression survives) is arm 3, which came out indeterminate. And even within this arm the measured disturbance is not what orders the curve: it saturates at 0.867 by $\kappa=0.5$, where ASR is still 0.022, and the jump to 0.487 happens at the next rung at the same measured disturbance, so the *dose* moves ASR here while the flip rate does not.

Ecological validity: the real transforms do not lie on the synthetic curve. Pre-registered caveat (iv) required this check, and it fails in an informative way. RFA $\rightarrow$ Krum sits at ASR 0.779 with measured disturbance 0.667, where the synthetic ladder’s nearest rung (0.600) gives 0.011: a residual of +0.767. NormClip $\rightarrow$ Reputation sits at 0.806 with disturbance 0.333, against 0.017 at the identity rung: +0.789. Two anchors do land: RFA $\rightarrow$ Reputation, 0.895 at disturbance 1.000 against 0.779 at 0.933 (+0.116), and both `cos_krum` anchors, which coincide with its curve (+0.000 and -0.044) because that statistic is invariant under all three transforms. This is not a range mismatch: read off the same live rounds, NormClip realizes ρ of mean 3.61 (max 4.90) and RFA mean 11.93 (max 17.38), both inside the ladder’s $\{1.00, 2.72, 7.39, 54.60\}$ span. The conclusion is about our abscissa: **the rate at which a transform flips a defense’s decision is not a sufficient statistic for what that transform does to the defense**: two transforms at matched flip rate differ by up to 0.77 in ASR. **And the axis they are unmatched on is now measured, not conjectured.** Adversary-correlated dispersion $\Delta_c := \mathbb{E}[c_i | i \in A] - \mathbb{E}[c_i | i \notin A]$ (§C) separates the transforms carrying those anchors from the ladder: RFA reaches $\Delta_c = -1.21$ and NormClip -0.72 , against -0.34 at the ladder’s most dispersed rung: $2\text{--}3.6\times$ more extremely targeted at ρ that is 3.7 and $13\times$ smaller, so **matched on the statistic, badly unmatched on the targeting, and ρ does not order Δ_c . This rescues no predictor**: Δ_c ’s magnitude does not order suppression either; rfa $\rightarrow$ krum still fails at 0.779 at $\Delta_c = -1.21$, while reputation, targeted harder still on both attacks, has rep $\rightarrow$ coord_median suppressing, and FoolsGold’s Δ_c is *positive* (+0.663). These figures are **post-hoc and exploratory**, computed with no ASR and rescoring nothing, and are tabulated in the supplement (§ S10). Arm 4 has no real-transform anchor at all, since `coord_median` was not on the metric-swap menu, so its ecological validity is untested.

What we now claim less of. C2’s causal content is established prospectively for one statistic (distance to the client median) and refuted for another (summed pairwise distance) at p up to 54.6; the pooled claim that measured disturbance predicts loss of suppression *across* defenses is refuted, and we no longer make it. Within-defense disturbance remains the only design the testability boundary admits, and §C.1’s general statement stands untouched by this outcome; it is a statement about admissible designs, and this is the admissible design, run and reported as it landed. What changes is the empirical status of C2: one confirmed arm, not a validated screening mechanism.

E.2 THE TARGETED INTERVENTION: ISOLATING C2 FROM THE ATTENUATION CHANNEL

Reading `cos_krum`’s entry in Table 1. Its -0.425 is the full-ladder endpoint, which lies below our accuracy floor; **the witness we rest on is instead its first, gate-passing rung** (0.173, §6.2). The rank discordance between $\Delta\Lambda_n$ and ΔASR across the two moving arms is §6.2’s refutation of our own predicted ordering. This section gives the emitter and the per-artifact provenance for every cell of that table.

Round counts and per-row provenance, relocated here from the table’s caption. Admission never moves in any measured round of any arm: 60 rounds per CIFAR-10 arm and 56 on the EMNIST-byclass replication, including the 48 per arm where reputation and `coord_median` carry adversarial mass at baseline, so the constancy is not an artifact of arms that never admit an adversary at all. The decision/admission dissociation is accordingly not one aggregator’s property: it holds for a selector, a cosine-invariant selector, a coordinate-wise order statistic and a weighted averager, which is what the table’s *Kind* column records. Row 5 replicates row 1 on a second dataset and architecture, with its channels measured *before* its ASR existed; row 6 is the pre-registered score-only control, row 1 with the magnitude channel closed, so its decision, admission and influence columns are row 1’s own. The table is *not* an out-of-sample test. **Row 5 is the frozen $n=3$ in every column, its ΔASR included**, because its three channel columns were measured at that seed count and were not re-measured: the same cell’s ΔASR at $n=5$ is -0.013 , reported with its provenance in disclosure (i) below, and substituting it here would put one $n=5$ cell in a row of $n=3$ ones.

What Mode S does to the ladder’s confirmation. Reputation was the ladder’s one confirmed arm (§E.1); Mode S shows most of that confirmation was attenuation. With the adversary’s share held fixed, reputation’s rise at $\rho=54.6$ shrinks from the ladder’s $+0.762$ to $+0.178$ (one-sided Welch $t=-5.22$, $p=0.0017$ that it is smaller), and what remains tracks a small *admission* change ($0.000 \rightarrow 0.020$) rather than the large decision change ($0.000 \rightarrow 1.000$), so H-admission is confirmed here too (Table A11). The confound is thus *explained*, not explained away: a genuine, much smaller, admission-driven effect survives it. The class-(a) control cuts the other way and refutes us: `cos_krum`’s selection is bit-identical at every rung (decision and admission both 0.000), yet ASR is 0.584 at the identity (it never suppresses) and *falls* 0.425 across the ladder ($p_{\downarrow}=0.0008$; the $\kappa=2$ rung is below the accuracy gate). Exact invariance of the statistic neither implies suppression nor holds it fixed; the magnitude channel moves it regardless. We report this as a refutation of Proposition 10’s practical content, not a confirmation of its algebra.

Mode A: the two regimes, and where they hold. The payload dose gives Theorem 8 a single falsifiable shape: ASR single-peaked at $r=1$, since an over-attenuated adversary carries no payload (Regime B) and an over-amplified one is discarded by a rank aggregator (Regime A). It is **confirmed for `coord_median`**, the rank aggregator the regimes are stated for: ASR rises $0.124 \rightarrow 0.315 \rightarrow 0.443$ into $r=1$ and falls $0.443 \rightarrow 0.296 \rightarrow 0.098$ past it, both sides significant ($p_{\downarrow}=3.0 \times 10^{-4}$ amplified, 6.4×10^{-4} attenuated) at accuracy ≥ 0.74 throughout (Figure A5). It is **refuted for reputation**, which is monotone: ASR is maximal at the most attenuated rung (0.894 at $r=0.135$) and near zero once $r \geq 1$, because an attenuated adversary reads as *consistent* to a reputation score and stops being down-weighted. Reputation is not a rank aggregator, so the two-regime shape was never licensed for it; the split is the theorem behaving exactly where it applies and not where it does not.

A pre-registered replication on a second downstream defense, and what it refutes. One arm cannot establish which channel matters, and the two Mode-S arms above move together on both candidate predictors: reputation is higher on *both* decision and admission change and also had the larger rise, so nothing in that pair separates them. `coord_median` under the pixel backdoor is *rank-discordant* and therefore does: of the interpretable Mode-S arms it has the **smallest decision change** (0.482) and the **largest admission change** (0.033), so the decision reading predicts its rise is the *smallest* of the three and the admission reading predicts the *largest*. It is also a structurally different defense, a coordinate-wise order statistic (Proposition 10(b)) against Krum’s selector (class (c)), and it suppresses the pixel backdoor at usable accuracy, so there is suppression to lose. Thresholds were frozen against the two published rises (confirm if $\Delta > +0.178$, refute if $\Delta < +0.150$, indeterminate between) at commit 06d9bce before results/dose.replication/ existed, with the identity rung imported bit-identical and 15 new runs spent (experiments/pre.registration.dose.replication.md).

The outcome is $\Delta = +0.098$: the admission ordering is refuted. The arm with the largest admission change rose *less* than an arm with a smaller one, so **admission magnitude does not order outcomes across**

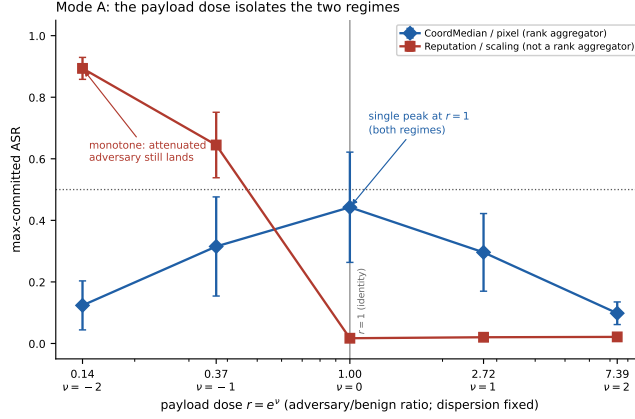


Figure A5: **Mode A, the payload dose: the two mechanism-preserving regimes, and where they hold.** With dispersion fixed and only the adversary/benign weight ratio r swept, `coordmedian` (a rank aggregator, the case Theorem 8’s regimes are stated over) is **single-peaked at $r=1$** as the theorem predicts (over-attenuated adversaries carry no payload, over-amplified ones are discarded) while reputation is monotone, off-spec and refuted for that shape. Mode A is the Δ_c -swept companion to Mode S’s Δ_c -pinned design (Proposition 12); Mode S itself is Figure 1. Bars are Student- t 95% intervals; numbers recomputed per seed by `analyze_targeted_dose.py`.

Table A11: Mode S (statistic-only), $n=5$ per arm ($n=8$ for `cos_krum`), per-attack mean ASR with the two measured channels. The decision channel moves far more than ASR throughout. $\dagger$ below the 0.35 accuracy gate. $\ddagger$ the pre-registered replication arm, *rank-discordant* by design (smallest decision change, largest admission change of any interpretable arm) and scored against thresholds frozen separately at 06d9bce before `results/dose_replication/` existed: $\Delta=+0.098$ falls below reputation’s $+0.178$, so the cross-arm admission *ordering* is refuted while the negative replicates on a different invariance class (§E.2). Verdicts otherwise scored against the readings frozen at 5130cec; per-seed values in the supplement (§ S8).

arm	channel	$\rho=1$	2.72	7.39	54.6	test	verdict
krum / scaling	ASR	0.062	0.016	0.023	0.036	$p_{\uparrow}=0.70$	H-admission confirmed, H-stat. refuted
	decision	0.000	0.533	0.800	0.733		
	admission	0.000	0.000	0.000	0.000		
reputation / scaling	ASR	0.017	0.019	0.025	0.195	rise $+0.178$ vs. ladder $+0.762$ ($p=0.0017$)	H-admission confirmed (mostly atten.)
	decision	0.000	0.933	1.000	1.000		
	admission	0.000	0.002	0.008	0.020		
cos_krum / pixel	ASR	0.584	0.411	0.295	0.159 $\dagger$	$p_{\downarrow}=8\times 10^{-4}$ $ \Delta =0.425$	refuted (Prop. 10 practical)
	decision	0.000	0.000	0.000	0.000		
	admission	0.000	0.000	0.000	0.000		
coord_median / pixel $\ddagger$	ASR	0.443	0.442	0.490	0.541	rise $+0.098$ < $+0.150$ ($p_{\uparrow}=0.069$)	ordering refuted; negative repl.
	decision	0.000	0.232	0.358	0.482		
	admission	0.000	0.005	0.013	0.033		

arms. Neither does the decision reading, which predicted the *smallest* rise of the three when $+0.098$ is the middle one, so this arm refutes a rank prediction from *each* candidate channel. The pooled model agrees: re-fit with these cells it is still not confirmed ($R^2=0.239$; the admission coefficient’s one-sided p moves $0.055 \rightarrow 0.041$, but the adversarial-share coefficient is negative, so the pre-specified conjunction fails). **What does replicate is the negative:** a decision change of 0.482 costs no suppression by the frozen margin, at mean accuracy 0.698, so the accuracy gate does not bind and the flagship result now holds on two downstream defenses in *different* invariance classes. Two qualifications, the first pre-specified in the freeze and the second post-hoc: this cell’s identity ASR is 0.443, so equivalence is to the *identity rung* and not to a low absolute ASR (an “every rung <0.5 ” criterion has no headroom here and was deliberately not used); and the rise is positive in 5/5 seeds (paired $t=3.39$, one-sided $p=0.014$, *not* pre-registered), so on this class the cost of disturbing the decision is **small and consistent rather than zero**. All numbers are recomputed per seed by `analyze_dose_replication.py`, which imports its rung builder and trend test from the frozen suite’s scorer.

The confounded design gets the causal sign wrong. This is the paper’s central empirical result. On this identical cell (matched aggregate displacement, decision change and clean accuracy, Table A3), **the confounded design does not lose power, it gets the sign wrong:** the outcome-gated ladder moves -0.272 ($p_{\downarrow}=0.0015$; $0.443 \rightarrow 0.170$) where the instrument moves $+0.098$. Closing the attenuation channel does not sharpen a weak positive effect, it **inverts it**, so a criterion calibrated against the confounded contrast would have been calibrated against the confound, and a cross-defense contrast would have read the fall as evidence about the statistic. That is why the identification boundary of §C.1 is not a technicality about power.

What the two arms jointly establish, stated formally. The arms above and the class-(a) control (`cos_krum`) are between them a *constructive* answer to the question C2 was written to settle. Both halves are witnessed by runs already reported, so the following formalizes measurements rather than adding a claim.

Proposition 11 (Statistic-level non-identifiability). *Within the positive per-client rescaling class there exist transform families T_1, T_2 and downstream defenses d_2 such that (a) T_1 and T_2 induce arbitrarily different statistics and decisions at d_2 while inducing the same admitted adversarial mass and the same suppression; and (b) T_1 and T_2 induce the same admitted adversarial mass (indeed the same statistic and the same decision) while inducing different suppression. Consequently no predicate of the downstream statistic alone determines the admitted adversarial mass, and no predicate of the admitted mass alone determines suppression.*

Proof. Both halves are exhibited by runs reported in this section. (a) Mode S into Krum under model-scaling, $\kappa=0$ against $\kappa=2$: ρ moves from 1 to 54.6 and Krum’s decision changes in up to 0.80 of rounds, while the support of the admitted adversarial mass changes in none of the 60 measured rounds (a floor on this arm: no adversary is admitted at any rung), the adversarial coefficient share is constant at 0.266667, and suppression shows **no evidence of a practically meaningful change under the pre-registered ± 0.15 practical-equivalence margin** ($|\Delta|=0.026$). (b) `cos_krum` under the pixel backdoor: by Proposition 10(a) its statistic is invariant, and its decision and admission are measured bit-identical (0.000) at every rung, yet ASR falls 0.173 by the first rung alone, at $\rho=2.72$, where the accuracy gate still passes ($0.599 \rightarrow 0.567$; paired $t=-1.976$, one-sided $p=0.044$, falling in 5/8 seeds). $\square$

The corollary is why the title says *beyond* statistic preservation: no metric defined *solely* on preservation of the downstream statistic can identify attack-suppression equivalence, because (a) exhibits statistic disturbance without suppression change and (b) exhibits suppression change without statistic disturbance. **Scope, stated immediately:** this is a non-existence claim *within the positive-rescaling class on these defenses and attacks*, established by exhibited counterexample: **not** a claim that no metric of any kind can order security, and not a claim about transforms outside the class.

Proposition 12 (ρ and Δ_c are independently controllable). *Within the same class, the statistic-level dispersion $\rho = \max_i c_i / \min_i c_i$ and the adversary-correlated dispersion $\Delta_c = \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$ are not functionally related: there is a family with $\Delta_c \equiv 0$ exactly while ρ ranges over $[1, 54.6]$, and a family with the statistic’s dispersion held fixed while Δ_c ranges over $[-1.138, +2.469]$. Hence neither quantity determines the other.*

Proof. By construction, with the realized values tabulated in the supplement (§ S10) and computed with no ASR involved. Mode S pins every adversarial coefficient at $c=1$ and spreads only the benign clients, giving $\Delta_c \equiv 0$ at every ρ up to 54.6; Mode A holds the benign weights uniform and sweeps only the adversary-to-benign ratio r , moving Δ_c over $[-1.138, +2.469]$ at fixed statistic dispersion. $\square$

This is why the two instruments were the right ones, and it makes the supplement’s §S10 warning precise: **a transform reported by ρ alone leaves Δ_c entirely free**, so ρ cannot be a sufficient statistic for what the transform does to a defense. It does *not* follow that Δ_c is a predictor: §E.1 reports that its magnitude fails to order suppression either.

The dissociation is not Krum-specific: all four aggregators, all four channels. Table A12 collects the endpoint change ($\kappa=0$ against $\kappa=2$) in each measured channel under Mode S for every downstream defense in the paper. **Recomputed, not transcribed:** every cell is emitted by `build_channel_table.py`, which only reads frozen artifacts under `results/`, `admission_measurement.json` and `femnist_admission.json` for the channel columns, the `targeted_dose/`, `dose_replication/` and `dose_femnist/` summaries for the ASR column; it also asserts that the identity rung displaces nothing ($0.00e+00$ for all six rows). The sixth row’s aggregate column comes from `displacement_decomposition.json` and its ASR from `score_only/summary.json`, both read-only; the emitter refuses to merge that arm into the ASR sources it shares cell keys with, so it cannot shadow the first row’s published number. The influence column is **aggregator-specific by necessity** (a selector has no weights) and its instantiation is: adversarial share of the Weiszfeld/reputation weight mass for the averager, adversarial share of selected coordinates for the order statistic, and the indicator that an adversarial update is selected at all for

the two selectors. The fifth row replicates the first row’s arm on a second dataset and architecture (EMNIST-byclass with `simple_cnn`) with **its channels measured prospectively, before its ASR existed**; the result is reported below. **What the Δ adm. column pools**: the four CIFAR-10 rows contribute 60 measured rounds each, 240 in total, and adversarial mass is present at baseline in 100 of those 240 rounds (including the 48 per arm where reputation and `coord_median` carry it) while its support moves in none of them, which is the count Figure 1’s annotation reports and why that annotation carries no single arm’s name.

Table A12: **The four channels, all four aggregators, under Mode S.** Endpoint change $\kappa=0 \rightarrow \kappa=2$. **The first column, Δ agg., is the relative displacement of the aggregate the defense emits**, $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$: what enters training, and **not** the value of a scoring statistic; the measured channels are aggregate displacement, decision, admission and influence. The dissociation between the *decision* channel and the *admission* channel is not a property of one aggregator: it holds for a selector, a cosine-invariant selector, a coordinate-wise order statistic and a weighted averager. The table does **not** order the arms ($\Delta\Lambda_a$ and ΔASR disagree in rank, which is §E.2’s refutation stated in one row-pair) and it is *not* an out-of-sample test: all rows are Mode-S ladders reported elsewhere in this paper. **The sixth row is the score-only control**: the same arm as the first with the magnitude channel closed, so its Δ agg. is the re-selection component alone and its decision, admission and influence columns are the first row’s own numbers, score-only leaving the selection untouched.

Aggregator	Kind	Δ agg.	Δ dec.	Δ adm.	$\Delta\Lambda_a$	Δ ASR
Krum	selector (pairwise distance)	0.892	0.733	0.000	0.000	-0.026
Reputation	weighted averager (cross-round state)	0.659	1.000	0.000	0.020	+0.178
Cosine-Krum	selector (cosine distance)	1.297	0.000	0.000	0.000	-0.425
Coord. median	coordinate-wise order statistic	0.670	0.482	0.000	0.033	+0.098
Krum (EMNIST-byclass)	selector (pairwise distance)	0.842	0.643	0.000	0.000	-0.017
Krum, score-only	selector, magnitude channel closed	0.986	0.733	0.000	0.000	-0.023

A second dataset and a second architecture, pre-registered. The fifth row’s arm is the first row’s arm re-run with EMNIST-byclass (62 classes, $1 \times 28 \times 28$) and `simple_cnn` in place of CIFAR-10 and `cifar_cnn`, everything else held: Mode S, Krum as d_2 , the committed model-scaling attack, the same four frozen rungs $\kappa \in \{0, 0.5, 1, 2\}$. The decision rule lives in `experiments/pre_registration_dose_femnist.md`, git-committed before any EMNIST-byclass ASR existed, and the runner refuses to start unless its `PREREG.COMMIT` matches that hash.

One naming correction, made here rather than left to be inferred. Earlier drafts labelled this arm “FEM-NIST” in the tables while the prose said EMNIST-byclass, and the tables were the wrong ones: the `femnist` key loads `torchvision` EMNIST (`split="byclass"`) and partitions it with this paper’s own label-Dirichlet split, **not** LEAF FEMNIST’s partition by writer. So the arm is a second dataset and a second architecture, and it is **not** a more federated partition than the CIFAR-10 arm; it is the same partition scheme on different data. Every table now reads EMNIST-byclass. The frozen artifact paths, the pre-registration filename and the JSON dataset key retain the `femnist` spelling because renaming a committed artifact would break the provenance the rest of this appendix rests on.

The premise was measured first, with no ASR anywhere. Krum’s selection changes on 0.643–0.714 of rounds across the three disturbed rungs while the support of the adversarial mass it admits is unchanged in all 56 measured rounds (again a floor: no adversary is admitted at any rung), so the arm has a real disturbance to test. Had the decision change come out near zero the arm would have been reported ineligible rather than run: that is the `cos_krum` failure mode, and this check is what distinguishes the two.

Suppression shows no evidence of a practically meaningful change. $\Delta = -0.017$ (mean ASR 0.027 $\rightarrow$ 0.010) against the frozen ± 0.15 practical-equivalence margin, beside CIFAR-10’s -0.026 on the same cell, with mean clean accuracy *rising* 0.767 $\rightarrow$ 0.835 so the accuracy floor binds on no rung. Under the frozen rule this is a **replication on a second dataset and a second architecture**: the pre-registration fixes that wording and explicitly forbids “generalizes”, which one further dataset does not license.

Three disclosures, each of them from the freeze rather than from hindsight. (i) $n=3$ seeds per rung, not 5, fixed in advance to match the EMNIST-byclass payoff matrix’s own trial budget. **The frozen verdict is that $n=3$ one, and the same ladder also reads at $n=5$ with no new run needed**: the comparability amendment of §E.3 ran seeds 45 and 46 of this exact cell at all four rungs, so pooling them with the frozen three, frozen rows winning and the frozen directory never written, gives $\Delta = -0.013$ beside -0.017 , mean ASR 0.024 $\rightarrow$ 0.011 at accuracy 0.758 $\rightarrow$ 0.841: the same label under the same unremoved margin. Two more seeds *widen* the interval across zero, $[-0.034, +0.009]$ against $n=3$ ’s $[-0.033, -0.001]$, so they remove an apparent small fall rather than confirm one, and **one of the five carries the opposite sign** (seed 45, $+0.015$, its identity rung landing at ASR 0.000018), which a 3-seed mean cannot show. Both counts are reported and neither replaces the other

(`analyze_dose_femnist.py --pooled`). Those two seeds were pre-registered in the comparability document before they ran and are post-hoc in this arm’s own. (ii) The secondary Jonckheere–Terpstra test was specified to detect a monotone *rise* (the outcome that would have refuted the flat prediction with attenuation already closed by construction) and the observed trend is a *fall* ($J=16.0$, $z=-1.563$, $p_{\uparrow}=0.941$; at $n=5$, $J=57.0$, $z=-1.211$, $p_{\uparrow}=0.887$, the same direction). We report it and we do **not** score it as support for flatness; the pre-registration states in advance that at $n=3$ a null from this test is not evidence of flatness, and two more seeds do not convert one into evidence. (iii) The two branches had unequal headroom. With the identity rung at ASR 0.027, refuting us required mean ASR($\kappa=2$) > 0.177 against a ceiling of 1 and was fully reachable, whereas the indeterminate branch ($\Delta < -0.15$) was arithmetically unreachable. **Equivalence here is to the identity rung, not to a low absolute ASR.**

Harness sanity, which is an identity check and not corroboration. The $\kappa=0$ rung is identity $\circ$ Krum, i.e. Krum alone, and it lands *bit-identically* on the EMNIST-byclass Krum figure from our earlier payoff-matrix sweep (0.0269784552 at accuracy 0.7670795973). Both pipelines are deterministic given a seed and both run three trials from base seed 42, so the agreement shows they execute the same protocol rather than a similar one: the strongest form the pre-registered check can take, and for the same reason **not** independent corroboration of that figure: the same computation run twice cannot corroborate itself. **The pooled seeds of disclosure (i) now carry the same check across two runners.** Having already read those two seeds from the comparability suite, we ran this arm’s own runner over the same eight (rung, seed) cells, pre-registered as an amendment at ad5479b before those runs: it reproduces all eight to ten decimal places, largest gap 0.00e+00, so the pooled reading is not an artifact of which suite measured it (`analyze_dose_femnist.py --pooled`). The same reading applies: identical protocols, not independent confirmation.

What the instrument does not hold fixed, named in full, and what it pins exactly. Mode S pins the *adversarial* coefficients; it does not pin the benign ones and was never claimed to. So the dose also moves (a) Krum’s pairwise distances, hence the statistic itself; (b) *which* benign client Krum selects, in 0.533/0.800/0.733 of rounds at $\rho=2.72/7.39/54.6$; (c) the **magnitude and geometry of the update actually aggregated**, since the selected client enters scaled by its own coefficient; and (d) the model trajectory, which diverges across rungs in consequence. Channel (c) is not a conjecture but a **number we already measure and print**: the Δ agg. column of Table A12 is $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$, the relative displacement of the aggregate the defense *emits*, and it reads **0.892** on this cell. The supplement § S9 splits that 0.892, as vectors, into 0.986 of *re-selection* (a different benign client is chosen, and in 0/11 of the rounds where that happens it is an adversary) and 0.973 of *rescaling* (the chosen client arrives scaled by its own coefficient); for `cos_krum` the displacement is *entirely* rescaling, since 0/15 selections change at any rung. The claim this arm licenses is therefore sharper than a claim about the statistic alone: **a 0.892 displacement of the emitted aggregate, together with decision changes in up to 80% of rounds, leaves suppression inside the pre-registered practical-equivalence margin, provided the adversarial coefficient share does not move. What is pinned is pinned exactly:** $c_{\text{adv}}=1.0$ at every rung with maximum deviation 0.00e+00 and the share constant to $3.5\text{e}-07$ on CIFAR-10, $3.9\text{e}-07$ on EMNIST-byclass: float32 norm read-back noise, not slack in the instrument (`measure_admission.py`). Naming (a)–(d) is what keeps the claim to the one channel the instrument closes by construction.

Closing channel (c) as well: the pre-registered score-only control. Channel (c) (the magnitude and geometry of what is actually aggregated) is the one confound Mode S leaves open, so we closed it too rather than argue about it. **Score-only Krum** scores on the transformed stack exactly as before and then aggregates the *untransformed* selected update: the dose can change *which* client contributes and nothing whatever about *what* that client contributes. Statistic and decision are disturbed identically to the uncontrolled arm (the selection is computed on the same transformed stack, so Δ dec. is the same 0.733) while Δ agg. is no longer 0.892 but the re-selection component alone, 0.986 relative to $\|u_{s_0}\|$ (the supplement § S9), the rescaling component being closed by construction. This is the reviewer-suggested control in its own terms: hold the selected update’s magnitude and direction fixed while perturbing the selection statistic independently.

Frozen before the arm existed. The decision rule is `experiments/pre_registration_score_only.md`, git-committed at 35788d9 before any score-only ASR existed, and the runner refuses to start unless its own `PREREG_COMMIT` matches that hash. *Primary:* $\Delta = \text{mean ASR}(\kappa=2) - \text{mean ASR}(\kappa=0)$ against the suite’s existing ± 0.15 practical-equivalence margin, on `krum / committed_scaling / CIFAR-10 / cifar_cnn`, the same four rungs and seeds 42–46. $\Delta > +0.15$ would have said that **the published flat result was partly an artifact of the magnitude channel** and that the paper’s central claim narrows to “with benign magnitudes free to move”: that branch is why the arm was worth three hours of compute. The $\kappa=0$ rung is imported rather than re-run, because at the identity the transform returns the stack unwrapped and score-only Krum *is* Krum alone; that is an exact claim, not an approximation, and it was verified bit-identically at one seed before the arm started.

The negative survives with the magnitude channel closed. $\Delta = -0.023$ (mean ASR 0.062 $\rightarrow$ 0.039) against -0.026 for the uncontrolled arm on the identical cell: **closing the magnitude channel moves the effect by 0.003.** The refuting threshold 0.212 is nowhere approached, and mean clean accuracy clears the 0.35 floor at every rung (0.565/0.629/0.624/0.630), so no rung is a collapsed model. **The central claim is**

therefore instrumented against two independent confounds (attenuation, closed by Mode S, and magnitude, closed here) and the sharpened statement is that a 0.986 displacement of the emitted aggregate together with decision changes in 0.733 of rounds leaves suppression inside the pre-registered practical-equivalence margin.

Three disclosures, from the freeze rather than from hindsight. (i) The secondary Jonckheere–Terpstra test was specified to detect a monotone *rise* (the outcome that would have refuted flatness with attenuation already closed) and the observed trend is a *fall* ($J=58.5$, $z=-1.110$, $p_{\uparrow}=0.867$, permutation $p_{\uparrow}=0.874$). We report it and do **not** score it as support for flatness; the pre-registration states in advance that at $n=5$ a null from this test is not evidence of flatness. (ii) The two branches had unequal headroom. With the identity rung at 0.062, refuting us required mean $ASR(\kappa=2) > 0.212$ against a ceiling of 1 and was fully reachable, whereas the indeterminate branch ($\Delta < -0.15$) was *arithmetically unreachable*: ASR cannot fall below 0, so the largest possible fall on this cell is 0.062. (iii) **Selectors only.** A weighted averager and a coordinate-wise order statistic emit no single selected client, so “the selected update” is undefined for them and the control does not cover them, which means the `coord.median` arm carrying the +0.098 rise is **not** magnitude-controlled, and neither is reputation’s +0.178. Score-only Krum is also an instrument and not a defense: it aggregates an update its own scoring stage never saw.

The corollary the pre-registration staked on `cos.krum`, and exactly what it settles. Section 5 of the same document asserted, before the run, that score-only `cos.krum` must be *bit-identical* across rungs: its statistic is exactly invariant under positive rescaling, so the selection cannot move, and with the emitted update untransformed the whole 50-round trajectory must repeat. It does (accuracy 0.528400 and ASR 0.003111 at all four rungs, $|\Delta| = 0.00e+00$ throughout) and the $\kappa=0$ value reproduces the published cell at that seed exactly, so the two pipelines agree at 0.000000 rather than approximately. **What the check does not do is confirm the arm’s published fall.** It runs one seed, as frozen, and that seed is unrepresentative of the arm: under the uncontrolled dose its own ladder *rises* +0.224 where the eight-seed mean *falls* −0.425. Closing the magnitude channel removes all 0.224 of that seed’s movement, which corroborates the *channel* (for `cos.krum` the emitted displacement is entirely rescaling, so no other route is open) but attributing the arm’s mean fall to magnitude rests on that decomposition rather than on this check, and settling it empirically would need the arm’s other seven seeds. Expanding a pre-registered seed count after seeing the data is precisely what pre-registration exists to prevent, so we report the invariance as confirmed and the arm-level attribution as **corroborated mechanically, not established at the arm mean.**

E.2.1 THE 2×2 CHANNEL FACTORIAL, AND THE DECOMPOSITION IT REFUSES TO LICENSE

Mode S opens two channels into a selector: the *statistic* channel (the dose moves Krum’s pairwise distances, so *which* client is selected changes) and the *magnitude* channel (the selected client enters scaled by its own coefficient). Score-only closes the second. The fourth cell closes the first instead: **emit-only Krum** scores the *untransformed* stack, pinning the decision to the identity rung’s, and aggregates the *transformed* selected update. The four cells are then a factorial, and if the two channels acted independently the three measured effects would satisfy $\Delta_{\text{full}} = \Delta_{\text{SO}} + \Delta_{\text{EO}}$. **They do not.** Rules were frozen in `experiments/pre_registration_emit_only.md` at b995f1b before any emit-only ASR existed, with the runner refusing to start unless its own `PREREG_COMMIT` matches that hash; the $\kappa=0$ rung is imported rather than re-run, because at the identity the transform returns the stack unwrapped and emit-only *is* Krum alone, an exact claim verified bit-identically before the arm started. All three harness checks the freeze required pass at 0.00e+00 across five published arms and the score-only control, so the refactor that added this cell perturbed no published computation.

Cell	statistic channel	magnitude channel	mean ASR	Δ ASR
$\kappa=0$ identity (imported)	closed	closed	0.062	—
Score-only	open	closed	0.032	−0.030
Emit-only	closed	open	0.107	+0.046
Full Mode S	open	open	0.023	−0.039
additive prediction $\Delta_{\text{SO}} + \Delta_{\text{EO}}$				+0.016
residual against measured Δ_{full} (frozen margin 0.05)				−0.054

The table above is the published $n=5$ design, retained as the pre-registration requires; the verdict it produced has since been withdrawn by our own top-up. Seeds 47–61 were added under rules frozen at 684b31e, and at $n=20$ $\Delta_{\text{EO}} = +0.010$ (95% CI $[-0.028, +0.047]$), so the magnitude channel is *established* inert rather than merely point-inside, while the residual is -0.019 ($[-0.061, +0.023]$), **inside** the additivity margin. **So the $n=5$ reading that the channels interact does not survive**, and what remains is *consistent* with separability and not an establishment of it, the interval reaching -0.061 and so not contained in ± 0.05 . Its half-width, 0.042, in fact came in *narrower* than the 0.057 that top-up’s own pre-registration projected, so what blocks containment here is the off-centre point estimate rather than the width the projection worried about.

Read the $n=5$ paragraphs below as the record of what was published and why it was fragile, not as the current verdict.

The primary rung is forced, and the additive model gets the sign wrong. At $\kappa=2$ mean clean accuracy is 0.333, below the 0.35 floor, so the primary contrast is reported at $\kappa=1$ and **that substitution is forced by the floor rather than chosen after seeing ASR**: Section 5 of the pre-registration named $\kappa=2$ as the rung at risk, and named it for this cell specifically, because emit-only aggregates the *rescaled* update and so degrades training in a way the other three cells do not. Seed-matched at $\kappa=1$, $\Delta_{EO} = +0.046$ sits inside the frozen inert margin of 0.05, so **the magnitude channel is inert on its own**, which is what we predicted. But $\Delta_{SO} + \Delta_{EO} = +0.016$ against a measured $\Delta_{full} = -0.039$: an additivity residual of -0.054 , outside the frozen 0.05 margin, and **the additive model does not merely mis-size the joint effect, it reverses its sign**. So **no per-channel decomposition of the uncontrolled arm is licensed**, and we withdraw any reading of the re-selection/rescaling split as a decomposition of suppression. That split remains sound as the thing it says it is: the supplement’s algebraic decomposition (§ S9) decomposes a *displacement*, and says in its own terms that it makes no claim about suppression.

Three reasons this verdict is fragile, all of them ours to report. (i) It is knife-edge on both margins at once, $|\Delta_{EO}|$ falling *inside* the inert margin by 0.004 and the residual *outside* the additivity margin by 0.004. (ii) Both pre-registered secondaries are null: the emit-only per-seed sign count is 3 up against 2 down, and Jonckheere–Terpstra across the four rungs gives $z = -0.437$, $p_{\uparrow} = 0.669$. (iii) One seed carries the result. Seed 42’s emit-only run collapses to clean accuracy 0.2613, below our floor *at that seed* though the rung mean of 0.487 clears it, and its Δ is $+0.272$ against -0.007 , $+0.007$, $+0.013$, -0.058 ; excluding it would give $\Delta_{EO} = -0.011$ and residual -0.003 , reading as separable. **We do not exclude it**: Section 5 fixes $n=5$ on seeds 42–46 and applies the floor to a rung’s *mean* rather than per seed, so no per-seed exclusion is licensed, and the pre-registered top-up settled it without one. **At $n=20$ the knife edge is gone and the verdict it produced is withdrawn**: $\Delta_{EO} = +0.010$, 95% CI $[-0.028, +0.047]$, so the magnitude channel is *established* inert, and the residual is -0.019 , $[-0.061, +0.023]$, inside the additivity margin, so the interaction reading does not survive. Seed 42 remains the only draw of 20 past either outlier threshold and is still excluded from nothing; realized sd falls $0.130 \rightarrow 0.080$. Separability is now *consistent with* and **still not established**, its interval reaching -0.061 and so not contained in ± 0.05 even though its half-width, 0.042, came in *narrower* than the 0.057 that top-up’s pre-registration projected at this variance, and dropping the one seed that produces the unwelcome answer is precisely what pre-registration exists to prevent. Section 6.1 of the same document declared, before the run, that at $n=5$ only a large $|\Delta_{EO}|$ would be decisive, and $|\Delta_{EO}|$ is not large. **We therefore report this arm as not decisive, and act on it only in the direction that withdraws a claim**: low power is a reason to stop asserting a decomposition, never a reason to assert one. Every number above is emitted by `experiments/analyze_emit_only_factorial.py`, which reads only frozen artifacts and asserts all four primary quantities reproduce from the per-seed cells to $< 10^{-9}$ before printing. **Selectors only**: the factorial is defined for `krum` and `cos_krum`, the harness raising for any other d_2 , so it is one selector family, one attack, one dataset, one architecture.

What this settles, and what it does not. The intervention delivers the controlled isolation of C2 the two cross-defense suites could not: within a single defense, with attenuation excluded by construction *and*, in the score-only arm, the magnitude of the aggregated update held exactly fixed as well, **substantial disturbance of the statistic d_2 reads need not produce a corresponding change in suppression once the adversarial coefficient share is held fixed** (on the adjudicating Krum arm under Mode S and, by pre-registered replication, both on a downstream defense in a different invariance class and on a second dataset and architecture) while *admission* of adversarial input is the channel the evidence points to. The statistic-level reading the framework was first written in is refuted; the admission-level reading is confirmed as a *direction* in these interventions rather than established in general. It does *not* rescue the framework’s positive direction wholesale: the pooled two-channel model over all 26 cells is **not** confirmed ($R^2=0.239$), and the replication refutes its cross-arm ordering outright, so admission is a better single predictor than the statistic without being a quantitative one. It addresses neither recall nor discovery (FG→RFA remains a false negative), and nothing in §E.1 is rescored: its pooled refutation, its two refuted arms and its indeterminate arm stand exactly as reported.

Mode M: coordinate masking, a second transformation class. Every other arm in this paper transforms client updates by a positive per-client scalar, the family Prop. 10 classifies and Thm. 8 bounds. **Mode M** is outside it. Benign client i has a random m fraction of the coordinates of each parameter tensor zeroed, drawn from an independent Bernoulli($1-m$) seeded by `default_rng([seed, round, slot])` so it is reproducible and uncorrelated with adversary status, and the surviving coordinates are then **renormalized to the update’s original L_2 norm**. Renormalization is load-bearing and was required in the pre-registration *before any ASR existed*: an unnormalized mask shrinks benign norms and re-opens the magnitude channel that §E.2’s control exists to close. The unnormalized variant was considered and rejected for that reason, and the rejection is recorded rather than omitted. **Adversarial updates are returned unmodified, exactly**: the maximum deviation of every adversarial coefficient from 1 is 0.00 with object identity preserved, a strictly stronger pinning than Mode S’s 3.5×10^{-7} float32 read-back. Rungs are $m \in \{0, 0.2, 0.5, 0.8\}$ and $m=0$ returns the

Table A13: **The same aggregators under two transformation classes, every prediction frozen before its measurement.** Prop. 10 classifies aggregators under *positive per-client rescaling* only; coordinate masking is outside that family, so Thm. 8 and Prop. 10 make **no** prediction there, which is what makes the second class a test rather than a second demonstration. Rescaling predictions were frozen at 5130cec, masking predictions at 684b31e, each before the corresponding run existed; `experiments/build_prediction_matrix.py` refuses to emit if any observed cell disagrees with its frozen document. **cos.krum is the adjudicating row:** exactly invariant under rescaling, its selection bit-identical at every rung, it loses the guarantee under masking on the same attack and the same seeds, which is what it means for the classification to be a statement about a transformation *class* and not about an aggregator. Masking ASR is measured for `krum` only (§ E.2); the other three masking cells are channel results and are marked ---.

aggregator	Prop. 10 class	Δ dec., rescaling		Δ dec., masking	
		predicted	obs.	predicted	obs.
<code>cos.krum</code>	(a) exactly invariant	unchanged	0.000	lapses	0.467
<code>krum</code>	(c) not invariant	changes	0.733	changes	0.467
<code>coord.median</code>	(b) conditionally inv.	partial	0.482	partial	0.700
<code>reputation</code>	(c) not invariant	changes	1.000	changes	0.533

update list unwrapped, so the identity rung is bit-identical to Krum standalone by construction; we check rather than assert this, the identity rung agreeing to 10^{-9} across all three directories that carry those seeds.

The channels were measured before any ASR existed, as the EMNIST-byclass arm’s were. Krum’s decision changes in 0.000/0.133/0.400/0.467 of rounds across the ladder while the adversarial mass it admits is 0.000 at every rung, at an aggregate displacement of 1.095 against Mode S’s 0.892 at a *larger* decision change. So the arm was established prospectively as a generalization test, and the pre-registration (684b31e) then split four ways, including two that would have refuted us: a monotone rise in m would confirm the statistic reading, and masking that made Krum admit the adversary would void the equivalence claim at that rung and confirm the *admission* reading. **Neither fired.** At $n=10$ (seeds 42–51) the paired contrast $m=0 \rightarrow m=0.8$ is $\Delta = -0.016$, 95% CI $[-0.051, +0.019]$, contained in the frozen ± 0.15 margin at a half-width 23.1% of it (TOST $p=5.3 \times 10^{-6}$), every rung mean clearing the 0.5 ceiling and the 0.35 accuracy floor, and Jonckheere–Terpstra finding no increasing trend ($p_{\uparrow}=0.895$). The mask analogue of the score-only control gives -0.018 , $[-0.047, +0.010]$, agreeing with the full ladder to 0.002: since adversarial updates are unmasked in *both* conditions the two ladders differ only when a benign client is selected, so their agreement localizes the absent effect to *which client Krum picks* rather than to the content it emits.

What this arm does and does not license. Only the flat-with-admission-unchanged outcome generalizes the negative, and that is the outcome observed; had admission moved, the arm would have been reported as a confirmation of the admission reading instead. **One mask is one transformation, not the class of non-rescalings:** rotations, projections, quantization and sign compression are untested, so the supportable claim is that the negative replicates on a second, structurally different transformation class, never that it holds for arbitrary upstream processing. Mode M has nothing to impose in 24/120 measured rung-rounds, identical to Mode S’s figure, and like Mode S it is an *instrument* and not a defense, since it reads adversary identity. The `cos.krum` mask ladder was considered and dropped for compute before any ASR existed, which is why the masking column carries one outcome and three channel-only entries.

E.3 SIX CELLS, AND A PRE-REGISTERED MECHANISM THAT DID NOT SURVIVE THEM

Table A3 contrasts the two designs on the one arm that motivated the instrument. That contrast is one cell, and a single cell cannot say whether the sign reversal is a property of the evaluation design or of `coord.median` under a pixel backdoor. We therefore ran the contrast on six cells: four that already existed in frozen artifacts and are **re-scored, never re-run**, and two that did not exist and were run for this purpose (60 runs, plus 8 under Amendment 2 below).

Both designs are the same computation. The two ladders differ in exactly two tokens, the family name and whether `generic.compose` receives an `adv_mask`; data partition, participant draw, attack, local training and seeds are identical. `run_comparability_cells.py` refuses to start unless it first reproduces a published `results/dose_response/cell` bit-identically ($|\text{diff}| = 0.000\text{e}+00$), so the new cells are the same experiment as the four they are compared against, and `analyze_comparability.py` refuses to print any pooled number unless all four published contrasts reproduce.

What was frozen, and when. Before any new run, we pre-registered a mechanism (`experiments/pre-registration.comparability.md`, frozen at c986ef4; two amendments at 1026a96 and f16083b; the three hashes are kept because every other pre-registration in this paper carries its own, and

Table A14: **The confounded ladder against the Mode-S instrument, on six cells.** Δ ASR from $\kappa=0$ to $\kappa=2$, paired 95% t intervals, each design at its own full n . **AGREE** := intervals overlap and point estimates share a sign, or both lie within ± 0.05 of zero; **DISAGREE** := intervals disjoint or signs opposite; **SIGN REVERSAL** := signs differ *and* both intervals exclude zero. The four unmarked cells are the training data the rule was read off. **The rule predicted AGREE for the EMNIST cell and the designs disagree, so it is refuted**; per the frozen clause it is withdrawn, the four-cell pattern is demoted to a description, and the table is reported in full. Every cell emitted by `analyze_comparability.py`.

cell	outcome-gated	Mode S	$\Delta\Lambda_a$	pred.	observed
krum / scaling	-0.008 [-0.089, +0.074]	-0.010 [-0.032, +0.012]	0.0000	AGREE	AGREE
cos.krum / pixel	-0.405 [-0.796, -0.014]	-0.425 [-0.732, -0.118]	0.0000	AGREE	AGREE
reputation / scaling	+0.762 [+0.658, +0.866]	+0.178 [-0.112, +0.469]	0.0196	DISAGREE	DISAGREE
coord.median / pixel	-0.272 [-0.422, -0.122]	+0.098 [+0.018, +0.178]	0.0332	DISAGREE	SIGN REVERSAL
coord.median / scaling*	+0.002 [-0.078, +0.081]	+0.196 [+0.112, +0.280]	0.0086	DISAGREE	DISAGREE
krum / scaling, EMNIST*	+0.157 [-0.175, +0.489]	-0.013 [-0.034, +0.009]	0.0000	AGREE	DISAGREE

* out-of-sample; the other four are the cells the rule was read off.

dropping them for brevity would make this the one prereg citation a reader cannot check): the two designs disagree exactly where the downstream defense admits adversarial mass, gated on the admission-level change $\Delta\Lambda_a$ at the top rung. The four existing cells separate perfectly under that rule, which is precisely why **they are training data and cannot also be evidence for it**. The rule mechanically generated a prediction for each new cell, committed in advance: DISAGREE for `coord.median/scaling` ($\Delta\Lambda_a=0.0086$, disclosed at the time as the weakest margin of any cell), AGREE for `krum/EMNIST-byclass` ($\Delta\Lambda_a=0.0000$). The document names in advance what would refute it and commits to reporting refutation rather than fitting a threshold.

The out-of-sample result: one confirms, one refutes (Table A14). `coord.median` under *model scaling* disagrees, as predicted, on the weakest margin in the set: the outcome-gated ladder reads +0.002 [-0.078, +0.081] where the instrument reads +0.196 [+0.112, +0.280]. This is the cell that matters for the headline, because it is the paper’s sign-reversal aggregator under the *other* committed attack, and it says **the disagreement is not specific to the pixel backdoor**. It is a disagreement, not a second sign reversal, and we do not call it one. The EMNIST-byclass cell then **refutes the rule**: $\Delta\Lambda_a$ is exactly 0.0000, the rule predicts the designs coincide, and they do not (+0.157 against -0.013, opposite signs).

We report the refutation as a refutation. $\Delta\Lambda_a$ does not explain when the two designs disagree. The four-cell separation that suggested it stands as a description of four cells and nothing more, and no threshold was fitted after the fact to rescue it. **The observation survives the mechanism’s death**: across six cells the two designs disagree on four, on two different attacks, two datasets and two architectures, and one of those disagreements is a sign reversal. What we do not have is an account of *when* to expect it, which is the honest state of this result and a direct limit on how far the protocol of §4 can be turned into a rule of thumb.

The short form first, because it is all the argument needs: we pre-registered a mechanism before the runs that tested it, one out-of-sample cell refuted it, and it is therefore withdrawn. **One process failure is disclosed below that, because it nearly reversed this section** and because a reader entitled to the first sentence is entitled to how nearly it went the other way. The EMNIST cell was first scored on non-identical seed sets ($n=5$ against a frozen $n=3$), the disagreement carried entirely by two seeds with no counterpart in the other design. We amended the pre-registration to complete the controlled ladder to the frozen seed set rather than resolve the cell on the three shared seeds, which give AGREE: choosing them would have chosen the answer. The completed comparison is the fully paired $n=5$ in the table and it still disagrees. That same amendment asserted that cross-design pairing was the convention the freeze had used. **It was not**, and what caught the error was not our judgement but the analyzer’s own gate: under cross-design pairing the four *published* cells stop reproducing (`krum/scaling`’s controlled ladder moves -0.010 at $n=20$ to -0.026 at $n=5$), and the script refused to print. A third amendment withdraws the claim, and the full sequence is left in the pre-registration.

F THE COMPOSITION THE SCREEN CANNOT CERTIFY: FOOLSGOLD→RFA

This section is the empirical witness for Proposition 7. FoolsGold→RFA has the *lowest observed committed* ASR of every composition evaluated here (0.093), and **our own criterion classifies it as a false negative**: neither constituent suppresses the attack alone (FoolsGold 0.732, RFA 0.845), so its suppression is *emergent* in the sense of Proposition 7 and C1 cannot hold. **We report it in detail because it is the sharpest illustration of the screen’s scope, not because it is a flagship**: any criterion that certifies inherited suppression must miss compositions of this kind, which is what caps recall *structurally*; with one emergent case among five low-ASR pairs that cap is 80%, and the 40% we measure sits below it for reasons other than emergence. The characterization below is therefore descriptive (what this composition does, measured at $n=30$) and is not evidence for the criterion. **The $n=30$ measurement corrected our own earlier figure downward in**

reliability, and we report the correction rather than the settled number alone: the per-seed distribution is right-skewed (median 0.060, mean 0.093, sample skew +2.07, 8/30 seeds at or above 0.10, maximum 0.380), and seeds 42–46 fall in its left tail, so the $n=5$ value we first led with understated the pair by a factor of 1.8. Any 5-seed figure for this composition should be read against that spread. We avoid calling it security-maximal: it also sustains the *highest* ASR under our adaptive adversary (0.167, versus 0.045 for the same pixel attack on the same five seeds and 0.064 at $n=30$), and it carries a 50% accuracy cost, so it is the right choice only where utility is nearly irrelevant. We validate at $n=30$ seeds with a full adaptive attack suite ($N=10$, $K=5$, $f=0.2$, 50 rounds), in Table A15:

Table A15: FG→RFA under base and adaptive attacks. Base: $n=30$, reported as median [IQR] with P95 tail; adaptive: $n=5$, mean±std. Clean accuracy: test-set accuracy on non-triggered inputs. No-attack baseline for RFA alone: 0.737.

Attack	Config	ASR	P95 / max	Acc	Δ_{mix}
<i>Base composition ($n=30$; median [IQR])</i>					
	scaling	0.060 [0.048, 0.096]	0.253 / 0.380	0.497	−0.778
	pixel	0.052 [0.035, 0.087]	0.127 / 0.200	0.492	−0.683
	oracle	0.056 [0.044, 0.090]	0.189 / 0.236	0.494	−0.839
<i>Projection adaptive ($n=5$)</i>					
	$\epsilon=2.0$ (best)	0.079 ± 0.042	− / 0.160	0.495	
<i>Criterion-aware joint-constraint ($n=5$)</i>					
	$\epsilon=1.0$, decorr	0.167 ± 0.069	− / 0.268	0.562	
	$\epsilon=2.0$, decorr	0.130 ± 0.030	− / 0.185	0.506	
	$\epsilon=2.0$, no decorr	0.050 ± 0.008	− / 0.059	0.494	
<i>Neurotoxin ($n=5$)</i>					
	top-20%, scale = 2	0.048 ± 0.009	− / 0.057	0.495	
<i>Consensus-shift ($n=5$)</i>					
	rate = 0.05	0.045 ± 0.012	− / 0.059	0.495	

Results. Median ASR across all base attacks ($n=30$): 0.056 [IQR 0.044–0.090]; P95 worst-case: 0.253 (committed_scaling). **Three of the six adaptive strategies beat the base attacks and we lead with the strongest, not the median.** All six are measured at $n=5$ on seeds 42–46, where the base composition reads 0.045 (pixel), 0.051 (scaling) and 0.050 (oracle), so that is the seed-matched comparison. Against it, the criterion-aware joint-constraint adversary with decorrelation reaches mean 0.167 at $\epsilon=1.0$ (median 0.135, max 0.268), which is $3.7\times$ the 0.045 of the pixel arm it is built from and the ratio every other section of this paper quotes, and 0.130 at $\epsilon=2.0$, and the projection adversary reaches 0.079; the other three sit level with the base (the same criterion-aware attack without decorrelation 0.050, Neurotoxin 0.048, consensus-shift 0.045). **So the composition does not stay under 10% against the strongest adversary we built,** and 0.167 is the figure the opening paragraph reports. What survives is weaker and still worth stating: no adaptive *mean* reaches the base composition’s own P95 tail of 0.253 (its worst single seed is 0.380), though the $\epsilon=1.0$ maximum of 0.268 does, and the $\epsilon=1.0$ gain costs the adversary nothing in clean accuracy (0.562 against 0.495), so it is a real gain rather than a utility trade. The accuracy tradeoff (0.50 vs. 0.74 for RFA alone) and Pareto alternatives are discussed below (Table A16).

Composition-aware adversary (end-to-end). We build an adversary that *knows* the full FG→RFA pipeline is deployed (Kerckhoffs) and optimizes against the complete composition, solving both constraints simultaneously: pairwise cosine similarity below FoolsGold’s down-weighting threshold, and ℓ_2 distance to the geometric median below RFA’s rejection radius. Any gradient-based optimizer faces the same tension (coordinating a backdoor raises FG-similarity, while staying near the median caps the perturbation and weakens the backdoor) and our analytical construction reaches the same boundary, the Weiszfeld iteration being contractive and the gradient at convergence determined by the final-iterate reweighting the joint constraint directly constrains.¹ The ablation is informative: without decorrelation ASR stays at baseline (0.050), RFA-evasion alone failing because FoolsGold is the binding constraint; with it the adversary gains partial signal (0.130–0.167), which Theorem 8 explains as bounded, each adversary’s block-restricted delta being too small to move the median outside the benign cluster.

Contrast with rep→tm. FG→RFA (median ASR 0.056) provides near-absolute suppression; rep→tm (oracle ASR 0.316) provides strategic dominance against all tested adversary policies but does not suppress committed_pixel (0.679). The two serve different roles.

¹A fully gradient-based optimizer differentiating through 50 Weiszfeld iterations would face identical binding constraints at convergence; the analytical formulation is equivalent without requiring a differentiable relaxation.

Accuracy–security tradeoff. FG→RFA achieves the lowest ASR of any tested composition but imposes a substantial clean-accuracy cost (0.50 vs. 0.74 for RFA alone). This cost is intrinsic to FoolsGold’s aggressive per-client downweighting, not to composition itself. Table A16 places FG→RFA on the accuracy–security Pareto frontier alongside alternative compositions:

Table A16: Accuracy–security Pareto frontier. Max ASR = worst-case mean across committed attacks, and Clean Acc is measured on the arm that attains it. All at $N=10$, $K=5$, $f=0.2$, cifar_cnn. The three composition rows are emitted by `experiments/analyze_headline_cis.py`: FG→RFA at $n=30$ (seeds 42–71), FG→CM and Rep→CM at $n=5$ (seeds 42–46). No *composition* dominates another: FG→CM buys 0.185 accuracy for +0.014 ASR. The single-defense rows are listed for reference and are not all frontier points; Rep alone dominates RFA alone on both axes.

Defense / Composition	Clean Acc	Max ASR	Regime
Temporal mix (NC/fedavg)	0.79	0.96	No security
RFA alone	0.74	0.91	Bimodal; unreliable
Rep alone	0.78	0.84	Scaling-only
Rep→CM	0.76	0.42	Moderate
FG→CM	0.68	0.11	Strong
FG→RFA	0.50	0.09	Min ASR (evaluated)

FG→RFA is the correct choice under a *security-first* objective (e.g., safety-critical FL where any backdoor is unacceptable). For operators with accuracy constraints, FG→CM (acc 0.68, max ASR 0.107) buys 37% more accuracy for suppression that is statistically indistinguishable on the worst-case arm the two share (Welch $p = 0.52$ on committed scaling, 0.093 against 0.107). It is *not* Pareto-superior and we do not claim it is: its point estimate is the higher of the two, and on the pixel arm, which is neither pair’s worst case, the gap does reach significance in FG→RFA’s favour (0.064 against 0.099, Welch $p = 0.022$). Rep→CM (acc 0.76, max ASR 0.423) offers near-baseline accuracy with moderate security. The criterion certifies FG→CM and Rep→CM; FG→RFA reaches the lowest ASR of the three but fails the tightened per-attack C1 and is therefore a false negative of our own criterion rather than a certified recommendation. The choice among them is an operator-specific utility decision, not a criterion failure.

G RELATED WORK

The contrast in one sentence. Prior work asks whether a defense withstands an attack, and the composition literature treats composition quality as an *outcome to predict*. We ask a different question: the **identifiability of mechanistic explanations** for that outcome, under the evaluation designs commonly used to compare defenses. Concretely, whether an evaluation criterion can causally attribute a composition’s security to preservation of the downstream mechanism.

Byzantine-robust aggregation, studied in isolation. Krum (Blanchard et al., 2017), Trimmed Mean (Yin et al., 2018), RFA (Pillutla et al., 2022), FoolsGold (Fung et al., 2020), FLTrust (Cao et al., 2021) and FLAME (Nguyen et al., 2022) are each proposed and evaluated as a *single* stage. We propose no new defense; we reason about *ordered pairs* of existing ones, which is a different object: the upstream stage’s output is the downstream stage’s input, so the question is no longer whether a mechanism works but whether it *survives* a transform applied before it. Ensemble adversarial training (Tramèr et al., 2018) *assumes* that upstream processing preserves downstream signals; we make that assumption explicit, checkable, and (as it turns out) **the wrong abstraction for security**.

Adaptive attacks are why an inherited-robustness guarantee is not a security guarantee. Optimized model poisoning defeats Krum, trimmed mean and FoolsGold once the attack is chosen *against* the defense (Shewalkar & Houmansadr, 2021; Bhagoji et al., 2019), and the same holds for distributed and durable backdoors (Xie et al., 2020; Wang et al., 2020; Zhang et al., 2022). Our C1 quantifies over a *committed* attack by construction, so this literature bounds what any inherited-suppression criterion (ours included) can promise. Our single whitebox composition-aware stress test (§F) is reported in order to *measure* that gap (0.045 → 0.167 seed-matched on the pixel arm, $3.7\times$), not to defend the criterion against it; a screening condition stated against an adaptive adversary is what the framework most lacks.

Evaluation methodology, which is where this paper sits. The adjacent precedent is adversarial robustness, where reported numbers proved unreliable until evaluations were required to be adaptive (Croce et al., 2021). Our point is methodological in the same way but about **identification** rather than attack strength. Wherever a screen gates candidate pairs on standalone effectiveness and then reads the composition’s outcome as evidence about the *mechanism*, Proposition 6 shows that inference cannot work at *any* sample size: the counterfactual is excluded by the gate, not undersampled. **We do not claim this is how the FL literature generally proceeds**, and we can name one external instance rather than a practice: Fenaux et al. (2025) pair a defense effective

against large-deviation updates with one effective against small-deviation updates, and validate the pairing by the combined defense’s attack success rate, which selects components by standalone effectiveness and then reads the composition’s outcome. Composition itself is named as an open problem in the field’s survey of them (Kairouz et al., 2021). The claim we do make is structural rather than sociological, and its clearest instance is *our own* screen: Lemma 1 shows $C0 \wedge C1$ *force* the downstream defense to suppress the attack alone, so the gate is constitutive of an inherited-suppression screen and not an incidental filter on it. Statements of a composition-preservation intuition in that literature are informal; what is new here is the *level hierarchy* at which the intuition fails (Definition 2) and the class of designs that can test it (§C.1, §E.2). **So what makes this a machine-learning contribution and not an application of standard causal-identification theory is a combination of four things, and the three-level form of the argument is in §3 rather than repeated here:** a structural identification boundary for a shape of evaluation, the preservation hierarchy that produces the failure, an interventional instantiation in FL that measures it, and the empirical non-implications $(P4) \not\Rightarrow (P3)$ and $(P4) \not\Rightarrow (P5)$ that the collider argument alone does not predict.

Attacks and defenses. Backdoor attacks (Chen et al., 2017; Bagdasaryan et al., 2020; Wang et al., 2020; Xie et al., 2020) and robust defenses (Pillutla et al., 2022; Sun et al., 2019; Blanchard et al., 2017) form the landscape. Beyond our norm-based menu, certified robustness (Xie et al., 2021) and filtering defenses (Nguyen et al., 2022; Fung et al., 2020) build stronger single mechanisms; we compose fixed defenses deterministically rather than strengthening a single mechanism, so these are complements to our object rather than baselines for it. Randomized transformations (Xie et al., 2018; Liu et al., 2018) in centralized settings are the conceptual neighbor of a *randomized-menu* question, which this paper does not ask: every claim here concerns deterministic joint application (§C). **An earlier version of this paper compared against a randomized menu and reported that deterministic composition dominates it; that comparison is not part of this submission and no claim here rests on it.**

Backdoor persistence and durability. The persistence mechanism we formalize has empirical precursors. Bagdasaryan et al. (2020) design model-scaling explicitly to embed triggers that survive gradient updates; Zhang et al. (2022) (Neurotoxin) engineer triggers to occupy “less-used” gradient directions, achieving multi-round persistence even under strong norm clipping. These works establish the phenomenon, and this paper *uses* it rather than modelling it: persistence is why a committed attack stays committed across rounds, which is what makes a single committed attack against a fixed defense pair the right setting for an identification question. **We make no formal claim about persistence.** An earlier version of this paper advanced a $(\gamma, \alpha, p_{\text{eff}})$ model of when persistence eliminates the value of defense switching, together with a heuristic boundary as $\gamma \rightarrow 1$; that material is not part of this submission, and we withdraw the claim rather than leave it asserted without the evidence that supported it. What remains is γ as a measured descriptive quantity (Appendix K) and the DBA persistence replication of §M.3, which returns an *intermediate* verdict and establishes no boundary.

Defense composition and ensembling. Composing defenses is well-studied in adversarial ML for centralized models: adversarial training combined with input purification (Crocce et al., 2021), detection-then-rejection pipelines, and ensemble adversarial training (Tramèr et al., 2018). These works implicitly require that upstream processing does not destroy downstream signals; our signal-preservation criterion formalizes this requirement explicitly with measurable margins. In FL the closest *structural* precedent is bucketing (Karimireddy et al., 2022), which restores existing robust aggregators under heterogeneous data by inserting an upstream reshuffling step ahead of them: this is the object we study, an upstream transform feeding a fixed downstream aggregator. Its target is convergence under heterogeneity rather than suppression of a committed attack, so it does not meet Proposition 6; what it shows is that the contrast our protocol demands is the natural one for this object, since establishing that an upstream stage helps at all requires comparing the same aggregator with and without it. In Byzantine-tolerant distributed systems, protocol composition (e.g., layered BFT) has a long history; the key distinction is that Byzantine agreement composes for fault tolerance among replicas, while our setting composes for poison resilience across client updates where the adversarial signal is a continuous gradient, not a binary fault. **A predictive criterion for FL composition does exist in recent work, and we withdraw the priority claim we previously made here.** Fenaux et al. (2025) categorize defenses by the deviation of adversarial updates from the mean of the updates and predict that pairing a large-deviation defense with a small-deviation one closes the gap either leaves alone. Our object is prior to theirs and different in kind: not which pairings succeed, but whether a comparison across defenses can identify *why* one did. Proposition 6 therefore applies to their criterion and to ours alike, and it constrains what validating evidence for such a criterion can establish rather than whether the criterion is correct.

H LIMITATIONS, IN FULL

(L1) Limited scope and architecture-conditional boundaries. Three attack families (model_scaling, pixel, DBA), two architectures (CifarCNN, ResNet18, plus `simple_cnn` on the replicated arm), and one dataset (CIFAR-10) for every composition suite but one. The collapse boundary is architecture-conditional (ResNet18 suppression at NC10/rep90: 0.865 vs. CifarCNN 0.097, a $10\times$ shift; per-seed values in §N.2), so results at other N , K or datasets may differ. **That gap is also evidence for this paper’s own thesis:** a swing of that size

under a change of architecture alone is why we state the preservation conditions as mechanisms to be measured per setting rather than as architecture-independent empirical laws.

Cross-distribution transfer is validated across architecture, scale ($N \in \{10, 100\}$) and adversarial fraction ($f \in [0.05, 0.6]$; §D.6(D)), and *dataset* transfer is now tested for the flagship Mode-S arm alone, EMNIST-byclass with `simple_cnn`, where the negative replicates at $\Delta = -0.017$ (§E.2).

For the screen itself the obstacle turned out to be sharper than an untested transfer: on EMNIST-byclass C1's precondition has no input on the committed pixel arm. C1 requires that some constituent suppress each committed attack *alone*, and under the identical composition protocol the pixel backdoor is essentially unsuppressed there: re-measured here, `coord_median` reaches 0.609 at clean accuracy 0.836 (seeds 0.717/0.493/0.618), and over the seven single-defense pixel cells measured there that 0.609 is the *minimum*, with `krum` next at 0.660 and the other five between 0.988 and 0.998 (six *distinct* aggregators, since `multi_krum` reduces to `fedavg` at $K=5$). **So no pair drawn from them can satisfy C1 there:** strategy B certifies nothing over them, every such pair is predicted HIGH, and the strategy comparison of §6.3 cannot be made on that dataset from these inputs. **The coverage is exactly those cells, and we state the gap rather than generalize past it:** `foolsgold`, `reputation` and the cosine mirrors have no EMNIST-byclass pixel cell, so what is established is C1's failure over the measured menu and *not* its unsatisfiability over the whole one. **The failure is the pixel arm's, and we do not escape it by dropping the other.** On the model-scaling arm `krum` does suppress there (0.027 at clean accuracy 0.767), so a scaling-only protocol would supply a C1 input and manufacture the comparison. We decline it for the reason §D.1 declines the mirror move: the committed protocol requires *each* attack, and choosing the arm that happens to have an input after seeing which does is attack-shopping regardless of the resulting labels. **On any of these readings it is a scope limit on the criterion's precondition and not a failed replication:** no prediction was made and refuted, and the question a screen answers does not arise where nothing suppresses the attack under test. We report it rather than lower f , subset the classes or lengthen training until a positive appeared, each of which is a new design axis chosen after seeing the obstacle. The `coord_median` figure is *bit-identical* to the corresponding cell of `results/femnist/payoff_results.json` (0.6091787658... at 0.8363207047...), which by the standard we apply to ourselves in §E.2 (the same computation run twice cannot corroborate itself) establishes protocol *identity* and not corroboration; what it does buy is that the matrix's other cells are protocol-comparable, so we did not spend 21 h re-measuring them (`experiments/measure_femnist_c1_inputs.py`, `results/femnist_c1_inputs/summary.json`). **We then screened the scaling arm for a further comparability cell and report the screen rather than a cell** (supplement § S13): of the five defenses with no scaling cell there, `fltrust` is excluded by argument (its admission is bit-identical under this rescaling) and we measured the other four, of which `foolsgold` and `cos_reputation` are the only two that would add a downstream defense and both fail the *accuracy* floor, $10\times$ rescaling collapsing the model to 0.046 and 0.100 clean accuracy, so **no defense outside Table A14's four is admissible on that axis.** `reputation` does meet the precondition (0.3465 at 0.704), but on seeds 0.9947/0.0152/0.0296, and a replication resting on a spread that wide is one we decline to run.

Dataset transfer therefore remains untested for every other composition suite: the preservation conditions are dataset-agnostic in mechanism (cosine similarity, geometric distance) while the downstream decision boundary may shift with output dimensionality. *The 2×2 channel factorial is narrower still:* it is defined only for the two selectors whose decision can be transplanted (`krum`, `cos_krum`) and the harness raises for any other d_2 , so it is one attack on one aggregator family, reported at $n=5$ as **not decisive** (§E.2.1) and acted on only where it withdraws a claim. *Two further gaps in the intervention:* neither DBA nor Neurotoxin was run through it, and the score-only control is a single arm, one aggregator, one attack. *And scale is a prediction, not a measurement:* the *pre-registered* PASS predictions hold at $N=100$ /ResNet18 (FG→RFA 0.069) but far from converged accuracy, so they bound nothing about converged training, and **the FAIL controls run at that scale proved uninformative**, both landing below 0.5 (§S4). *What transfers:* the qualitative PASS/FAIL prediction and the mechanism. *What does not:* the exact suppression ASR. A practitioner at $N=10^4$ should use the criterion to select candidates and then pilot at their target scale: it identifies *what to try*, and architecture-specific calibration determines *how well it works*.

(L2) The composability criterion is an evaluation-prioritization rule, not a security screen and not a discovery tool. It has no established predictive validity: it orders candidates for evaluation, and the pooled model that would let it predict is unconfirmed ($R^2=0.239$) with its cross-arm ordering refuted (§E.2). The criterion gives *sufficient* conditions (C1∧C2, with C3 as a margin diagnostic): it reliably identifies mechanism-preserving compositions in the common case where one defense is known-effective, but it cannot certify synergistic compositions where no single defense suppresses the attack alone. The conservative bias is by design: false negatives are operationally cheap to recover from (try the composition anyway) while false positives would deploy an insecure stack. Accuracy is $16/18 = 89\%$ on the development set, $22/24 = 92\%$ on the pre-registered held-out set (90.5% combined 2-way) and $9/10 = 90\%$ on pre-registered 3-way triples (supplement § S5). **None of those tallies should be read as discrimination:** only 5 of the 42 two-way pairs are low-ASR, so a constant HIGH call scores $37/42 = 88.1\%$ and the combined 90.5% beats it by one pair. Precision (2/2) and recall (40%) are the quantities that carry information, and they are why we present the criterion as a prioritizer. All

clean $C1 \wedge C2 \wedge C3$ PASS predictions are correct; the four misses are either the predicted intermediate behavior of rank-based aggregators (Proposition 9: dev-set $NC \rightarrow cm$, rfa $\rightarrow cm$; held-out fedavg $\rightarrow cm$) or one speculative LOW call the criterion does not license (held-out fg $\rightarrow NC$, a C1 failure). Formalization is partial: Theorem 8 proves $C2 \wedge C3$ for the weighting $\rightarrow rank/geometric$ class with explicit margins, the remaining families are empirically validated, and C3’s threshold condition depends on d_2 ’s discrimination boundary, which in practice requires calibration runs.

(L3) Per-round aggregator switching only. We test per-round defense selection. Randomized *internals* (DP noise, subsampling smoothing à la CRFL, random participation masks) modify the embedding mechanism directly and may retain value even under persistence. Three adaptive attacks are tested against composition: projection-based against rep+tm (ASR saturating at 0.639 for $\epsilon \geq 2$, 0.597 at $\epsilon=1$; the same attack against fg $\rightarrow rfa$ reaches only 0.079, Table A15), Neurotoxin-style (Zhang et al., 2022) top- k gradient projection, and a two-phase consensus-shifting attack against rep+tm that biases updates toward the target at rate r for 25 rounds before injecting full backdoor updates (App. M). At shift_rate $\in \{0.01, 0.05\}$: mean ASR 0.684 ± 0.138 ($n=10$) and 0.667 ± 0.137 ($n=8$), comparable to committed.pixel (0.679) and below the temporal mix (0.915). At $f=0.2$ the honest 80% anchor the coordinate-wise median, limiting Phase-1 drift to $\sim r/(1-f)$ per round, so Phase 2 degenerates to standard committed.pixel: a mechanism that weakens monotonically with f , and at $r \leq 0.05$ over $T \geq 200$ rounds cumulative drift may eventually overcome it. The robustness claim is therefore specific to the tested regime ($f=0.2$, $T=50$).

(L4) The falsifiable prospective positive evidence is one cell, and C0 was added retrospectively. This is the sharpest limitation we can state about our own design, and we state it rather than let a reader find it. Until §D.5 there was none at all, and one cell is what changed. Of the prospective suite’s 8 predicted-LOW pairs (§D.1), 6 are $\cdot \rightarrow FLTrust$ near-identities: FLTrust gates on a scale-invariant cosine and renormalizes every update to the server-gradient norm, discarding exactly the magnitude information an upstream positive rescaling modifies (verified bit-identical per seed for pass-through and non-clipping d_1), so Proposition 10(a) permits no other outcome and these confirmations cannot falsify it; the other 2 have FLTrust as d_1 , where suppression is inherited and C0 puts them out of scope. C0 was itself formulated *after* seeing this suite, so its 7 exclusions are retrospective. The three matched contrasts that appeared to probe Proposition 10 most directly (a $19-20 \times$ ASR gap at fixed d_1) are **withdrawn**: two ran against MultiKrum, which at $K=5$ selects all five participants and is bit-identical to FedAvg, and the genuine selector leaves a gap that measures C1 failure rather than disturbed invariance (§S2).

The decisive experiment does not exist, and that is structural. We had named it as a downstream defense with a rescaling-invariant admission signal that is *weak in isolation*; Lemma 1 shows no such pair can be certified in scope. §D.2 runs the achievable substitute, holding the suppression *principle* fixed while varying only the invariance class, and it did not deliver a positive test either, standalone effectiveness being unholdable here because the magnitude information the mirrors discard is what the originals use to suppress a magnitude attack (8/8 frozen labels, TP = 0). So the strongest C2-specific evidence is the *post-hoc* within-defense contrast of §D.2, which holds C1 exactly fixed but was not pre-registered and is resolved on only one of its two cells at $n=3$.

Three pre-registered attempts followed, and each refuted something of ours. §E delivers exactly one falsifiable prospective positive (arm 2, reputation under model-scaling, $0.017 \rightarrow 0.779$ at $p = 1.7 \times 10^{-6}$ with clean accuracy nearly unchanged) while refuting two of its own four predictions and its pooled prediction, leaving the *destruction* half of C2’s mechanism confirmed and the preservation half’s control indeterminate (§E.1). The targeted intervention of §E.2 closes the attenuation channel by construction and isolates C2 within a single defense: substantial disturbance of the statistic does not change suppression *on the adjudicating Krum arm under Mode S*, with admission the channel the evidence points to there, and it *refutes* the preservation-half reading, the exactly-invariant `cos_krum` neither suppressing nor holding ASR fixed. The third was designed to be the falsifiable prospective *positive*: a replication arm chosen because the statistic and admission readings order its rise *oppositely*, frozen in git before the arm ran. It replicated the negative in a different invariance class and **refuted the positive ordering**, the arm with the largest measured admission change rising *less* (+0.098) than one whose admission change was smaller.

The census of §D.5 is the one place this changed, and it changed by exactly one cell. We closed the 3×3 region where Prop. 10(a) settles C2 algebraically, registering its two unspent cells at 96f8856; the single certified pair was registered LOW with *low* confidence and an explicit *uncertain* flag, and measured LOW at 0.263. Unlike every predicted-LOW pair above it is neither a near-identity (0 of 3 seeds identical to d_2 alone, on both arms) nor an inherited suppression, since *reputation* holds only model-scaling and `cos_krum` only the pixel backdoor. **That is a falsifiable prospective positive, and it is one cell at $n=3$ with a per-seed spread of 0.000 to 0.681**: it establishes that the criterion’s precision survives a blind test in the one region our invariance result decides exactly, and nothing quantitative. A validated *positive* predictor therefore remains absent: the pooled two-channel model is unconfirmed over 26 cells ($R^2 = 0.239$) and its cross-arm ordering refuted outright, so admission is *more causally proximal* than the statistic without being a quantitative predictor.

(L5) Theorem 8’s conditions need a pilot, and (1)’s is not operationally checkable as stated. Neither $\rho < r^*$ nor $\Delta_{\text{sep}}^l > 2(R_B + \delta')$ is readable off a (d_1, d_2) definition pair: both need a pilot measurement. **So two kinds of cheapness have to be kept apart, and the paper has previously run them together.** Prop. 10’s classes are *algebraically* checkable: (a) and (c) are decided from the definitions of d_1 and d_2 alone, with no data, no runtime cost and no per-deployment work. Theorem 8’s conditions are closed-form inequalities and are *still* none of those things, because their terms have to be instantiated by measurement on the deployment they are being checked for. **Algebraically checkable does not imply deployment-time cheap**, and only the first of the two is something this paper delivers. Regime A is worse than that. Its ordering-survival condition is $\rho < S$ with S the attack’s coordinate extremeness against the benign scale. Building the dose–response ladder (§E) required us to *measure* S rather than stipulate it, and it is not a stable per-attack constant: on model-scaling it ranges 0.00–12.59 across the 12 measured rounds that sampled an adversary (median 0.57), on the pixel backdoor 0.61–1.63 (median 0.92). The $S=10$ of §C.3 is an illustrative stipulation, not a measurement. Two consequences we state rather than work around. First, a practitioner cannot check $\rho < S$ from a single round: the quantity swings by orders of magnitude round to round, and on both committed attacks its median is below 1, so Regime A’s condition fails even at the identity rung under this estimator. Second, we therefore built *no* prediction in §E on S , resting the ladder on directly measured disturbance of d_2 ’s statistic instead. The margin form of the theorem remains valid as a statement about ρ and S ; what is not established is that S admits a stable estimator at the round granularity a deployment would have to check it at, and a usable Regime A needs an S estimator with characterized variance, which we do not provide. **And where the geometric condition is measurable it stops being informative in the regime that matters most:** the margin it rests on is exactly what heterogeneity closes, RFA’s all-round mean falling from 3.24 to 1.01 at Dirichlet $\alpha=0.1$, the limit at which Corollary 2 degrades to $n_a/(n_b+n_a)$ and certifies nothing (§C.3, §D.6(E)). So under extreme heterogeneity the conditions are not merely awkward to check but empty, which is the sense in which §2 calls the bound vacuous there.

(L6) Measured disturbance is not a transferable predictor, and our dose instrument varies the wrong quantity for three of its four arms. The pre-registered ladder (§E.1) refutes the pooled claim it was built to test ($\rho_s = +0.351$, $p = 0.091$) and refutes two of four per-arm shapes, with ASR *falling* as the dose rises in three arms. Two distinct faults, both ours. First, the abscissa: at matched measured disturbance the real upstream transforms sit up to +0.79 ASR above the synthetic curve, and Krum’s within-margin arm does not reproduce the very contrast (§D.2) we attributed to disturbance of its statistic, so the rate at which a transform flips a defense’s decision is not a sufficient statistic for what the transform does: *which* clients are reweighted matters, and ρ does not measure that. We now measure the quantity that does, Δ_c (the supplement § S10): the transforms carrying the two anchors that miss are targeted far beyond any ladder rung (−1.21, −0.72 against the ladder’s −0.34) at ρ that is 3.7 and 13× smaller. That names the axis of the mismatch without supplying a replacement predictor: Δ_c ’s magnitude does not order suppression either. Second, the instrument: mean-1 normalization fixes the aggregate’s scale but not any client’s relative weight, so with the dose assigned independently of adversary status a large ρ mostly dilutes whichever client carries the poison (measured: the ladder’s Δ_c drifts monotonically negative with the dose, $0.000 \rightarrow -0.140 \rightarrow -0.240 \rightarrow -0.336$), and three arms end up measuring Theorem 8’s attenuation regime rather than C2 disturbance. The adversary-targeted dose that separates the two, recorded as not run before the freeze (caveat (iii), the supplement § S7), is run in §E.2, and in retrospect Mode S is the Δ_c -**pinned** instrument (Δ_c exactly 0 at every rung, ρ up to 54.6) and Mode A the Δ_c -**swept** one (−1.14 to +2.47): with every adversary’s weight pinned, Krum’s statistic is disturbed in up to 80% of rounds while the admission of adversarial input does not move and ASR stays inside the pre-registered practical-equivalence margin, so *which* clients are reweighted, not how often a decision flips, is indeed the operative variable, and the ladder confound is *explained* (reputation’s rise shrinks from +0.762 to +0.178 once the adversarial share is held fixed). That isolates C2’s causal content within a single defense; it does not rescue the pooled positive direction ($R^2 = 0.239$ over 26 cells, cross-arm ordering refuted by the replication).

I EVIDENCE STATUS: WHAT IS PROVED, WHAT IS SUPPORTED, WHAT IS NOT

Proved: the invariance classes of Proposition 10; sufficient conditions for mechanism preservation on the weighting→aggregator class (Theorem 8, Lemma 2, Corollary 2); the identification boundary for outcome-gated screening (Proposition 6, with Lemma 1 its FL instance); the recall bound on inherited-suppression screens (Proposition 7); and, formalizing the interventions, statistic-level non-identifiability (Proposition 11) and the independent controllability of ρ and Δ_c (Proposition 12). **The two boundary results are general** (they are statements about the shape of a screen, not about FL), so what carries beyond this paper is the design they license, not the screen. **Empirically supported:** that substantial disturbance of the downstream statistic need not produce a corresponding change in suppression when the adversarial coefficient share is held fixed, with adversarial admission the channel the evidence favours, *on the adjudicating Krum arm under Mode S* and, by pre-registered replication, both on a downstream defense in a *different* invariance class and on a *second dataset and architecture* (§E.2); that the screen behaves as a conservative prioritizer across 42 pairs; and that in the one region where Prop. 10(a) settles preservation *exactly*, a closed 9-cell census whose two unspent

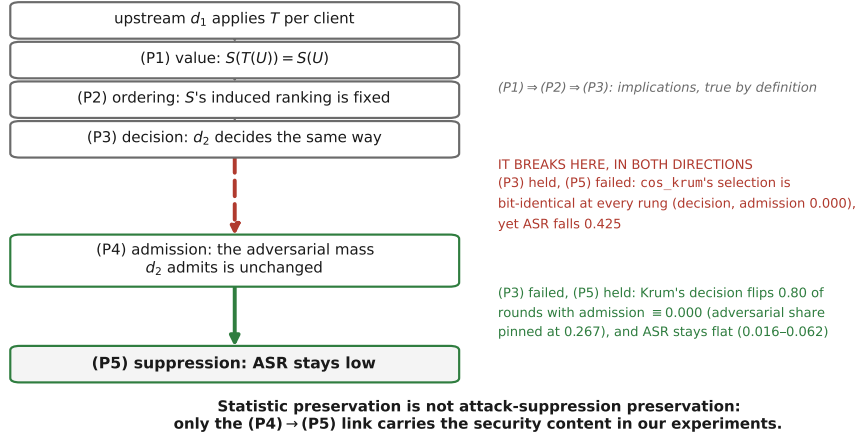


Figure A6: **The preservation chain, and where it breaks.** (P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) holds by definition; **below (P3) the chain breaks, in both directions**, and both witnesses come from the oracle intervention that pins every adversary's coefficient and so closes Lemma 2's attenuation channel by construction (§E.2). The pooled statistic-level prediction we had frozen is separately **rejected** by a pre-registered ladder ($\rho_s = +0.351$, one-sided $p = 0.091$, 16 cells; §E), and a pre-registered replication in a different invariance class reproduces the negative but **refutes the cross-arm admission ordering** ($\Delta = +0.098$; the pooled two-channel model stays unconfirmed, $R^2 = 0.239$ over 26 cells), so **admission is more causally proximal than (P1)–(P3) here, and not a validated predictor**. Every number in the figure and in this caption is recomputed from the scoring scripts' own JSON and printed by `story_chain_figure.py`.

cells were registered blind contains a single low-ASR composition and it is the one the criterion certifies, at 100% precision and 100% recall for a tenth of the runs against statistic preservation's 11% (§D.5, one cell at $n=3$, gated reading). Those two 100%s are contingent on the frozen seed count twice over, and disclosure (v) of §D.5 states the second contingency: the C1 input that certifies the pair reads 0.300 over seeds 42–44 and climbs with the seed count, to 0.490 at $n=5$ (still below the threshold, by 0.010) and to 0.584 at $n=8$, where C1 fails and the criterion certifies nothing in that census. **Not established:** the positive predictive direction of the criterion, since the pooled two-channel model is unconfirmed over 26 cells ($R^2=0.239$) and the replication refutes its cross-arm admission ordering, so admission is a direction and not a quantitative predictor; any ability to discover emergent synergy; and generalization beyond star-topology FL, or beyond CIFAR-10 for anything but the one arm replicated on EMNIST-byclass, where no single defense measured suppresses the committed pixel backdoor alone, so C1's per-attack precondition fails over the menu measured there and the strategy comparison is out of scope there rather than merely untested ((L1), which also states what the seven do *not* cover). **The open problem**, which the sections above refer to by pointing here: every reading of (P4) in this paper comes from an instrument that knows which clients are adversarial, and no deployed defense knows that, so whether the adversarial contribution a downstream mechanism admits can be estimated *without* adversary identity is open. Nothing here bounds it, and the design we license identifies that channel rather than screening on it.

J THE PRESERVATION CHAIN IN ONE FIGURE

The lower arc, and why the last two links must be broken by intervening rather than comparing. T can reach ASR by attenuating whichever client carries the poison (Lem. 2), **without passing through d_2 's statistic at all**. Mode S cuts that arc by construction, pinning every adversary at $c=1$. **The three annotations in panel (a) are three scopes, not one experiment:** Krum's flip rate; the admission support *pooled* over Table 1's four arms, so it names no arm and Krum's own zero is a floor; and `cos_krum/pixel` at its first gate-passing rung (§6.2).

Table A17: **TOST against the frozen ± 0.15 margin at the seed counts already on disk**, on the paired per-seed difference (top rung minus identity). Equivalence requires the 90% interval inside the margin, which at $\alpha=0.05$ is $\max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$. Post-hoc; no runs added. Emitted by `experiments/analyze_tost_existing.py`.

Arm	n	paired Δ	95% CI	90% CI (TOST)	verdict
Krum / scaling (flagship)	5	-0.026	$[-0.078, +0.026]$	$[-0.066, +0.014]$	equivalent
Krum / scaling (score-only)	5	-0.023	$[-0.108, +0.061]$	$[-0.088, +0.042]$	equivalent
Krum / scaling (EMNIST-byclass)	3	-0.017	$[-0.033, -0.001]$	$[-0.028, -0.006]$	equivalent
the same arm, pooled	5	-0.013	$[-0.034, +0.009]$	$[-0.029, +0.004]$	equivalent
Reputation / scaling	5	+0.178	$[-0.112, +0.469]$	$[-0.045, +0.402]$	not est.
<code>cos_krum</code> / pixel	8	-0.425	$[-0.732, -0.118]$	$[-0.671, -0.179]$	not est.
<code>coord_median</code> / pixel	5	+0.098	$[+0.018, +0.178]$	$[+0.036, +0.160]$	not est.

K NOTATION

symbol	meaning
c_i	upstream coefficient applied to client i 's update, $T(u_i) = c_i u_i$, $c_i > 0$
A	the set of adversarial participants in a round
ρ	statistic-level dispersion, $\max_i c_i / \min_i c_i$: <i>how far apart</i> the coefficients are
Δ_c	adversary-correlated dispersion, $\mathbb{E}[c_i i \in A] - \mathbb{E}[c_i i \notin A]$: <i>who</i> receives them
r	adversary-to-benign weight ratio, $w_a / \max_b w_b$; the Mode-A dial, $r = e^\nu$
S	the attack's coordinate extremeness (Theorem 8(1))
m	RFA separation margin, $\Delta'_{\text{sep}}/2(R'_B + \delta')$
Λ_a	aggregate adversarial influence: adversarial share of the Weiszfeld mass
κ	the Mode-S / ladder dispersion dial, $\rho = e^{2\kappa}$
γ	attack <i>persistence</i> , $\text{ASR}(t+1)/\text{ASR}(t)$ after a clean round (≈ 0.66); never a weight

L EQUIVALENCE TESTING AT THE EXISTING SEED COUNTS: WHERE THE FROZEN MARGIN PASSES AND WHERE IT ONLY APPEARS TO

Why this is here and what it is not. The pre-registered rule scores a *point* estimate against a margin: $|\text{mean ASR}(\kappa=2) - \text{mean ASR}(\kappa=0)| < 0.15$ and every rung < 0.5 . A point estimate inside a margin is consistent with a negligible effect *and* with an effect the sample cannot locate, and those are different claims. Table A17 scores the same frozen margin as a two-one-sided- t (TOST) equivalence test on the *paired* per-seed differences, declaring equivalence only when the 90% interval lies inside $(-0.15, +0.15)$. It is **post-hoc**, reads only frozen artifacts, **adds no runs and revises no published verdict**; the emitter is `experiments/analyze_tost_existing.py`. Pairing is not a modelling choice: every rung of an arm runs at the same seeds, so each difference holds one data partition fixed and the seed-to-seed spread that dominates this suite drops out of the contrast.

What it shows, including against us. The **flagship passes**, and passes at $n=5$: $p=0.0014$, the pairing holding sd to 0.042, so *the equivalence claim on the adjudicating arm does not depend on a larger seed count*. So do the score-only control ($p=0.0071$) and the EMNIST-byclass replication ($p=0.0004$, $n=3$, $\text{sd}=0.007$, and $p < 0.0001$ at the pooled $n=5$ of §E.2, where the interval widens across zero but stays well inside the margin). The two arms whose suppression genuinely moves correctly **fail** it, which is the positive control working: reputation (+0.178) and `cos_krum` (-0.425). **And one arm passes the frozen rule while failing the test:** `coord_median`'s +0.098 sits inside ± 0.15 , but its 90% interval reaches +0.160 and TOST gives $p=0.073$. We therefore report that arm's equivalence as **not established** rather than as inherited from the point estimate. **This is the one place the paper's frozen scoring rule was more permissive than the evidence warrants, and we say so rather than leave the width implicit.** `cos_krum` additionally sits above the 0.5 ceiling the frozen rule conjoins with the margin (0.584 at the identity), which the body already states as its never suppressing.

Where ± 0.15 came from, reconstructed and not pre-registered. The margin was frozen at 5130cec with *no written rationale*, which we state plainly rather than supply a derivation after the fact. What the pre-freeze record does support: the effect the design was built against was Round 11's own +0.762 rise at $\rho=54.6$, five times the margin, and the margin never stands alone, the frozen rule conjoining it with *every rung* < 0.5 . So the margin is deliberately generous and **the conjunction is what stops a generous margin from certifying a high-ASR arm**; on the flagship cell every rung's *mean* in fact sits below 0.07, the quantity the frozen rule scores.

M COMPOSITION AND ADAPTIVE-ATTACK DETAILS

M.1 COMPOSITION BASELINE DETAILS

Simultaneous composition (NormClip $\tau=5$ then reputation every round), $N=10$, $K=5$, $f=0.2$, 50 rounds, cifar_cnn, 5 seeds. Per-seed results:

Seed	model_scaling		backdoor_pixel	
	Accuracy	ASR	Accuracy	ASR
42	0.783	0.766	0.774	0.837
43	0.788	0.783	0.774	0.792
44	0.794	0.835	0.784	0.828
45	0.762	0.889	0.771	0.931
46	0.780	0.763	0.766	0.823
Mean $\pm$ std	0.781 ± 0.011	0.807 ± 0.048	0.774 ± 0.006	0.842 ± 0.047

Composition partially suppresses scaling (ASR 0.807 vs. pure NormClip ~ 0.95 , and 0.961 under the NE3 temporal-mixing policy defined in §M.2) but does not approach pure reputation (0.017). Norm clipping preserves the attack direction while reducing magnitude; the clipped update then passes reputation’s consensus-distance filter. Pixel attacks are similarly persistent under composition (ASR 0.842). Both attack families remain effective despite simultaneous application of both defenses.

M.2 DBA AT $f=0.4$: PER-SEED DETAILS

Three run labels are carried over from an earlier framing and used here only to identify runs, not to make any equilibrium claim: *NE3* is the mixed server policy that framing computed, which randomizes over defenses across rounds; *reputation eq* is the pure-reputation policy; and *BR* is the adversary’s best response to the policy it faces. NE3 BR on CIFAR-10 with DBA in place of model scaling at $f=0.4$ (raising adversarial co-participation to $K \cdot f = 2.0$ from the $f=0.2$ value of 1.0). $N=10$, $K=5$, 5 seeds 42–46, 50 rounds. Per-seed realized ASRs:

seed	DBA realized ASR	backdoor_pixel realized ASR
42	0.222	0.972
43	0.103	0.945
44	0.176	0.913
45	0.603	0.993
46	0.077	0.934
mean $\pm$ std	0.236 ± 0.190	0.952 ± 0.028

The DBA per-seed spread is $7.8\times$ between min and max; only seed 45 clearly persists, three (0.08, 0.10, 0.18) are below the threshold (≥ 0.20) we would call active persistence, and one (0.22) is borderline. Pixel persists cleanly at $f=0.4$ (0.952 ± 0.028), so the persistence mechanism survives an adversarial-fraction shift on the original attack. DBA’s seed-variability reflects per-round co-participation alignment (whether ≥ 2 of the 4 adversarial clients land in the same round to coordinate the distributed trigger) rather than the structural argmax-collapse mechanism we describe for scaling. The cleanest evidence for the headline persistence claim therefore remains model scaling; DBA at $f=0.4$ is suggestive but not a robust second-attack demonstration.

M.3 DBA AT $f=0.6$: PER-SEED DETAILS AND BIMODALITY REPRODUCTION

Follow-up to Appendix M.2, designed to test whether the DBA bimodality at $f=0.4$ is purely a per-round co-participation effect or whether it persists as a per-seed admissibility latent at higher f . At $f=0.6$, expected adversarial co-participation per round is $K \cdot f = 3.0$, making ≥ 2 adversarial co-participation per round near-certain. NE3-style BR with committed adversary playing DBA; 5 seeds (42–46), 50 rounds, $N=10$, $K=5$, cifar_cnn. Two deployed policies: *NE3 mix* (FedAvg 26% + NormClip 74%) and the *reputation equilibrium* (NormClip 83% + reputation 17%).

seed	DBA ASR (NE3 mix)	DBA ASR (reputation eq)
42	0.360	0.339
43	0.295	0.278
44	0.386	0.415
45	0.734	0.705
46	0.154	0.130
mean $\pm$ std	0.386 $\pm$ 0.192	0.373 $\pm$ 0.190

Observations. (i) Mean DBA ASR improves over $f=0.4$ (0.236) to ~ 0.38 on both policies, but does not reach the ≥ 0.85 persistence band scaling achieves. (ii) Per-seed bimodality persists at $f=0.6$: seed 45 alone reaches > 0.7 in both policies; the remaining four seeds cluster in $[0.13, 0.42]$. (iii) Crucially, *the same seed is the strong-persistence seed under both policies* (NE3 mix: 0.734; reputation eq: 0.705), and *the same four seeds are the weak-persistence seeds under both*: the per-seed cross-policy correlation rules out per-round co-sampling alignment as the dominant source of bimodality. The bimodality is a per-seed training-initialization latent in DBA itself: whether the distributed trigger embeds at all in the early rounds. This is the DBA analogue of the model-scaling binding-cell Bernoulli, but at a much lower mean (admissibility prevalence $\approx 1/5$ for DBA at $f=0.6$ vs. $\approx 4/5$ for scaling at $f=0.2$).

Pre-specified verdict: INTERMEDIATE. Overall mean 0.380 falls between PARTIAL (> 0.5) and NEGATIVE (< 0.3). Reading: DBA at $f=0.6$ does *not* replicate the persistence collapse with the cleanliness scaling achieves; the trigger-alignment mechanism has its own admissibility constraint we do not control by raising f . The cleanest headline-persistence evidence remains model scaling under norm-based menus, as acknowledged in L3.

N SCALE, ARCHITECTURE, AND EXTERNAL VALIDITY

N.1 SCALE VALIDATION ($N = 100$) AND DEFENSE-INFERENCING CHECK

Scale check at $N = 100$ (4×5 game, 5 seeds). Residual material from an earlier framing, recorded rather than dropped. Nothing in the screen, the identification result or the dose-response evidence depends on it. Write VoPD (value of policy diversity) for the gain the server’s equilibrium strategy achieves over its best *pure* defense, so VoPD = 0 exactly when the equilibrium is pure. The equilibrium is pure on seeds 43 to 46 (model scaling vs. RFA, VoPD = 0) and mixed on seed 42 (pixel + scaling vs. FedAvg + RFA, VoPD = 0.012); per-seed values in Table A18.

Table A18: Per-seed NE, CIFAR-10 $N = 100$ (4×5 game). Seeds 43–46 identical.

Seed	Type	Adversary	Server	VoPD
42	Mixed	Pixel + Scaling	FedAvg + RFA	0.012
43–46	Pure	Model scaling	RFA	0.000

N.2 RESNET18 ARCHITECTURE-REPLICATION MINI-SWEEP: PER-SEED DETAILS

Per-seed details for the ResNet18 architecture-replication mini-sweep behind the architecture-conditional collapse boundary in §H. `resnet18` with BatchNorm replaced by GroupNorm (8 groups), conv1 adapted to 32×32 (kernel 3, stride 1, maxpool $\rightarrow$ Identity); 11.2M parameters. Identical FL parameters to `cifar_cnn`: $N=10$, $K=5$, $f=0.2$, 50 rounds, $\tau = 5$. Committed-scaling BR, 3 mix points $\times$ 5 seeds = 15 runs at ~ 100 min/run on MPS.

mix (NC/rep)	s42	s43	s44	s45	s46	ResNet18 mean	cifar_cnn (R50)
90/10	0.926	0.929	0.965	0.974	0.962	0.951 $\pm$ 0.020	0.945 $\pm$ 0.021
50/50	0.924	0.971	0.813	0.920	0.903	0.906 $\pm$ 0.052	0.815 $\pm$ 0.104
10/90	0.917	0.733	0.783	0.975	0.915	0.865 $\pm$ 0.091	0.097 $\pm$ 0.090

NC-dominant endpoint replicates within 0.01; mid-mix is $\sim 9\%$ higher on ResNet18; the reputation-dominant endpoint shows an order-of-magnitude gap (no CI overlap). Seed 45’s accuracy collapses to 0.153 at NC10/rep90 (the persistent attack effectively breaks the model) while trigger ASR remains 0.975. **Verdict: MEDIUM** (monotonic ordering preserved, endpoint criterion fails). The `cifar_cnn` collapse boundary at ~ 70 – 90% reputation is therefore a property of the smaller-architecture regime. This sweep bounds the architecture-conditional claim of §H and establishes no boundary of its own; we draw no conclusion here about which mixture of the two defenses survives, since mixture policies are outside this paper’s scope (§C).